

A Theory of Payments-Chain Crises*

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First Released Draft: December 2021

Abstract

This paper introduces an endogenous network of payments chains into a business cycle model. Motivated by evidence of linked payments in Costa Rica, we present a model where agents order production in bilateral relations. Some payments are executed immediately. Other payments, chained payments, are delayed until other payments are executed. Because production starts only after orders are paid, chained payments induce production delays. In equilibrium, agents choose the amount of chained payments given interest rates and access to internal funds or credit lines. This choice determines the payments-chain network and aggregate total-factor productivity (TFP). The paper characterizes equilibrium dynamics and their innate inefficiencies. Agents internalize the direct costs of their payment delays, but do not internalize the costs induced onto others. This externality produces novel policy insights and rationalizes permanent reductions in TFP under excessive debt.

Keywords: Payments, Networks, Business Cycles

JEL Classification: E32, E42, G01

*We thank Luigi Bocola, Ricardo Lagos, Jennifer La'O, Ezra Oberfield, Michael Peters, Guillaume Rocheteau, Gary Richardson, Pierre-Olivier Weill, and Yuliy Sannikov as well as seminar participants at UC Irvine, UCLA, CEMLA, CESifo, for early discussions. Sebastian Merkel provided a thoughtful discussion. We are in debt with Ken Miyahara and Luis Yepez for their outstanding assistance, multiple discussions, and their creativity in tackling proofs. The views expressed herein are those of the authors and do not necessarily represent the views of the Central Bank of Costa Rica (BCCR). Results have been reviewed by the BCCR to ensure no confidential information is disclosed.

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Says A: *I could use some of B's goods; but I have no cash to pay for them until someone with cash walks in here!*
Says B: *I could buy some of C's goods, but I've no cash to do it 'till someone with cash walks in here.*

From the book *Stamp Scrip*, Irving Fisher, 1933

1. Introduction

During financial crises, there are visible declines in credit variables. Though harder to measure, there is also a widespread perception that the normal, seamless flow of transactions and goods slows down. In ways still not fully understood, productive resources appear to remain idle—not because of technical constraints or lack of demand, but because agents must wait for others to pay, who in turn are waiting on others.

A simple example illustrates the idea. Consider a general contractor hired to build a commercial property. The contractor relies on a network of subcontractors—electricians, plumbers, and carpenters—who agree to begin or continue work only after receiving partial payment. The contractor also places orders with material suppliers, who may customize some of the materials and are unwilling to ship them until payment is completed or, at minimum, a proof of funds is provided. However, the contractor is waiting for a payment from the project's developer, who is waiting for a loan installment or the proceeds from the sale of another property. During crises, banks may delay loan payments and property sales may slowdown, disrupting the developer's funding sources, and therefore, the entire chain: the contractor cannot pay the electrician or the supplier; the electrician postpones work; the supplier holds off on shipping; and the project stalls. The slowdown is not due to a lack of labor, materials, or technical capacity, but rather to each party waiting to be paid by someone else. These cascading delays can leave productive resources idle, and the national accounts will record the outcome as a decline in productivity.

Despite how natural they seem, such transactional delays are rarely featured in macroeconomic models. Production and trade are typically centralized, with goods exchanged on spot markets and payment frictions largely abstracted away. While supplier financing and credit constraints are sometimes modeled, the usual assumption is binary: agents are either constrained or unconstrained. What is missing is a tractable framework in which economic activity can stall simply because agents must wait for other transactions to clear before making their own payments. In such settings, even solvent and willing producers may remain idle—not because they face traditional financial constraints, but because payment is delayed somewhere upstream.

This paper proposes a model of payments chains and studies their macroeconomic implications, particularly during financial crises. Developing a theory of payments-chain disruptions is important for understanding observed dynamics in productivity. Since the onset of modern business cycle analysis, economists have argued that fluctuations in total factor productivity (TFP) often reflect credit-market conditions (e.g., [Summers, 1986](#)). This view becomes especially salient during crises in developing economies, where contractions are frequently driven by large declines in measured TFP. Financial crisis models often explain these declines through input misallocation. However, replicating large observed TFP drops remains challenging, particularly given the relatively smooth labor flows during such episodes.¹

Payment-chain disruptions offer an alternative explanation for this co-movement. Delays in the timing of payments can propagate through production networks, leading to widespread idle time and reduced output, even when inputs are available. Through this mechanism, it is not the quantity or allocation of inputs that constrains production, but the timing: the slowdown of payments delays the utilization of resources.

These forces are salient empirically. Section 2 leverages detailed microdata on the universe of interbank transfers in Costa Rica, along with transaction-level details for each sale in the economy, to present the first evidence that many firms depend on payments made to them to execute further payments. In particular, about one-half of firms consistently execute payments on the exact day they receive a payment. We further provide on the impact of chained payments given a financial disruption by exploiting a natural experiment in which bank accounts were suddenly frozen for a subset of firms, and analyze how the consequences of this payment disruptions largely depend on whether impacted firms were part of a payments-chain.

Guided by this evidence, and to model payment-chain disruptions and their impact on production, we introduce a payment-chain production network in Section 3. This framework explicitly links the timing of economic transactions to the timing of production. In the network, production takes place through random bilateral relationships, where customers place orders with suppliers. Some of these are spot orders, which are paid upfront and trigger immediate production. Others are chained orders, which are

¹The puzzle of large drops in TFP and capital utilization alongside small labor movements was first noted by [Meza and Quintin \(2005\)](#). It has since appeared in several contexts: [Meza and Quintin \(2007\)](#), [Mendoza \(2010\)](#), [Oberfield \(2013\)](#), and [Karabarbounis et al. \(2021\)](#) document similar patterns in the Chilean banking crisis of the early 1980s, the Mexican and East Asian crises of the 1990s, and the recent Greek crisis. These studies cite numerous additional examples.

only paid once a preceding order has been paid. Since funds are transferred with a delay, chained orders introduce production lags that can cascade through the network. Chained transactions correspond to situations like the construction example above: a subcontractor, such as a carpenter, is scheduled to work but waits to begin work until the general contractor receives payment from the developer. Each chained payment depends on the clearance of prior payments along the chain.

A payments-chain network consists of many chains, each representing a linked transaction sequence. The first payment in each chain is a spot payment, financed by resources external to the network—such as savings, working capital credit, or accounts payable—and initiates the sequence. All subsequent payments are chained and rely on the completion of earlier payments in the chain. Thus, chained orders are contractual obligations whose fulfillment is contingent on the settlement of prior transactions.

A key implication of the model is that the further downstream an order lies in a payment chain, the longer the production delay, since each chained transaction introduces a lag in transaction times. The fraction of chained orders in the economy determines the overall structure and length of these chains: as the share of chained orders rises, the distribution of chain lengths shifts to longer chains, and average waiting times increase. These delays, in turn, reduce aggregate productivity. Specifically, measured TFP in this environment is given by: $\mathcal{Y}(\mu; \delta) = (1 - \mu) + \mu \frac{\delta - \delta\mu}{\mu - \delta\mu} \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) < 1$, where μ is the fraction of chained orders relative to total orders and δ is a parameter that captures production/transactions delays.

The payments-chain network developed in Section 3 is a stand-alone, mechanical production block where transactions are taken as given. However, it is portable to a range of contexts where agents decide to make spot or chained transaction agreements. In the second part of the paper, Section 5, we embed this network into a tractable, deterministic business-cycle model.

In that setting, a new payments-chain network is formed in each period. There is a natural borrower and a natural saver which, for simplicity, we model as households. By construction, the saver always has funds to place spot orders. By contrast, the borrower carries outstanding debt. The borrower can obtain funds to place spot orders by borrowing from exogenous credit lines. However, her credit lines are limited by her outstanding debt. When outstanding debt impairs her access to short-term funds, she can still place chained orders. However, placing chained orders is privately costly, as goods bought through chained orders are more expensive. Critically, whereas borrow-

ers internalize that chained orders are more expensive, they do not internalize that by placing chained orders, they delay other transactions.

The formula for TFP $\mathcal{Y}$ showcases how credit conditions translate into TFP declines. When short-term funds are limited, borrowers place chained orders to maintain a consumption level. This tightening of short-term funds increases the fraction of chained orders, μ . Measured TFP falls as labor inputs remain idle.

In the environment, the evolution of debt impacts TFP by impairing credit lines. In turn, TFP influences the desire to accumulate debt. There are three critical regions of the state space. For debt levels below an efficient-debt threshold, the economy is in a steady state without delays. For moderate debt levels, the economy features a *temporary* payments-chain crisis. Production delays last until the economy transitions toward an efficient steady state. For such moderate debt levels, crises are only temporary because borrowers have incentives to repay debts—to access short-term funding in the future. When debt is above a threshold, the economy features hysteresis, permanent payments-chain crises. Hysteresis occurs when borrowers decide to maintain high debt levels because the benefits of deleveraging happen too late into the future and require an excessive sacrifice of current consumption. The economy remains with permanent production delays in this debt overhang region of the state space.

The environment leads to novel policy implications presented in Section ???. We study a Ramsey planner that has instruments to influence the path of debt, but cannot tax spot and chained expenditures differentially. This Ramsey planner respects the payments technology but, as opposed to households, internalizes the effects of chained orders on TFP. The exercise illustrates that transitions are constrained inefficient for two reasons: savers spend too little in spot orders, and borrowers spend excessively in chained orders. Because the inefficiency is two-sided, debt may be excessively high or excessively low relative to the social optimum during a payments-chain crisis.

We also revisit fiscal multipliers. We allow the government to make spot chained expenditures to tax receipts. Although all forms of government spending are waste, government spending can produce positive multipliers during a payment-chain crisis. A novel insight is that multipliers are positive only if the government makes spot expenditures. If government expenditures are chained to tax income, they are detrimental. The reason for positive multipliers is not the stimulus of aggregate demand, a conventional view regarding the benefits of fiscal policy. Rather, government expenditures stimulate output by speeding up payments, a monetarist reinterpretation.

Literature Review. The literature on financial crises is vast. This paper connects with theories that underscore the sharp declines in aggregate TFP. The link between financial crisis and TFP is not at all obvious because financial crises can manifest themselves through distorting labor-market outcomes, not productivity. One branch of the literature explains declines in aggregate TFP through increased misallocation—see [Pratap and Urrutia \(2012\)](#) or [Oberfield \(2013\)](#). However, a common finding is that heterogeneity can only partially explain TFP declines once models are disciplined with data on input use and heterogeneity. Here, the channel is utilization. Other studies also explain declines in TFP as aggravated financial conditions increase the cost of utilizing capital—see [Meza and Quintin \(2007\)](#). The mechanism here is different: financial conditions impact TFP through a slowdown of payments.

Beyond the focus on financial crises, the paper falls at the crossroads of several areas. Namely, the monetary-payments literature, the economic-networks literature, and the literature on aggregate-demand externalities. The issue of how payment instruments affect production is a classic theme: [Lucas and Stokey \(1987\)](#) analyzes a stochastic cash-in-advance economy; [Kiyotaki and Wright \(1989\)](#) studies trade with indivisible tokens; [Lagos and Wright \(2005\)](#) a model with divisible money and explicit trading arrangements.² Recent work focuses on how the distribution, and not the instruments per se, affects production—see [Lippi et al. \(2015\)](#), [Rocheteau et al. \(2016\)](#), and [Brunnermeier and Sannikov \(2017\)](#). Like this literature, the distribution of funding affects allocations, and the main distinction is that we focus on delays in sequential payments.

Sequential payments appear in many other studies. The payments-chain network is inspired by the model in [Kiyotaki and Moore \(1997a\)](#). [Kiyotaki and Moore \(1997a\)](#) focus on finding micro-foundations of production disruptions in a single chain. The contribution here is to present a network of transactions whose structure depends on payment decisions. Furthermore, we embed the payments-chain network into a standard business-cycle model. Other models of sequential payments include [Townsend \(1980\)](#), which studies sequences of payments with spatial separation, [Freeman \(1996\)](#) and [Green \(1999\)](#), which study sequential transactions in overlapping generation environments, [La’O \(2015\)](#), which studies a circular flow of transactions, or [Guerrieri and Lorenzoni \(2009\)](#), which studies sequential transactions in a [Lagos and Wright](#)-type environment. Recent work by [Hardy et al. \(2022\)](#) and [Bocola \(2022\)](#) contrasts payments

²See [Shi \(1997\)](#); [Lagos et al. \(2011\)](#); [Lagos and Rocheteau \(2009\)](#); [Li et al. \(2012\)](#); [Nosal and Rocheteau \(2011\)](#); [Rocheteau \(2011\)](#) for many other directions in that area.

funded externally against trade credit.³ Relative to these papers, there are two distinctions: here, transactions are formed in a network, and the network is endogenous to expenditure-savings decisions. The importance of this body of theoretical work is substantiated by a body of empirical evidence found in a number of recent papers: [Boissay and Gropp \(2007\)](#), [Jacobsen \(2015\)](#), [Barrot \(2016\)](#), and [Costello \(2020\)](#) among others.

With respect to the economic networks literature, the paper connects with models with endogenous network formation. The contribution relative to that literature is modest, as network formation is not strategic. By contrast, in [Oberfield \(2018\)](#), a network is formed through strategic partnerships. In [Kopytov et al. \(2022\)](#) and [Elliott et al. \(2022\)](#) firms form strategic links, being aware of possible supply-chain breakdowns. Here, the network is randomly formed, but the distribution of chains is endogenous to financial decisions. Like in [Elliott et al. \(2014\)](#), [Alvarez and Barlevy \(2021\)](#) and [Taschereau-Dumouchel \(2022\)](#), there are externalities here too. In those models, externalities occur when individual defaults provoke subsequent defaults. Here, externalities occur through payment delays. [Bigio and La'O \(2013\)](#) considers the propagation of financial shocks that induce misallocation in a production network. Here, there is no misallocation in the production network, but there are production delays.

Finally, the paper connects with models of aggregate demand externalities. An early model of these externalities is [Diamond \(1982\)](#) where, via search, consumption decisions affect output. In most of the literature, demand externalities result from nominal rigidities. There has been a recent interest in coupling nominal rigidities with financial constraints—for example, [Eggertsson and Krugman \(2012\)](#) and [Guerrieri and Lorenzoni \(2017\)](#). Recent papers have further introduced sequential transactions into environments with nominal rigidities—for instance, [Woodford \(2022\)](#) and [Guerrieri et al. \(2022\)](#). In those models, demand externalities occur when agents cut back on any form of expenditure. The nature of demand externalities here is different. In particular, the type of expenditures by the private or public sector matters: spot orders may stimulate output, but chained orders depress it. The demand externalities provoked by the slowdown of payments is part of a classic narrative. Almost a century apart, the opening quotation taken from [Fisher \(1933\)](#) and the renowned “babysitting co-op” analogy of [Krugman \(1998\)](#) belong to that same tradition. The rest of the paper attempts to provide a convenient analytic formulation.

³See also [Biais and Gollier \(1997\)](#).

2. Empirical Motivation

This section documents empirical evidence on payment chains, their widespread prevalence, and their substantial effects on firm-level productivity.

Data. We leverage two exceptionally rich data sources. The first is Sinpe—Costa Rica’s National Electronic Payments System—which processes and settles every interbank transfer in the country.⁴ In 2024 alone, Sinpe recorded over 62 million transfers, capturing, for each transaction, the precise timestamp, unique identifiers of sender and receiver, the banks involved, and the exact amount transferred. This unparalleled coverage yields a complete universe of interbank activity for 243,234 distinct firms, with an average transaction size of USD 4,507.⁵ The second dataset derives from Costa Rica’s VAT (Value-Added Tax) reporting system, which mandates an electronic invoice for every formal sale. For each invoice, the registry records the sale price, product code, and quantity sold, and flags whether payment was settled on the spot or was deferred. Deferred payments are recorded, in particular, because as long as the seller has not received full payment, classifying the payment as deferred allows her to defer the VAT payment on the unpaid amount until the payment is due, for up to 90 days. For deferred payments, the system also records the contractual repayment window. This feature provides direct, time-stamped insight into short-term financing arrangements and payment timing at a level of granularity rarely available.

Documented Regularities. We exploit the universe of interbank transfers among firms to test for systematic payment-chain behavior. Absent such chains, there should be no systematic relationship between injections of liquidity and the timing of payments of a firm. Thus, to empirically detect chained payments, we first examine if firm j is more likely to issue a payment within 24 hours of receiving one.⁶ Specifically, we estimate:

$$sent_{jt} = received_{j,t-1} + \lambda_t + \lambda_{j \times d} + \varepsilon_{jt}, \quad (1)$$

⁴Sinpe is an acronym for Costa Rica’s National Electronic Payments System (*Sistema Nacional de Pagos Electrónicos*), and is operated by the Central Bank of Costa Rica to process and settle all electronic interbank transactions.

⁵For more details on Costa Rica’s firm network, see Méndez and Van Patten (2025); Argente et al. (2025). A related paper is also Alvarez et al. (2023), that instead focuses on a particular branch of Sinpe—Sinpe Movil—which only pertains to peer-to-peer mobile money transfers.

⁶We begin with this narrow window to strengthen the inference of a direct link between payments received and payments sent; later, we extend our analysis to wider intervals.

where $sent_{jt} = 1$, if firm j made a payment on business day t , and $received_{j,t-1} = 1$ if firm j received a payment in the preceding 24 hours.⁷ To prevent this relationship from being purely mechanical, we include both time fixed effects and firm $\times$ day-of-week fixed effects.⁸

Table 1 reports the results. Column (1) shows that, on average, firms are significantly more likely to send a payment within 24 hours of receiving a payment. Reassuringly, payments are only triggered when the amount sent is *less* than the amount received, as shown in Table 3. Columns (2)–(4) reveal heterogeneity: the payment–chain effect is strongest among small firms (5 workers or less) and attenuated for firms with higher asset-to-sales ratios or greater loan access relative to sales.

Table 1: Triggered Payments Across Firms

Dependent variable: Probability that firm j makes a payment at time t

	(1)	(2)	(3)	(4)	(5)	(6)
$received_{j,t-1}$	0.030 (0.0005)***	0.010 (0.0008)***	0.039 (0.0008)***	0.038 (0.0008)***		0.030 (0.0005)***
$Small_j \times received_{j,t-1}$		0.017 (0.001)***				
$\frac{Assets_j}{Sales_j} \times received_{j,t-1}$			-0.019 (0.001)***			
$\frac{Liabilities_j}{Sales_j} \times received_{j,t-1}$				-0.018 (0.001)***		
$received_{j,t-5}$					0.011 (0.0004)***	0.010 (0.0004)***
Adjusted R ²	0.483	0.446	0.483	0.483	0.483	0.483
Observations	10,001,248	5,290,408	8,399,599	8,399,599	10,001,248	10,001,248
Clusters	222,306	74,566	174,896	174,896	222,306	222,306
$t, j \times d$ FE	Yes	Yes	Yes	Yes	Yes	Yes

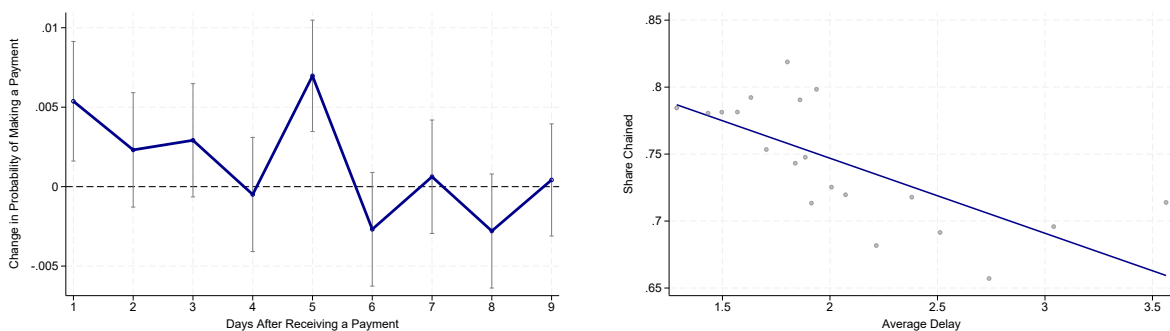
Notes: The independent variable in column (1) is an indicator that equals one if firm j received a payment from another firm in the last 24 hours. This variable is interacted with firm j 's size, assets-to-sales, and liabilities-to-sales in columns (2), (3), and (4), respectively. Column (5)'s independent variable is an indicator that equals one if firm j received a payment within 5 days of time t . Column (6) includes both indicators described above. Robust standard errors, adjusted for clustering by firm, are in parentheses. We include time and firm $\times$ weekday fixed-effects. Data is daily and spans 2024.

⁷Payments received on weekends or holidays are assigned to the next business day.

⁸Time fixed effects capture calendar-driven patterns—e.g., firms that always pay or receive on month-end or the 15th. Firm $\times$ day-of-week fixed effects absorb idiosyncratic weekly rhythms—e.g., some firms systematically pay on Thursdays, others on Fridays—thereby isolating only the abnormal deviations linked to incoming payments. The latter fixed effect would also discipline variation, for instance, from a firm that tends to make payments every day, and instead would force the variation to come from out-of-the-ordinary responses correlated with receiving a payment.

Columns (1)–(4) of Table 1 focus on a one-day window to approximate chained-payment responses. To trace the temporal profile of this effect for all firms, we implement a local projection exercise (Jordà, 2005; Jordà and Taylor, 2025). Panel (a) of Figure 1 illustrates that, while many firms issue triggered payments within one business day, there is also a second peak—nearly as large as the initial response—around five business days (one week) after receipt of funds. Moreover, coefficients in columns (1), (5), and (6) are remarkably stable, suggesting that payments unfold in two orthogonal waves—an immediate response and a delayed one around business day five.⁹

This two-wave structure reflects both the speed of Costa Rica’s modern payment infrastructure and residual frictions. In 2024, when Sinpe processes most transfers electronically, same-day responses are technologically feasible.¹⁰ However, even with a rapid payments network in place, a subset of firms continues to rely on paper checks, which take several business days to clear. This check-clearance lag produces the pronounced second spike at business day five. Consistent with this mechanism, firms exhibiting the five-day payment lag are roughly 60% more likely to pay by check than the rest. When we exclude these check-using firms, the day-five peak in Panel (a) of Figure 1 disappears, confirming that check-clearing times underlie the longer delays.



(a) Triggered Payments: Local Projection (b) Chained Payments vs. Average Delays

Figure 1: Dynamics of Chained Payments

Notes: Panel (a) shows a local projection (Jordà, 2005; Jordà and Taylor, 2025), where the dependent and independent variables have 13 lags, as determined by a VAR-based optimal lag selection criteria. Panel (b) shows the average share of firms classified as chained by sector against the average payment delay in the sector. Sectors correspond with ISIC sections. Observations are weighted by sales. Data is monthly and spans 2024.

⁹We also examine the intensive margin in Table 4, which yields results consistent with the extensive-margin analysis: firms are more likely to make payments once they receive transfers.

¹⁰By contrast, in the 1990s, most payments settled by check, and checks used to require up to 15 days to clear—so that the fastest possible response was about two weeks (Cerdas and Melegatti, 2014).

Placebo Exercises These relations are not mechanical. We conduct placebo tests, where we change the timing so that the independent variable represents payments received h days *after* making a payment, with $h \in \{1, 2, 3\}$. As shown in Table 5, results are consistently negative or insignificant, speaking against spurious temporal correlations.

Constructing Payments-Chains Leveraging the comprehensive ACH and wire transfers data, we aim to identify firms where the receipt of a payment triggers the issuance of subsequent payments. Our goal is to classify firms into three distinct groups: (i) those that tend to make chained payments without delay; (ii) those that tend to make chained payments with delay; and (iii) those that tend to make spot payments.

We proceed in two stages. First, to isolate firms chained without delay (type (i)), we compute the monthly frequency with which receiving any payment leads to issuing a payment *within one business day* for each firm. As shown in Panel (a) of Figure 13, this distribution has large masses of firms concentrated in a value of one (firms with perfect same-day response rates) and in a value of zero (firms with no same-day responses). Given this bimodal distribution, we then use k -means clustering to split the sample into two groups: firms type (i) and other firms.

Second, among those firms that were *not* classified as type (i), and informed by the local projection in Panel (a) of Figure 1, we consider the distribution of monthly occurrences in which receiving a payment led a firm to make a payment five days later. This distribution, shown in Panel (b) of Figure 13, shows that, in fact, few firms in the economy which are chained with delay, and allows us to further split this subsample into two groups: firms concentrated in zero, which correspond with type (iii)—spot, for whom payments sent are uncorrelated with payments received, and a few firms which are concentrated in one and are therefore type (ii)—chained with delay. This two-stage clustering yields a firm-month classification into three types; although firm types are highly persistent, our approach allows for type-switching from month to month.¹¹

Stylized Facts on Payments-Chains Once firms are classified, we construct payments-chain networks. Several remarks are in order. First, on a given month, 39% of firms tend to make spot payments, while the remaining 61% of firms engage mainly in chained payments. Among the chained firms, 80% make payments without delay and 20%

¹¹As an alternative, one can estimate each firm's fitted probability of triggered payments from the specification in column (6) of Table 1 and then apply k -means clustering with $k = 3$. The resulting spot vs. chained classification highly correlates with our baseline method, with a correlation of 0.89.

make payments with delay.¹²

In other words, chained firms are prevalent in the economy, a prevalence likely driven by the generally short delays most firms incur. If payment delays raise the cost of participating in a chained network, it is natural to ask whether longer delays deter firms from becoming chained. To explore this, Panel (b) of Figure 1 plots each sector's share of chained firms against its average payment delay. For instance, sectors in which firms tend to rely more on paper checks would have a longer average delay. The figure reveals a strong, negative relationship: sectors with longer delays host proportionally fewer chained participants.¹³

A second observation is that interactions across firm types are common. As shown in panel (a) of Figure 2, 26% of all links in the payments network are “mixed,” involving both a spot and a chained firm.¹⁴ Third, if we focus on transactions between same-type firms, network structures differ significantly. On the one hand, Panel (b) of Figure 2 shows the network for transactions between firms classified as chained, blue (green) dots representing firms chained without (with) delay; about 70% of all links. These links are particularly relevant as consecutive chained links ultimately form the payment chains. The network structure shown is a dense core with isolated nodes. The core cluster represents a critical mass of firms in tight transactional relationships and a circular dependency loop, while the outer nodes are less connected to same-type firms. Firms in this network's core are interdependent, which would relate to high systemic risk: if one core firm delays or defaults, it could ripple through the network. Finally, Panel (c) shows the remaining transactions, among firms classified as spot.¹⁵ The network structure is more diffused and suggests a decentralized transaction ecosystem.

With firm classifications in hand, we construct payment chains as network paths that begin and end at firms making spot payments, with chained firms potentially occupying the intermediate nodes. For each pair of spot firms connected through chained firms, we define the chain's length as the number of links (firms) separating them.¹⁶ Figure 14 displays the distribution of chain sizes for December 2024 with an

¹²The k -means clustering partitions firms at a fitted probability threshold of 0.42 when considering chained without delay vs. other firms, and of 0.15 when considering chained with delay vs. spot firms.

¹³Panel (b) of Figure 1 categorizes sectors by their one-digit ISIC sections. These broad groupings are chosen deliberately to have an input-output matrix with strongly dominant diagonals.

¹⁴This representation is based on December 2024. However, these shares are very stable: across all months in 2024, these shares are statistically equal and vary less than one percentage point from month to month.

¹⁵These red dots look less numerous than in Panel (a), as spot firms trade mostly with chained firms.

¹⁶To avoid double counting of overlapping fragments and to maintain computational feasibility at a

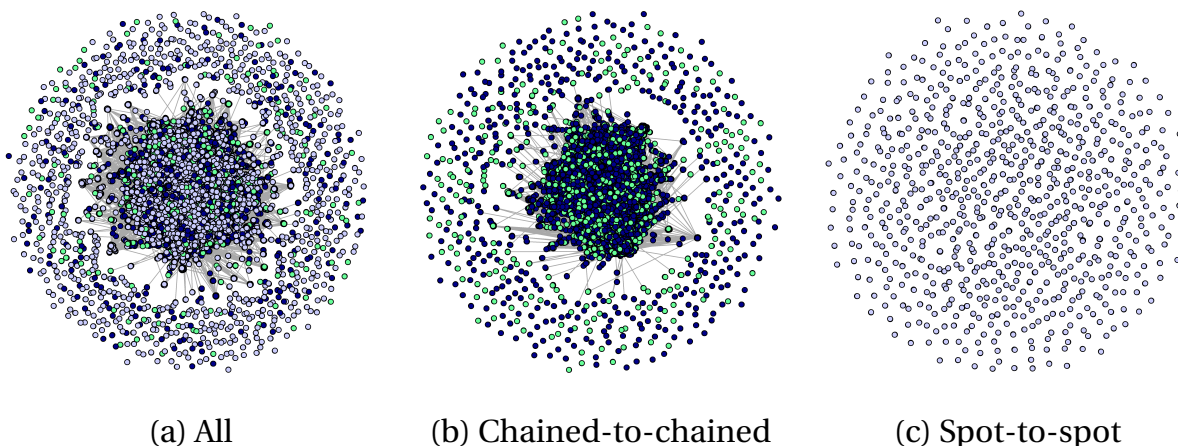


Figure 2: Network Structure by Firm Type

Notes: The figures show the network structure by link type. Panel (a) shows all links between firms. Panel (b) shows links between firms classified as chained. Panel (c) focuses on links between firms classified as spot. Data is collapsed for December 2024 to construct the network representation.

average chain of 5 links; results are nearly identical in other months, exhibiting only minimal month-to-month variation.

We further exploit the VAT registry of electronic invoices—linked at the firm level to Sinpe transfers—to observe prices and timing of payment—immediate or deferred, and if deferred, for how long—for firms classified as either chained or spot. Because sellers must issue an electronic invoice upon receipt of any partial payment, deferred transactions in the VAT system naturally capture chained-payment behavior. Indeed, firms classified as chained are 26% more likely to engage in deferred payments than spot-payment firms, with an average deferral of 34 days. Leveraging this rare level of detail, we find that, for a given product, invoice prices are 21% higher when payment is deferred, and that this premium is even larger when the purchasing firm is one that regularly makes chained payments. Appendix E.5 presents details of these results.

We also leverage daily data on deferred payments and itemized invoices to study how payment timing affects product delivery. Focusing on retail firms—where the items sold correspond exactly to the items purchased without any in-house transformation—allows us to match each input purchase directly to its downstream sale.¹⁷ For these firms, we isolate new-product transactions with deferred payment and ask: does

monthly frequency, we measure chain length by the shortest path between two spot firms in the payments network.

¹⁷For example, an automaker buys tires and parts but transforms them into cars, so input and output dates don't align neatly.

the retailer start selling the product as soon as the invoice is issued (potentially when receiving a partial payment), or only once the full payment arrives (recall the average deferral is 34 days)? For 69% of cases, sales of the product start only *after* the upstream seller has received payment in full. The timing of such transactions is shown in Figure 15—an event study centered on the date when *full* payment is due where the plotted coefficients measure store-level sales (our proxy for delivery).

A Payments-Chain Crisis A payments-chain crisis may arise when shocks to short-term liquidity propagate through chained transactions and impair downstream payments. We exploit a natural experiment to assess the empirical relevance of such crises.

In August 2024, Costa Rica’s *Financiera Desyfin*—a financial institution catering mainly to enterprises and business clients—was abruptly and unexpectedly intervened by the financial regulator, who immediately froze all customer accounts due to insolvency concerns. The regulator then declared the institution unviable and placed it into a resolution process under the Deposit Guarantee Fund law, leaving those funds inaccessible for several months.

Using Sinpe data, we can identify the directly shocked firms and trace how disruptions ripple through their surrounding payment networks. To build intuition, we will provide an example. Consider the pre-freeze network in panel (a) of Figure 3, in which arrows represent flows of payments. Firm D is classified as a spot payer, issuing transfers to chained firms downstream (to its right in the figure) with no systematic link between payments it received and sent. After the Desyfin freeze, however, firm D’s sent payments started happening only after receiving a payment, so that it “switched” to being classified as chained, as depicted in panel (b).

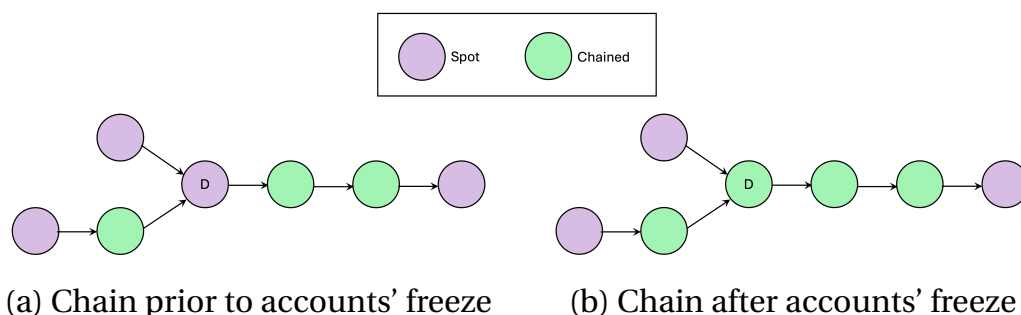


Figure 3: Example: Firm D with a Frozen Account “Switches” from Spot to Chained

Notes: The figures show an example of how the accounts’ sudden freeze might have impacted a chain in which one, formerly spot, firm “switched” to being classified as chained. Arrows represent payment flows, so circles positioned farther to the right correspond to firms that are further downstream.

With this example in mind, let $Desyfin_i = 1$ if firm i maintained an account at Financiera Desyfin during 2024, and let $Shock_t = 1$ for $t \geq \text{August 2024}$, when the freeze occurred. We also define $Switch2chained_{it} = 1$ if firm i changes from being classified as spot to being classified as chained in month t .¹⁸

We first explore if firms which had an account at Desyfin become more likely to “switch” to being classified as chained after the freeze. Column (1) of Table 2 reports that this occurs; the probability of switching from spot to chained increased in 0.008 pp—equivalent to a 17% rise relative to the low 0.05 baseline. These estimates include province×industry and time fixed effects to absorb regional or sectoral shocks.

Next, we examine how a “switch” to chained status impacts a directly shocked firm (D, in our example above). Namely, we consider the indicator $Switch2chained_{it}$ as an independent variable and study the behavior of a firm i that had an account at Desyfin *and* switched to being classified as chained. We find that firm i made 18% fewer purchases after the shock, which would align with a hindered ability to make payments, as shown in column (2) of Table 2. The same column shows that the term $Desyfin_i \times Shock_t$ has, on its own, a positive coefficient, consistent with positive pre-shock selection into Desyfin, which catered to corporate clients. In line with fewer purchases, column (3) shows sales fell and column (4) confirms a fall in value added of about 20%.

We finally ask how chained suppliers fare when selling to a directly shocked firm. Define $Selling2Desyfin_{jt}$ as an indicator equal to one if chained firm j was *selling to* a directly shocked “Desyfin” firm i . Panel (B) of Table 2 shows an 18% reduction in purchases after the freeze, along with a fall in sales and value added of a similar magnitude (columns 5-7), *but only when selling to a firm that switched to chained payments and held a Desyfin account*. These results underscore how payment chains can undermine economic resilience. Just as a project developer’s lack of funding can trigger cascading delays for the general contractor, subcontractors, and so on, a shock—such as a bank account freeze—that disrupts an initial spot payment can cause delays and hinder production for all firms down the payments-chain.

This effect on “one-link away” firms is persistent. While none of these firms was directly hit by a shock, Figure 4 shows that the effect remains at least four months after the shock occurred—the figure again focuses on firms selling to a Desyfin firm *which switched* to being classified as chained; i.e., we plot the interaction $Selling2Desyfin_{jt} \times$

¹⁸For this construction, recall that the classification of firms by type is month-specific and allows for type-switching across time. While the classification is flexible, firm types tend to be highly persistent.

Table 2: A Payments-Chains Crisis: Frozen Bank Accounts

A. Directly Shocked Firm i				
	$Switch2chained_{it}$	$\ln Purchases_{it}$	$\ln Sales_{it}$	$\ln Value Added_{it}$
	(1)	(2)	(3)	(4)
$Switch2chained_{it} \times Desyfin_i \times Shock_t$		-0.176 (0.064)***	-0.233 (0.086)***	-0.197 (0.109)***
$Switch2chained_{it} \times Shock_t$		-0.004 (0.577)	0.019 (0.010)**	-0.017 (0.012)
$Desyfin_i \times Shock_t$	0.016 (0.007)***	0.159 (0.048)***	0.195 (0.037)***	0.204 (0.048)***
% Δ w.r.t. mean [mean dep. var]	9% [0.18]			
Observations	1,200,404	1,192,436	939,984	636,485
Province $\times$ industry, t FE	Yes	Yes	Yes	Yes
B. Firm j Selling to a Directly Shocked Firm i				
		$\ln Purchases_{jt}$	$\ln Sales_{jt}$	$\ln Value Added_{jt}$
		(5)	(6)	(7)
$Selling2Desyfin_{jt} \times Switch2chained_{it}$ $\times Desyfin_i \times Shock_t$		-0.176 (0.029)***	-0.189 (0.033)***	-0.130 (0.038)***
$Switch2chained_{it} \times Shock_t$		0.038 (0.016)**	0.017 (0.019)	0.000 (0.023)
$Selling2Desyfin_{jt} \times Switch2chained_{it}$		0.651 (0.040)***	0.669 (0.044)***	0.661 (0.047)***
$Desyfin_i \times Shock_t$		0.035 (0.016)**	0.027 (0.019)	0.000 (0.023)
Observations		704,087	555,098	382,319
Province $\times$ industry and t FE		Yes	Yes	Yes

Notes: The dependent variable in column (1) is an indicator equal to 1 if firm i switched from being classified as a spot firm to a chained firm in period t . Columns (2), (3), and (4) investigate the impact of the freeze on firms who held accounts at Desyfin, before and after the freeze, differentiating between impacts if the firm switched to being classified as chained or not. Columns (5), (6), and (7) instead focus on firms that were selling to a firm directly connected to Desyfin, before and after the shock. Robust standard errors, clustered by firm, are in parentheses. We include province $\times$ industry (ISIC-4) and time fixed-effects. Data is monthly and spans 2024.

$Switch2chained_{it} \times Desyfin_i \times Shock_t$. The effect is also persistent along a chain. As documented in Figure 7, the effect propagates at least 3 links upstream from a directly hit Desyfin firm that switched to chained.¹⁹

¹⁹The sample size becomes smaller as we increase the number of links of separation, because fewer observations feature chains that long (see Figure 14). Thus, results beyond 3 links are too noisy to be conclusive.

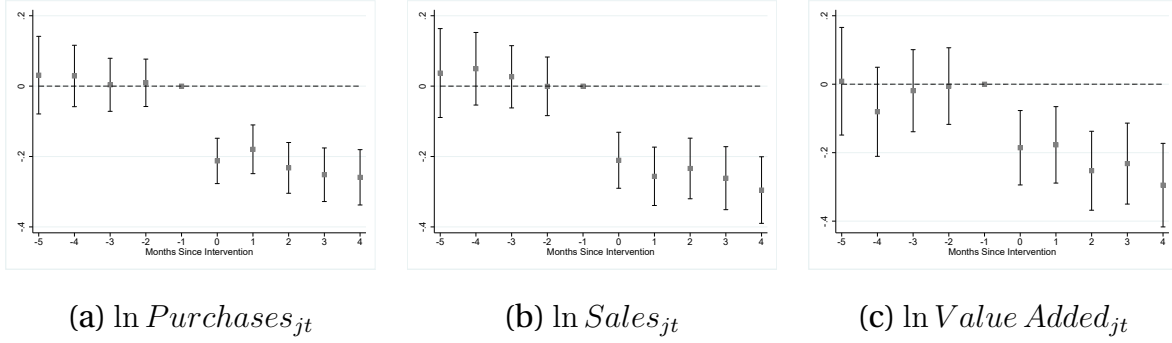


Figure 4: Persistent Effect Along the Chain

Notes: The figure shows an event study where time zero corresponds with the month when Desyfin accounts were frozen. The dependent variables in panels (a), (b), and (c) are the (log) of purchases, of sales, and of value added, respectively, of firm j in month t . The plotted coefficients correspond with the triple interaction of $Selling2Desyfin_{jt} \times Switch2chained_{it} \times Desyfin_i \times Shock_t$, and therefore captures the impact of being one link upstream from a Desyfin firm and switching to being classified as chained *after* the freeze. Robust standard errors are clustered by firm. We include province $\times$ industry (ISIC-4) and time fixed-effects. Data is monthly and spans 2024.

3. Payments-Chains and Productivity

Motivated by the empirical evidence of a payments-chain network, this section presents an approach to map a set of income and expenditures in the economy into a payments-chain network. When transaction delays entail an economic loss, the payment-chain network translates into TFP losses. We then embed this module into a dynamic business-cycle model.

Bilateral Relations. To fix ideas, assume that production is organized through bilateral agreements in which a customer orders a product from a supplier owning one production input. The agreement is exclusive: only the agent placing the order values the output once produced. In turn, production units are exclusively dedicated to producing for a specific client.²⁰ As examples of such relations, we can think of a home renovation project, a medical service, manufacturing an engineered product with a specific blueprint, or commissioning a piece of art.

There are two types of orders: *spot* and *chained* orders. Spot orders are paid immediately. Chained orders are contingent on previous payments; they are paid after the client receives a payment from another transaction. There are N production units. In turn, there are N^s spot orders and N^x chained orders. There are equal numbers of production units and orders, $N = N^s + N^x$. In the following section, we work with a

²⁰The agents' identity in these agreements is not explicit at this stage. Identities are explicit, and these matter for the economy's dynamics in the following section.

limit for $N \rightarrow \infty$ and recast this condition as a market-clearing condition. Each production unit is assigned an identifier, $i \in \mathcal{N} = \{1, 2, \dots, N\}$. Likewise, each order is assigned a unique identifier, $i \in \mathcal{N}$. We partition $\mathcal{N}$ into two sets, $\mathcal{N}^s$ and $\mathcal{N}^x$, to denote the set of identifiers of spot and chained orders, respectively.²¹ Also, we work under the assumption that payments are identical across orders, a condition we explain in the following section. We also provide an interpretation of spot and chained once we introduce financial decisions in that section.

We define two relations that together define the payments-chain network. First, $\mathcal{P} : \mathcal{N} \rightarrow \mathcal{N}$, is a one-to-one assignment from an order to a production unit. By assumption $i \neq \mathcal{P}(i)$. The interpretation is that $\mathcal{P}(i)$ is the unit that produces order i . The assignment $\mathcal{P}$ is entirely random: any unit can be assigned to any order with equal probability. Second, a chained income-expenditure relation associates a chained order with a production unit. This relation is the identity function defined on $\mathcal{N}^x$. That is, order i uses the payments to production unit i if $i \in \mathcal{N}^x$, as a source of funds to execute order i . The idea is that although order i is not externally funded, the customer who places order i owns production unit i . Hence, order i can be funded after order $j = \mathcal{P}^-(i)$ executes a payment. The assumption that the income-expenditure relation is the identity is innocuous.²²

To anticipate how the two relations induce a payments-chain network, consider production unit j assigned to order i , $\mathcal{P}(i) = j$. The client placing order i must pay for j 's production. This creates a payment link from i to j . In turn, if order j is spot, $j \in \mathcal{N}^s$, the funds used to pay for order i are unused in further payments. However, if order j is chained, the funds are used again to pay unit $k = \mathcal{P}(j)$. In other words, when $j \in \mathcal{N}^x$, there is a flow of payments from i to j , and from j to k . If order k is also chained, the same funds are used to pay unit $\mathcal{P}(k)$, and so on. The payments chain continues until a final order in a chain is placed on some production unit i not associated with a chained order. Since every order is paired with a production unit, the economy features an entire network of transactions forming a collection of payment chains.

The payments network is inconsequential if there are no delays in transferring funds.

²¹ $\mathcal{N}^s$ and $\mathcal{N}^x$ are a partition of $\mathcal{N}$: $\mathcal{N}^s \cap \mathcal{N}^x = \emptyset$ and $\mathcal{N}^s \cup \mathcal{N}^x = \mathcal{N}$. The number of elements is $N^s = \mathcal{N}^s$, $N^x = \mathcal{N}^x$.

²²Formally, we can define the chained income-expenditure relation $\mathcal{X}$ as the identity function on $\mathcal{N}^x$, that is $\mathcal{X} : \mathcal{N}^x \rightarrow \mathcal{N}^x$ such that $\mathcal{X}(i) = i$. The idea is that $i \in \mathcal{N}^x$ obtains funds from production unit, $\mathcal{X}(i) = i$. Indeed, the identity function $\mathcal{X}(i)$ can be replaced by any injective function $\mathcal{X} : \mathcal{N}^x \rightarrow \mathcal{N}$ so that production units and associated chained orders do not have the same identifiers. Changing identifiers does not affect the results.

Indeed, if every payment was executed immediately, the transfer of funds in the entire network would be instantaneous. Moreover, even if transfers are not immediate, the network would be inconsequential if no resources were lost as transactions are delayed. Of course, transaction delays can occur for many reasons: technology restrictions, inattention, or financial frictions. The economic consequences can be various: inventory depreciation, time-discounting, or utilization.

Here, we provide an example of how the network is consequential. We introduce two financial frictions that delay payments and, consequently, production. The first friction is that production can only start once the buyer shows proof of funds. If an order is spot, by definition, proof of funds is immediate since the order is funded externally. By contrast, chained orders can provide evidence of funds only after the buyer receives a payment from another transaction and the money enters their account. Production requires proof of funds to deter fraudulent behavior; otherwise, buyers could promise payments they know will never occur. Since production is customized, this would lead to a disadvantageous ex-post renegotiation.

We need a reason to delay transactions: here, we invoke limited commitment. The idea is that after the customer shows proof of funds, the funds are only released after the fraction $1 - \delta$ of the output is inspected. Without inspection, The producer could produce a good for himself, taking away the funds. Since there is no way to verify a customization, inspection is necessary. The delay in transactions provokes production losses. Namely, the payments-chain network determines production: production occurs within a unit time interval. Each order starts at some endogenous time $\tau \in [0, 1]$ when the payment is executed. Once production starts, it cannot stop. Production is linear in time: if production starts at τ , production is $1 - \tau$. The delay in the time to execute payments then translates into idle resources which map into TFP losses.

Placing these frictions together, proof of funds and limited commitment, we obtain a production structure where the greater the number of chained orders, the more payment delays and the longer the production delays. To produce predictions about aggregate TFP, we first formalize the definition of the payments chain network and derive statistical properties that help us map it into production losses.

Payments-Chain Network. Once we enumerate transactions, we can construct a network of transactions. A sequence of payments in the network that starts from a single original source of funds defines a **payments chain**. Naturally, every chain must start

with a spot order, followed by a sequence of chained orders in which those initial funds are transferred from one transaction to another. The number of chained orders defines the **chain length**. A **payments-chain network** is the collection of all payment chains, encompassing the universe of transactions in the production time interval. Formally:

Definition 1. A **payments-chain network** $\mathcal{K}$ is an acyclical directed network with degree 1 with nodes $\mathcal{N} = \{1, 2, \dots, N\}$ and links $\mathcal{V} = \{(i, j) \mid \mathcal{P}(i) = j, j \in \mathcal{N}^x\}$, $\mathcal{K} = (\mathcal{N}, \mathcal{V})$. A **payments-chain** of length n is a finite sequence of nodes $\{i_k\}_{k=0}^n$ such that the sequence starts with some $i_0 \in \mathcal{N}^s$ and $\forall k \in \{1, \dots, n\}$, $\mathcal{P}(i_{k-1}) = i_k \in \mathcal{N}^x$. By convention, if $i \in \mathcal{N}^s$ and $j = \mathcal{P}(i) \notin \mathcal{N}^x$, i and j define a chain of length zero.

Nodes corresponding with the payment chain network represent both orders and production units. The directed links represent the direction of the flows of funds. As production is bilateral, one or no links stem from each node. A link from i to j indicates that i orders from production unit j and that j is a chained order, $\mathcal{P}(i) = j \in \mathcal{N}^x$. The source of funds of order j is the funds paid in order i . In turn, if a node does not receive a link, it represents a spot order. Furthermore, if no links are directed toward order $j = \mathcal{P}(i)$, then order j is also spot—in which case i and j form a zero-length chain. In this construction, each order has a source of funds. Any (longest) path of links defines a payment chain. The collection of payment chains is the payment-chain network.

Examples. Let us provide an example. Set $\mathcal{N} = \{1, 2, \dots, 8\}$ and let the subset of spot orders be $\mathcal{N}^s = \{1, 3, 7\}$. Also, define the production relation as follows: let $\{i_n\}_{n \in \mathcal{N}} = \{1, 5, 7, 4, 6, 2, 3, 8\}$ such that $i_{n+1} = \mathcal{P}(i_n)$ and $i_1 = \mathcal{P}(i_N)$. Thus, the links in this payments-chain network are $\mathcal{V} = \{(1, 5), (7, 4), (4, 6), (6, 2), (3, 8)\}$.

Several graphs are associated with this example. The left panel of Figure 5 depicts the chained income-expenditure relation. In that panel, we split each node into counterparts: the production units $\{u_n\}$ and production orders $\{o_n\}$ for nodes, $n \in \{1, 2, \dots, 8\}$. The links represent the flow of funds from production units to their corresponding chained orders, defined by the chained income-expenditure relation.

The middle panel adds the links to the flow of payments for production, corresponding to $\mathcal{P}$. That is, the links in that panel add the payments production units. Adding the links of the chained income-expenditure and production relations allows us to trace funds. Notice that the links from orders to production units with the same color share a common source of funds.

The right panel depicts the payment chain network. In this example, there are three chains. The first chain is of length one—from 1 to 5. The second chain starts at 7, and links nodes 7, 4, 6, and 2. Since 4, 6, and 2 are chained orders, the length of the second chain is three. The third chain links node 3 with node 8 and is also of length one.

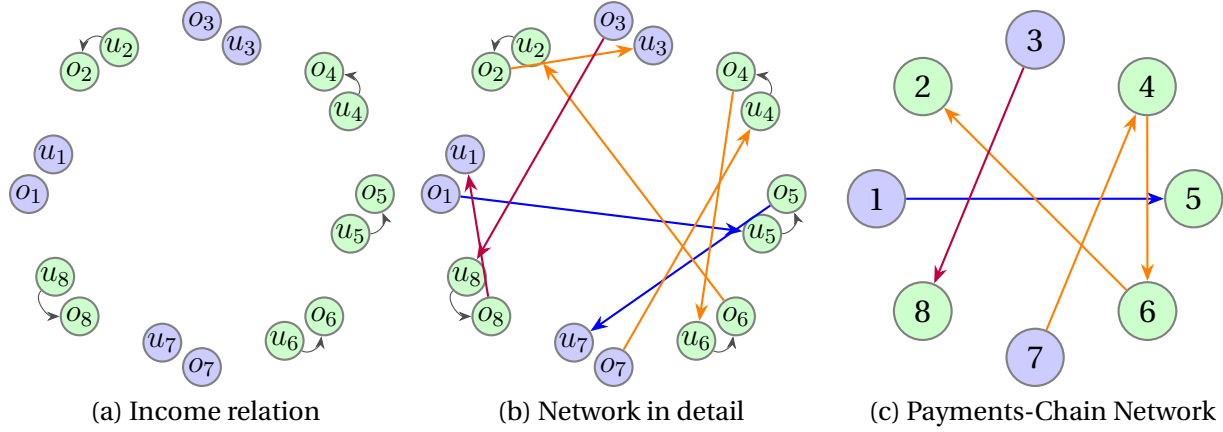


Figure 5: Components of the Payments-Chain Network

Chain-Length Distribution. We now derive the distribution of payment-chain length as the number of transactions increases, $N \rightarrow \infty$. As the number of transactions increases, we keep the ratio $\mu \equiv N^x/N$ constant. This fraction is a parameter of the probability distribution $G(n; \mu)$ chain length in the network.

Proposition 1. Let $n \in \{0, 1, 2, \dots\}$ be the length of a payment chain in the payments-chain network. Then, n is a random variable with probability mass function (p.m.f.) $G(n; \mu)$ where $G(n; \mu)$ is the *geometric p.m.f.* with parameter μ ; $G(n; \mu) = (1 - \mu) \mu^n$.

Proof. Recall that the production relation is random. As $N \rightarrow \infty$, a node has a link directed toward it if it corresponds to a chained order. Thus, a node is linked with probability μ . A node does not receive a link if it is spot, with probability $1 - \mu$. Furthermore, recall that each chained order is funded and thus belongs to specific payment chain which, in turn, starts with a specific spot order. Thus, there is a one-to-one relation between each payment chain and a spot order. Hence, there are N^s chains in total, which we can also index by $i \in \mathcal{N}^s$. If chain i forms a chain of length zero, it must be that $\mathcal{P}(i) \in \mathcal{N}^s$. This, happens with probability $1 - \mu$ —because each production link indeed occurs with equal probability. Likewise, chain i is of length 1 if $\mathcal{P}(i) \in \mathcal{N}^x$ but $\mathcal{P}(\mathcal{P}(i)) \notin \mathcal{N}^x$. This happens with probability $\mu \times (1 - \mu)$. In a chain of length two, there

are two consecutive chained orders followed by a spot order. This occurs with probability $\mu^2 (1 - \mu)$. Proceeding by induction, we arrive at the geometric distribution. $\square$

The length distribution is convenient for obtaining a measure of economic losses. For that, we turn to our specific micro-foundation.

TFP. We use this distribution to obtain aggregate production: we first derive the production in a chain of arbitrary length. In a chain of length zero, there is always a spot order whose production begins immediately. In a chain of length one, the production of the first spot order begins immediately, but there is a delay in the second order. For the second order, the funds are received after $1 - \delta$ of the production of the first order is finished. This happens at time $\tau = 1 - \delta$. Hence, this leaves only δ time to produce the second order in the chain. For the second order, production is δ of which the $1 - \delta$ fraction must be inspected. If there is a third order in the chain, the transfer of funds occurs $(1 - \delta) \delta$ time after the first transfer at $1 - \delta$. Adding these consecutive delays, the production of the third order can only start by $(1 - \delta) + \delta (1 - \delta) = 1 - \delta^2$. This leaves δ^2 time to produce that third order.

We can deduce a pattern by forward induction.²³ In a chain of length n , the production vector of $n + 1$ consecutive orders in a chain is $\{1, \delta, \delta^2, \dots, \delta^n\}$. Using the distribution of chain lengths to compute expectations, we obtain aggregate production.

Proposition 2. (Output per worker): *Given μ and δ , the average output corresponding to chained orders converges, as $N \rightarrow \infty$, to:*

$$\mathcal{A}(\mu; \delta) = \frac{1 - \mu}{\mu} \frac{\delta}{1 - \delta} \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) < 1. \quad (2)$$

$\mathcal{A}$ is strictly concave, decreasing, and satisfies $\lim_{\mu \rightarrow 0} \mathcal{A}(\mu; \delta) = \delta$, $\lim_{\mu \rightarrow 1} \mathcal{A}(\mu; \delta) = 0$, $\lim_{\delta \rightarrow 0} \mathcal{A}(\mu; \delta) = 0$, and $\lim_{\delta \rightarrow 1} \mathcal{A}(\mu; \delta) = 1$. Mean output is $\mathcal{Y}(\mu) = (1 - \mu) + \mu \mathcal{A}(\mu) \leq 1$.

This theorem is key as it yields a formula output: To obtain output, we first compute the average production among chained orders, $\mathcal{A}(\mu; \delta)$.²⁴ $\mathcal{A}$ is strictly concave and de-

²³If the k -th node initiates production at time $1 - \delta^{k-1}$, the delay from its inspection is $(1 - \delta) \delta^{k-1}$, which added to previous delays leads to a transfer of funds only by time $1 - \delta^{k-1} + (1 - \delta) \delta^{k-1} =: 1 - \delta^k$. This, leaves δ^k time for production to the subsequent unit. Since we computed the delay for $k = 1, 2, \dots$ the productions in a chain of length n are $\{1, \delta, \delta^2, \dots, \delta^n\}$.

²⁴To obtain output, we first compute the average production of chained orders in an n -length chain, $(n^{-1} \sum_{i=1}^n \delta^i)$. With the average production of each chained orders, we can compute the expected value of production in chains with at least one chained order. Integrating across all possible lengths, we obtain

creasing in μ and has well-behaved limits.²⁵ Total output $\mathcal{Y}(\mu; \delta)$ is constructed by noting there are $(1 - \mu)$ spot orders for which production is 1 and μ for chained orders whose average production is $\mathcal{A}$. Since labor is in fixed supply, $\mathcal{Y}$ is both output and TFP. However, since efficiency losses are fully driven by utilization among chained orders, we refer to $\mathcal{A}$ as TFP directly. To illustrate the formula for $\mathcal{A}(\mu; \delta)$ and its construction, the left panel of Figure 6 plots the distributions of chain lengths—corresponding to two values of μ . For the higher μ , the distribution shifts mass to chains of greater length. The right panel graphs $\mathcal{A}$ as a function of μ , for two values of δ . Productivity falls as the fraction μ increases. In turn, the lower δ , the greater the delays and the lower TFP.

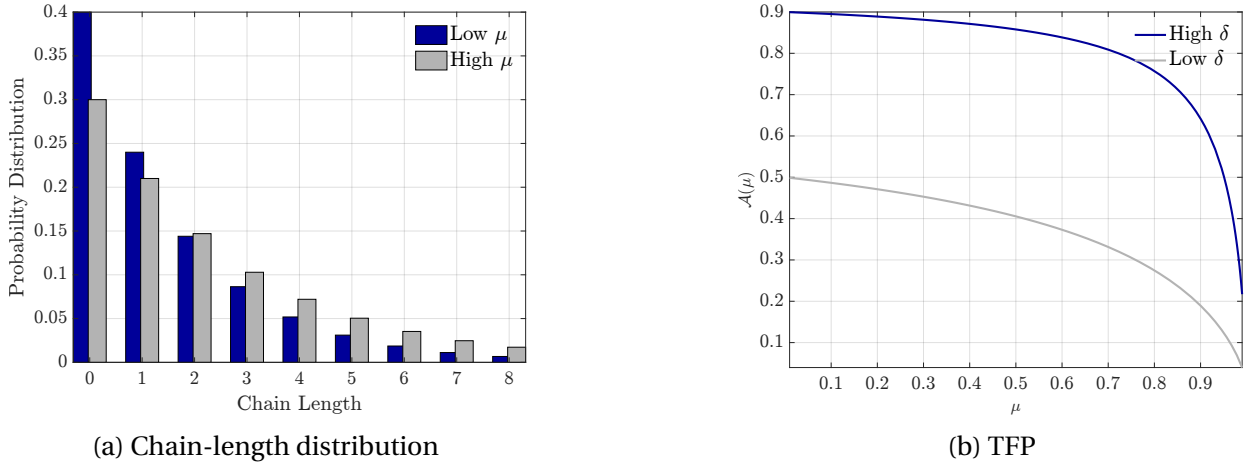


Figure 6: Chain-Length Distributions and TFP

Notes: Left panel graphs $G(n)$ for $\mu = \{0.6, 0.7\}$. Right panel $\mathcal{A}(\mu)$ for $\delta = \{0.5, 0.9\}$.

The main insight is that TFP losses result from payment delays. It is worth distinguishing the sources of these losses from other environments. Unlike search models, there are no congestion externalities—the production assignment is one-to-one. Unlike sticky-price models, the fact that payments are the same across orders does not have implications for production (input uses do not depend on prices). The source

$\mathcal{A}(\mu; \delta)$, the average production for chained orders. For that, we use the discrete probability $(1 - \mu) \mu^n / \mu$. This is the distribution of chain length conditioned on $n > 0$, obtained from $G(n; \mu)$. Integrating across all possible lengths, we obtain $\mathcal{A}(\mu; \delta)$, the average production for chained orders.

²⁵With respect to the limits of $\mathcal{A}$, as $\mu \rightarrow 1$ the chains become larger but their average production decreases to zero since the additional production of chained order decreases exponentially. On the other hand, when $\mu \rightarrow 0$ the chain length tends to 1 and thus per-worker productivity tends to δ since average delay times converge to one period. In turn, when $\delta \rightarrow 0$ transfers take the full production period leaving not time for production. Conversely if $\delta \rightarrow 1$, chained orders feature no delays. Interestingly, $\mathcal{A}(\mu; \delta)$ resembles an entropy function. We are unaware of a connections between the geometric distribution, a discounted sequence, and the entropy function.

of TFP losses are the production delays caused by payment delays. The economy is at full capacity when all transactions are spot. Importantly, TFP losses are magnified by the random assignment of orders. Holding fixed the number of spot orders, if payment chains could be reorganized to be all of the same length, the economy would still feature TFP losses, but less so than under random assignment (Jensen’s inequality).

Discussion: Reduction of Economic Complexity. The payments-chain network here is simple. In practice, economies involve more complexity than exhibited here: Transactions differ by size and are coupled in more intricate production networks. Moreover, production ends by the end of the period and orders are not withdrawn.²⁶ Studying these richer dimensions would make the problem more realistic, but complicate the analysis.²⁷ Despite the crude assumptions, the payments-chain network illustrates the effects of delays. A virtue is that we can obtain a mapping from a financial quantity, μ , to measurable TFP. Because μ depends on agent decisions in the following section, we can study policy exercises immune to the Lucas critique.

3.1 Extensions

We now extend the baseline micro model to accommodate several empirically motivated features that affect chain-length distributions and measured productivity. Each extension modifies the expected duration of chained transactions or the output recovered from them, while preserving the core logic of the payment-chain framework.

Correlated Order Types and Micro-Level Frictions. Rather than independent order assignment, we model payment types as a Markov process with correlation parameter $\rho \in [-\mu(1 - \mu), \min\{\mu, 1 - \mu\}]$. This generates the empirically observed clustering in payment practices (Figure 5), where $\rho = 0$ recovers the baseline and $\rho > 0$ creates persistent payment chains.

We also introduce three sources of chain disruption that limit realized output: (i) *default truncation* at position η , (ii) *heterogeneous delays* where each agent delays with

²⁶To allow withdrawals, we would have to find an endogenous maximal chain-length provided that we obtain a relative price for funds and production, as we do in the following section. In that case, the production in a chain would drop to zero for orders above a given position. An analog, but more complicated, formula to TFP, (2), can be derived for the case.

²⁷Statistical physics handles combinatorial problems that lead to this complexity. Potentially, such methods can be used in a similar environment to the one here. In particular, statistical physics offers tools to calculate statistical properties of the system.

probability q , and (iii) *immediate transfers* with probability $1 - q$. These frictions preserve geometric chain-length distributions while reducing per-chain productivity in empirically plausible ways.

Generalized TFP Results. Let $p = \mu + \rho$ denote the effective chaining probability. The key productivity relationships become:

(i) With correlation only: $\mathcal{A}^{\text{corr}}(p) = \mathcal{A}(p)$ (no direct TFP effect).

(ii) With heterogeneous delays:

$$\mathcal{A}^i(\mu; q) = (1 - q) + \frac{1 - p}{p} \cdot \frac{\delta}{1 - \delta} \log \left(\frac{1 - p(1 - q + \delta q)}{1 - p} \right).$$

(iii) With truncation: Complex formula involving Lerch transcendent functions (see Appendix B.2).

The heterogeneous delays case is particularly relevant empirically, showing how partial payment failures reduce TFP beyond pure waiting costs.

Alternative Mechanisms. To match empirical chain-length distributions, we allow agents beyond position η to seek alternate suppliers, creating right-truncated distributions that shift mass toward shorter chains. This process follows a Fokker–Planck equation whose properties depend on the resampling mechanism details (Appendix B.4).

When agents can redirect idle labor to backup activities worth $\alpha < 1$ units, TFP losses scale as $\mathcal{Z}(\mu) = \alpha + (1 - \alpha) \cdot \mathcal{A}^i(\mu; q)$. This captures the empirical reality that payment delays need not result in complete output loss.

Full mathematical derivations for all extensions are provided in Appendices B.1–B.3.

4. Endogenous Savings and Expenditure Decisions

4.1 From Payment-Chain Networks to Portable Budget Constraints

The payment-chain network developed in the previous section provides a mapping from the fraction of chained orders μ to aggregate productivity $\mathcal{Y}(\mu; \delta)$. This framework is designed to be modular and portable across different economic environments. We now demonstrate how the payment-chain structure manifests itself in reduced-form

budget constraints that can be embedded into a broad class of models without needing to refer back to the underlying network details. Encoding the inefficiencies provoked by the payment-chain network into budget constraints allows us to study how individual agent decisions regarding spot versus chained orders aggregate into macroeconomic effects, while maintaining analytical tractability.

Timing. Consider a dynamic model with discrete time periods $t \in \{0, 1, 2, \dots\}$. Expenditure and savings decisions are made at the start of each period. All transactions occur within each unit time interval, following the payment-chain network structure described earlier, in the times $\tau(n) = (1 - \delta)\delta^n \in [0, 1]$ corresponding to position $n \in \{0, 1, 2, \dots\}$ in a payment chain. The numéraire is in goods.

We focus on a single agent, which could be a household, a firm, that has y goods available for sale in the network at the beginning of each period. This endowment is composed of N individual units, each of size $1/N$, that must be sold through the payment network. The agent also begins the period with financial wealth W and must decide how to allocate expenditures between spot orders (requiring immediate payment) and chained orders (paid contingently upon receiving income from sales). Let E^s and E^x denote the agent's expenditures on spot and chained orders, respectively. These correspond to N^s spot orders and N^x chained orders.

Income Flows and Transaction Timing. Recall that each of the N goods the agent sells is randomly assigned to a position within a payment chain. Index each good by $j \in \{1, 2, \dots, N\}$ and let $u_j \in \{0, 1, 2, \dots\}$ denote the random position at which good j is sold, where position 0 corresponds to the good being sold in a spot sale. The distribution of positions is governed by the chain-length distribution $G(n)$. Since all goods are sold at the same price, one of numéraire $1/N$, total income equals the endowment:

$$y = \frac{1}{N} \sum_{n=0}^{\infty} \sum_{j=1}^N \mathbb{I}[u_j = n]. \quad (3)$$

Similarly, each chained order faces random execution timing. Enumerate each chained order by $i \in \{1, 2, \dots, N^x\}$ and let $o_i \in \{1, 2, 3, \dots\}$ denote the random position at which chained expenditure i is executed. The goods obtained from chained order i is $(1 - \delta)\delta^{o_i}$

per unit, due to the delay, so the total goods obtained from chained orders is:

$$X = \frac{1}{N} \sum_{i=1}^{N^x} (1 - \delta) \delta^{o_i}, \quad (4)$$

where X is a random variable as o_i is random. Summing across positions, total chained expenditures equal:

$$E^x = \frac{1}{N} \sum_{n=1}^{\infty} \sum_{i=1}^{N^x} \mathbb{I}[o_i = n]. \quad (5)$$

Within-Period Sequential Budget Constraints. Wealth evolves sequentially within each period as transactions are executed at different times $\tau(n)$. Let $W_{\tau(n)}$ denote wealth immediately after all transactions at position n have been completed. Spot expenditures and spot sales both occur at $\tau(0)$, so that

$$W_{\tau(0)} = W + \frac{1}{N} \sum_{j=1}^N \mathbb{I}[u_j = 0] - E^s. \quad (6)$$

This budget adds the income from goods sold in spot transactions and subtracts spot expenditures from the initial wealth W , yielding $W_{\tau(0)}$. At each later position n , the agent receives income from goods sold and pays for executed chained orders:

$$W_{\tau(n)} = W_{\tau(n-1)} + \frac{1}{N} \sum_{j=1}^N \mathbb{I}[u_j = n] - \frac{1}{N} \sum_{i=1}^{N^x} \mathbb{I}[o_i = n] \quad \forall n \geq 1. \quad (7)$$

The first term represents carried-over wealth, the second term is income from sales at position n , and the third term is payments for chained orders executed at position n .

From Payment-Chain Network Budgets to Period Budgets. To derive the period budget constraint, we sum across all sequential budgets. The wealth remaining at the end of the period, W' , satisfies:

$$W_{\tau(\infty)} = W + \underbrace{\frac{1}{N} \sum_{n=0}^{\infty} \sum_{j=1}^N \mathbb{I}[u_j = n]}_{=y} - E^s - \underbrace{\frac{1}{N} \sum_{n=1}^{\infty} \sum_{i=1}^{N^x} \mathbb{I}[o_i = n]}_{=E^x}. \quad (8)$$

This gives us the fundamental intertemporal budget constraint:

$$\frac{W'}{R} + E^s + E^x = y + W, \quad (9)$$

where $W'/R = W_{\tau(\infty)}$ represents the present value of wealth carried to the next period at the between-period interest rate R .

This budget encodes the evolution of transactions within the period. An agent placing many orders faces uncertainty about the actual goods X obtained from chained purchases. However, as the number of orders becomes large ($N \rightarrow \infty$), the law of large numbers eliminates this risk. Taking the limit as $N \rightarrow \infty$ while holding total expenditures fixed:

$$\lim_{N \rightarrow \infty} X = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^{N^x} (1 - \delta) \delta^{o_i} = E^x \sum_{n=1}^{\infty} (1 - \delta) \delta^n G(n) = \mathcal{A}(\mu) E^x, \quad (10)$$

where $\mathcal{A}(\mu)$ is the aggregate productivity factor and μ is the economy-wide fraction of chained orders. The budget constraint becomes:

$$\frac{W'}{R} + S + q(\mu)X = y + W, \quad (11)$$

where $q(\mu) \equiv \mathcal{A}(\mu)^{-1}$ is the effective price of goods obtained through chained relative to spot orders. The units of goods available for consumption C_t (or investment I_t) are

$$I_t + C_t = S_t + X_t.$$

This formulation implicitly relies on a “big family” assumption in the tradition of [Lucas and Stokey \(1987\)](#), where individual transaction risk is diversified away but income from spot transactions is not pooled to finance pending chained orders. This maintains the sequential payment structure that drives the productivity effects while ensuring analytical tractability. The costs of placing chained orders is thus fully captured by the *endogenous pricing*, in the formula for $q(\mu)$. Exploiting its portability, we showcase several implications of the payments-chain network in a couple of economic environments.

4.2 Consumption-Saving Decisions

The previous section shows how, given an amount of chained expenditures, E^x , an agent obtains a number of goods from chained transactions that depend on the net-

work structure and the ratio of economy-wide chained transactions μ . In that analysis, the share of chained orders is taken as primitive. We now turn to behavior. This section develops two dynamic decision problems in which agents choose whether to place spot or chained orders, or to save, while taking as given the distribution of payment chains. A common theme will emerge: when short-term means of payment (e.g., credit lines) are scarce, agents optimally resort to chained orders despite their higher cost. Moreover, unlike in standard models of financial frictions, agents here do not always have strict incentives to save or delever to escape their constraints.

The partial-equilibrium analysis of these problems serves as a prelude to the general equilibrium setting where we embed these decision problems in a dynamic framework à la Kiyotaki and Moore (1997b).

Entrepreneur Problem. We now study the problem of entrepreneurs who buy and sell goods in a payment chain network. Time is discrete time, $t = 0, 1, 2, \dots$. Each period, entrepreneurs receive a unit endowment of non-storable goods that are available for sale and purchase goods produced by others for consumption. Entrepreneurs have log-utility over consumption goods. At the start of each period, entrepreneurs place and receive purchase orders, as above. There is perfect foresight. The network's implications are summarized by an exogenous sequence $\{q_t\}_{t \geq 0}$, the average unit price of goods obtained via *chained* orders (spot price normalized to 1). Also, the sequence of interest rates, $\{R_{t+1}\}_{t \geq 0}$, is taken as given.

At the start of period t , the entrepreneur chooses *spot* purchases $S_t \geq 0$ (requiring immediate funds) and *chained* goods $X_t \geq 0$ (purchased through chained orders). These goods add to the total goods that enter the entrepreneur's utility, C_t , and its period expenditures, E_t :

$$C_t = S_t + X_t, \quad E_t = S_t + q_t X_t.$$

The entrepreneur enters period t with debt B_t and carried next-period debt $B_{t+1} \leq \bar{B} \equiv 1/(1 - \beta)$. Importantly, spot purchases must be financed with up-front funds. A *spot borrowing line* (SBL) $\tilde{B}_t$ limits intraperiod liquidity:

$$S_t \leq \bar{S}_t \equiv \max\{\tilde{B}_t - B_t, 0\}.$$

The entrepreneur chooses its total expenditures and its split, between spot and chained expenditures, to maximize its utility over consumption streams $\sum_t \beta^t \log(C_t) =$

$\sum_t \beta^t \log(S_t + X_t)$. Given total expenditures E_t , since $q_t \geq 1$, by cost minimization, the entrepreneur makes as many spot expenditures as the SBL permits. Thus, given E_t , spot and chained expenditures are respectively:

$$S_t = \min \{ \bar{S}_t, E_t \}, \quad X_t = (E_t - \min \{ \bar{S}_t, E_t \}) / q_t. \quad (12)$$

Given this optimal split, by the principle of optimality, the entrepreneur's objective of maximizing intertemporal utility can be cast into a dynamic programming problem:

Problem 1. (Entrepreneur's Problem): *Given B_0 and $\{ \tilde{B}_t, R_{t+1}, q_t \}_{t \geq 0}$, the entrepreneur chooses debt holdings, $\{ B_{t+1} \}_{t \geq 0}$, that solve:*

$$V_t(B) = \max_{B' \leq \bar{B}} \log(S + X) + \beta V_{t+1}(B')$$

where S and X are given by (12) and total expenditures by $E = B' R_{t+1}^{-1} + 1 - B$.

The time index in the value function reflects the evolution of $\{ R_{t+1}, \tilde{B}_t, q_t \}$. The Bellman equation expresses the entrepreneur's problem as a choice over intertemporal debt. The accumulation of debt determines the period's expenditures and the optimal split (12) yields S and X given the level of expenditures induced by the path of debt.

Interpretation of Financial Constraints. The spot borrowing line (SBL), $\tilde{B}_t$, captures short-term working-capital capacity such as credit lines, overdrafts, and supply-chain finance facilities, including trade credit. These instruments differ from carried-over debt and provide intraperiod liquidity to make spot purchases²⁸ In contrast, chained payments can be understood as outstanding accounts receivable against goods to be delivered, representing agreements where minimal payment precedes delivery.

The SBL constraint, $\tilde{B}_t$, differs fundamentally from the hard debt limit, $\bar{B}$, that governs intertemporal borrowing B_{t+1} . While $\bar{B}$ represents the maximum sustainable debt level, $\tilde{B}_t$ applies only to within-period short-term funding. Critically, agents can roll over SBL usage into future debt B_{t+1} without immediate principal repayment. This distinction provides a natural way to model credit crunches: financial intermediaries can tighten short-term credit lines without forcing costly defaults via sudden principal re-

²⁸Examples include Business Credit Lines, Standby Letters of Credit, and Supplier Finance Programs. Under a Supplier Finance Program, buyers defer payment while suppliers receive immediate cash from banks against receivables.

payment demands. This approach reflects the reality that banks often prefer to restrict new lending rather than trigger widespread defaults in periods of financial stress.

Characterization. To characterize the entrepreneur's consumption-savings behavior, we identify two debt thresholds that determine whether they rely on chained purchases voluntarily to increase their period consumption or are forced into them as the only form of expenditures, given the lack of short-term funds. Around these thresholds, *marginal prices* and the notion of *marginal inflation*—the effective price change between current and future consumption—play a central role. The two critical debt thresholds that govern the entrepreneur's expenditure decisions are described below.

Lemma 1 (Expenditure Threshold Points). *Define the efficiency threshold, $B_{t+1}^* \equiv R_{t+1}(\tilde{B}_t - 1)$. Then: - $S_{t+1} = 0$ if and only if $B_{t+1} > \tilde{B}_{t+1}$ (no spot purchases next period) - $X_t > 0$ if and only if $B_{t+1} > B_{t+1}^*$ (chained purchases this period)*

The efficiency threshold B_{t+1}^* marks the debt level above which entrepreneurs begin using chained purchases in the current period. The spot borrowing line $\tilde{B}_{t+1}$ is the debt level above which entrepreneurs lose access to spot purchases in the following period.

Because debt levels affect the availability of spot expenditures, the relevant price at which goods can be bought at the margin varies with the level of debt. The threshold debt levels determine the relevant prices for marginal decisions:

Definition 2 (Marginal Prices and Inflation). *The marginal expenditure price is $\tilde{q}_t^E(B') \equiv 1 + (q_t - 1)\mathbb{I}_{[B' \geq B_{t+1}^*]}$. The marginal borrowing price is $\tilde{q}_{t+1}^B(B') \equiv 1 + (q_{t+1} - 1)\mathbb{I}_{[B' > \tilde{B}_{t+1}]}$.*

Marginal prices capture the marginal cost of financing expenditures. If current debt forces chained purchases, reducing debt requires sacrificing $1/\tilde{q}_t^E$ units of consumption. If future debt exceeds the spot borrowing line, savings translate into chained rather than spot purchases, yielding only $1/\tilde{q}_{t+1}^B$ additional goods per unit of debt reduction. These marginal prices capture the consumption-saving tradeoffs that determine the entrepreneur's debt trajectory via the following generalized Euler equation, that adjusts the usual intertemporal condition with discontinuous marginal price shifts:

Proposition 3 (Entrepreneur's First-Order Condition). *Any solution $\{B_{t+1}\}_{t \geq 0}$ to the entrepreneur's problem satisfies:*

$$\frac{E_{t+1}}{E_t} \frac{Q_t}{Q_{t+1}} = \beta \frac{R_{t+1}}{\Pi_{t+1}(B_{t+1})} \quad \text{if } B_{t+1} \neq B_{t+1}^*$$

$$\text{and} \quad \beta q_t R_{t+1} \geq \frac{E_{t+1}}{E_t} \geq \beta R_{t+1} \quad \text{if } B_{t+1} = B_{t+1}^*$$

Despite its unconventional form, this Euler equation retains a familiar interpretation: marginal utilities are equated over time, corrected for price differences between spot and chained consumption. The left side represents the ratio of marginal rates of substitution (expressed in expenditures over average prices), while the right side captures the risk-adjusted return, deflated by “inflation” in terms of marginal prices.

This Euler condition is necessary but not sufficient. The value function is not concave in B' , so multiple sequences of future debt may satisfy the condition starting from an initial state. This non-convexity opens the door to hysteresis in general equilibrium.

Stationary Dynamics. We now characterize the stationary dynamics of the consumption-saving problem, i.e., when parameters are constant, $\tilde{B}_t = \tilde{B}_{ss}$ and $R_t = 1/\beta$ with $\tilde{B}_{ss} > 1$. Let $B_{ss}^* \equiv B^*(1/\beta, \tilde{B}_{ss})$. We obtain the following:

Proposition 4 (Stationary Entrepreneur Problem). *In a stationary solution: If $B_0 \leq B_{ss}^*$, then $B_t = B_0$ for all t . Moreover, there exists $B^h > \tilde{B}_{ss}$ such that:*

If $B_0 < B^h$, then $B_t \rightarrow B_{ss}^$ in finite time. If $B_0 > B^h$, then $B_t = B_0$ for all t .*

Figure 7 brings together the theoretical insights of the proposition to illustrate how the entrepreneur’s problem generates multiple steady states and path dependence. To illustrate the forces at play, we overlay the entrepreneur’s value function with two benchmark cases: $\bar{V}$, the value function when the spot borrowing limit is unbounded, ($\tilde{B}_t = \infty$), and, $\underline{V}$, where no liquidity is available ($\tilde{B}_t = 0$).

Panel (a) demonstrates how the value function deviates from both perfect liquidity ($\bar{V}$) and zero liquidity ($\underline{V}$) benchmarks. For debt levels below B^* , the value function V coincides with $\bar{V}$: entrepreneurs are unconstrained and consume the annuity value of their endowment net of interest payments entirely via spot purchases, showcasing that for sufficiently low debt levels, the SBL is irrelevant. When debt exceeds the high threshold B^h , the value function lies on $\underline{V}$, indicating that entrepreneurs consume exclusively via chained purchases forever. Crucially, even though these highly indebted entrepreneurs could potentially save to access better prices in the future, they choose not to because the benefits of deleveraging come too far in the future, requiring an excessive short-run sacrifice.

Panel (b) shows the non-monotonicity in the optimal debt policy function. Entrepreneurs with debt below B_{ss}^* maintain constant debt levels, enjoying efficient spot

transactions. Likewise, entrepreneurs with debt above B^h remain permanently at their initial debt levels. Only those with moderate debt in $(B_{ss}^*, B^h]$ gradually deleverage toward the efficient threshold B^* .

Panel (c) illustrates how consumption policies differ across debt regions. When debt is below (above) B_{ss}^* (B^h), the entrepreneur consumes the annuity of his total wealth annuity spot (chained) consumption. For debt levels slightly below B^h , we observe a reduction in overall consumption, which is how entrepreneurs escape the expensive chained consumption of those high debt levels. The chain-saw pattern of the optimal policy is a feature: it reflects a trade-off between smoothing consumption and shortening the number of periods to arrive at the efficient debt level.

Panel (d) traces the transition dynamics that emerge from the non-monotonic policy function for debt. Starting from different initial debt levels, entrepreneurs either start and remain in debt levels that allow consuming the period income in spot consumption, remain permanently consuming the period income in expensive chained goods, or temporarily delever to unconstrained debt levels. The appendix provides a complete characterization of B^h and the periods needed to arrive at B_{ss}^* . The properties have important general equilibrium consequences.

4.3 Investment Decisions

We now incorporate investment and production into the entrepreneur's problem. Here, both consumption and investment must be carried out using intermediate goods purchased via the payment network. The problem is as before, except that the capital purchases determine tomorrow's goods available for sale:

Problem 2. (Entrepreneur's Problem with Production): *Given B_0 and $\{\tilde{B}_t, R_{t+1}, q_t\}_{t \geq 0}$, the entrepreneur chooses debt-holdings and investment $\{B_{t+1}, K_{t+1}\}_{t \geq 0}$, the solution to:*

$$V_t(B, K) = \max_{B' \leq \tilde{B}, K' \geq 0} \log(S + X - K') + \beta V_{t+1}(B', K')$$

where S and X are given by (12) and total expenditures by $E = B'R_{t+1}^{-1} + f(K) - B$.

Here, $f(K)$ is production with the input (e.g., capital or inventory stocks) K acquired in the payment-chain network the previous period. The currently purchased goods, $S + X$, are distributed between consumption, C , and inputs acquired for production the next period, K' , assuming full depreciation of inputs (capital) for simplicity.

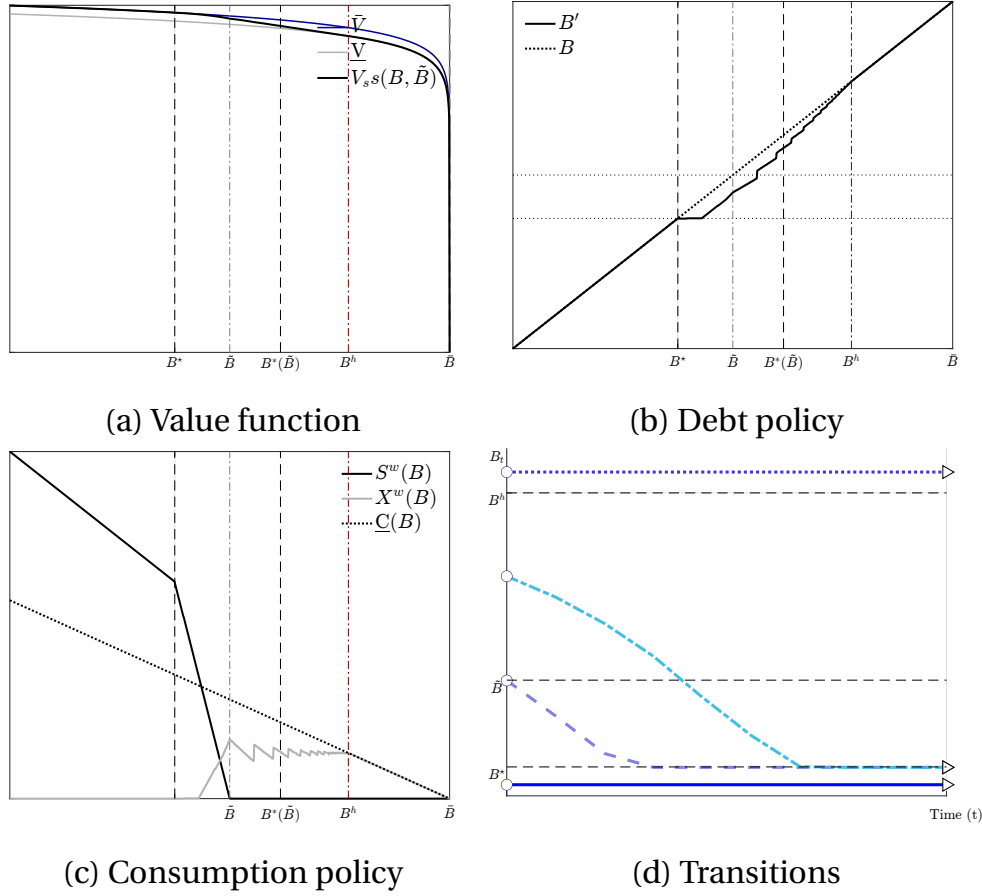


Figure 7: Bellman Equation and Policy Functions: Farmer Problem

Note: Figures computed using value function iteration with $\beta = 0.999$, $q = 1.001$, and $\tilde{B} = 1.001 \cdot B$.

In a problem without intermediate inputs, the entrepreneur's stationary problem reaches two steady-state debt levels: one efficient, characterized by spot purchases and full output realization, and another disrupted, where all transactions are chained. With investment and production, the same two steady states emerge. However, when entrepreneurs are constrained by the SBL, the higher cost of goods also distorts the optimal investment scale of investment. In particular, if debt is low, income net of financial expenses can be spent on spot purchases, consuming C_{ss} and investing K_{ss} , without distorting investment: $f'(K_{ss}) = R = \frac{1}{\beta}$. As in frictionless economies, the return on investment (in inputs) equals their marginal product, which also equals the rate of return on capital. However, if debt is sufficiently high, the entrepreneur ends up in a steady state debt level, making only chained purchases. The effective price of intermediate inputs exceeds one: $q > 1$. In this case, the first-order condition for inputs becomes: $f'(K_{ss}) = qR = \frac{q}{\beta}$. Hence, in this steady state, the marginal product of capital

exceeds the rate of return on saving and investment, inducing insufficient investment.

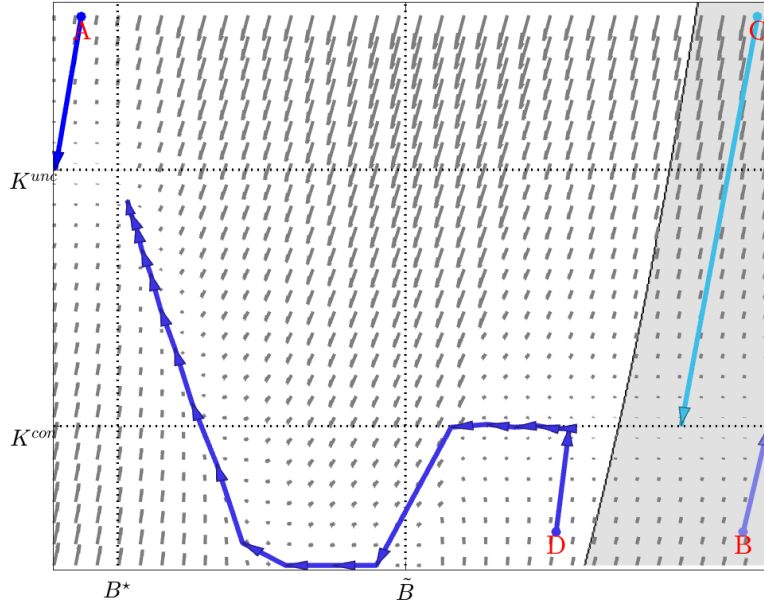
The same incentive to reduce debt and confront better prices, particularly for moderate debt levels, is also present in the production problem. The key insight remains: payment-chain frictions create two domains of attraction, and the inclusion of investment decisions does not alter this fundamental property. Instead, it amplifies the welfare costs by distorting not only consumption but also capital allocation. The investment wedge $f'(K_{ss}) - R$ quantifies the additional steady-state impact on productive efficiency resulting from delays in the payment-chain network.

Figure 9 illustrates these dynamics in the two-dimensional state space of debt and capital. Panel (a) shows the phase diagram with arrows indicating the direction of change for both state variables. The shaded region represents the domain of attraction toward the disrupted steady state with only chained expenditures and inefficient investment. Four representative paths demonstrate the range of possible dynamics. Path A begins with low debt and excess capital: convergence to the efficient steady state occurs immediately—inputs fully depreciate in this problem—as excess income is used to reduce debt, reaching equilibrium with spot expenditures and undistorted investment in one shot. Paths B and C start with high initial debt but different capital levels—both converge immediately to the disrupted steady state with underinvestment. The most interesting case, Path D, starts with intermediate debt and low capital. Initially, the entrepreneur accumulates both debt and capital, but once debt falls sufficiently to enable some spot purchases, investment is reduced to accelerate deleveraging—consistent with the incentive to escape the high marginal prices of chained transactions. Panels (b) and (c) trace these debt and capital paths over time, confirming that the same two-domain pattern holds with production: entrepreneurs either converge quickly to efficiency or remain trapped in a state of persistent underinvestment.

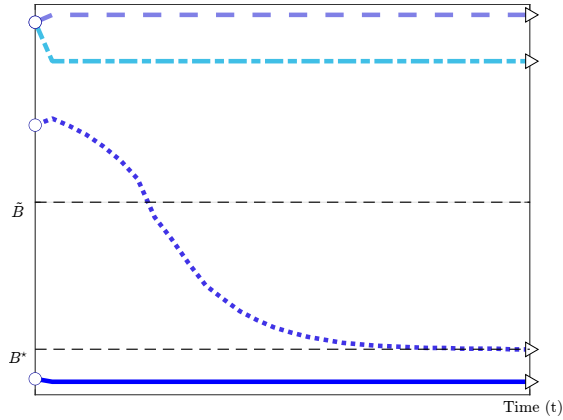
This section showed that entrepreneurs' incentives to remain at high debt levels with chained expenditures also emerge in a problem with production. Having shown this pattern, we now move to the general equilibrium implications of these properties for the simplest case where we abstract from investment.

5. General Equilibrium

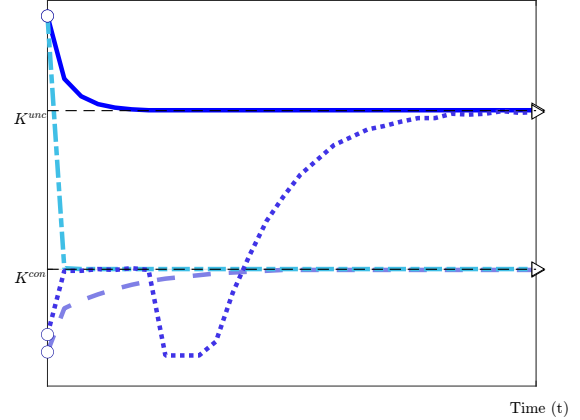
The previous section analyzes how entrepreneurs choose between spot and chained purchases given prices and interest rates. We now embed these decisions into a gen-



(a) Phase Diagram



(b) Path of Debt



(c) Path of Capital

Figure 8: Investment and Leverage Decisions

Note: Figures are calculated using value function iteration with $\beta = xxx$, $\delta = xxx$, and $\tilde{B} = xxx \cdot \bar{B}$. Panels (a) and (b) xxx. The shaded area in panel (c) corresponds to the hysteresis region.

eral equilibrium where prices and interest rates are endogenous. This transition reveals how individual payment choices create aggregate externalities: when one agent uses chained purchases, they contribute to network congestion that affects the effective prices faced by others. These externalities, absent from partial equilibrium analysis, drive the model's key insights about payment-chain crises and hysteresis.

Environment We embed the payments-chain network into a dynamic equilibrium setting à la Kiyotaki and Moore (1997b). The economy consists of two groups of agents who interact through both goods and financial markets.

Entrepreneurs are the agents analyzed in the previous section. They receive a per-period endowment of $y = 1$ units of goods and finance consumption through a combination of current income and borrowing. As shown above, when facing smooth borrowing constraints, entrepreneurs optimally choose chained purchases.

Creditors are new agents who hold positive financial wealth but receive no endowment. They serve as the economy's savers, providing the funds that entrepreneurs borrow. Creditors always have sufficient liquidity to make spot purchases, so they never use chained transactions. The timing remains identical to the partial equilibrium problems above. Each creditor maximizes lifetime utility subject to their budget constraint. At the start of period t , the creditor holds real deposits D_t , which earn gross return R_{t+1} between periods. The creditor has logarithmic preferences and solves:

Problem 3 (Creditor's (Recursive) Problem). *Given D_0 and $\{R_{t+1}\}_{t \geq 0}$, the creditor chooses deposit holdings $\{D_{t+1}\}_{t \geq 0}$ to solve:*

$$V_t(D) = \max_{D' \geq 0} \log(C) + \beta V_{t+1}(D')$$

subject to the intertemporal budget constraint: $C + R_{t+1}^{-1}D' = D$.

The solution is standard for logarithmic preferences: $C_t = (1 - \beta)D_t$ and $D_{t+1} = \beta R_{t+1}D_t$ for all $t \geq 0$. Since creditors always have positive wealth, they exclusively make spot purchases. Their total expenditure each period is $C_t = (1 - \beta)D_t$.

Equilibrium. Market clearing in the asset market requires the creditors' deposits exactly equal entrepreneurs' debt: $D_t = B_t$. Spot expenditures come from both creditors (who only make spot purchases) and entrepreneurs' spot purchases: $E_t^s = C_t + S_t$. Chained expenditures come only from entrepreneurs: $E_t^x = q_t X_t$. Total expenditures must equal the economy's endowment, since this is the numeraire. Since entrepreneurs receive $y = 1$ and creditors receive nothing: $E_t^s + E_t^x = 1$.

From the payments-chain technology (Theorem 2), the fraction of expenditures that are chained is $\mu_t = E_t^x$, which determines the effective price through $q_t = \mathcal{A}^{-1}(\mu_t)$.

Definition 3 (Symmetric Competitive Equilibrium). *Given a sequence of borrowing limits $\{\tilde{B}_t\}_{t \geq 0}$, sequence $\{B_t, C_t, S_t, X_t, R_t, q_t\}_{t \geq 0}$ is a **symmetric competitive equilibrium** if:*

1. **Optimization:** Given $\{R_t, q_t\}_{t \geq 0}$, the sequences $\{B_t, S_t, X_t\}_{t \geq 0}$ solve the entrepreneur's problem and $\{C_t\}_{t \geq 0}$ solves the creditor's problem.
2. **Asset and Goods market clearing:** $D_t = B_t, C_t + S_t + X_t = \mathcal{Y}(q_t X_t)$ for all t .
3. **Price consistency:** $q_t = \mathcal{A}^{-1}(q_t X_t)$ for all t .

The definition of equilibrium is standard, except that q_t solves a fixed-point equation given chained expenditures and output in terms of final goods, $\mathcal{Y}(q_t X_t)$, which depends on this price. This equilibrium concept encodes the transactions and delays that take place in the payments-chain network.

This general equilibrium structure captures the key externality driving payments-chain crises. When individual entrepreneurs face tight borrowing constraints, they rationally choose chained purchases to smooth consumption. However, each entrepreneur's use of chained purchases contributes to network congestion, raising the effective price q_t that all agents face. This creates a vicious cycle: higher debt levels force more chained purchases, which increase q_t , making chained purchases even more costly and potentially trapping the economy in an inefficient equilibrium.

The creditors play a crucial role beyond simply closing asset markets. Their constant access to spot purchases provides a “liquidity buffer” that prevents the entire economy from relying solely on chained transactions. The relative size of this buffer—determined by the equilibrium level of B_t —governs whether the economy operates efficiently or falls into a payments-chain crisis. Since creditors hold positive deposits and entrepreneurs are in debt, we have $D_t = B_t > 0$. From this point forward, B_t represents the single endogenous aggregate state variable capturing both creditor wealth and entrepreneur debt. An equilibrium is characterized as a mapping from B_t to R_{t+1}, q_t , current quantities, $\{C_t, S_t, X_t\}$, and future debt B_t .

Equilibrium Dynamics: Recursive Representation. A virtue of this setting is that it renders a tractable way to characterize the equilibrium dynamics through a recursive representation in the state variables $\{B, \tilde{B}, \tilde{B}'\}$. Under this representation, we obtain the equilibrium allocation given $\{B, \tilde{B}\}$. Since only B is endogenous, we seek a mapping $\mathcal{B}$ that determines tomorrow's debt as a function of today's state: $B' = \mathcal{B}(B, \tilde{B}, \tilde{B}')$.

The key equilibrium allocation can be expressed as functions of the state variables, exploiting the individual optimization conditions and market clearing. From the creditor's logarithmic preferences, their period expenditures are a linear function of current

debt B ; namely, $C(B) = (1 - \beta)B$. As total expenditures equal total income (which equals 1) the entrepreneurs' expenditures can be represented functionally as a residual of total income minus the creditors' expenditures; $E(B) = 1 - (1 - \beta)B$. In turn, the entrepreneurs' spot expenditures are the maximal expenditures share allowed by their SBL; namely, $S(B, \tilde{B}) = \min\{\max\{0, \tilde{B} - B\}, E(B)\}$. The functions $C(B)$, $E(B)$, $S(B, \tilde{B})$ determine chained purchases $\mu(B, \tilde{B}) = E(B) - S(B, \tilde{B})$. In turn, the price q satisfies $q(B, \tilde{B}) = \mathcal{A}^{-1}(\mu(B, \tilde{B}))$ and chained expenditures $X(B, \tilde{B}) = \mu(B, \tilde{B})/q(B, \tilde{B})$. Finally, the average price solves $Q = E(B)/(S(B, \tilde{B}) + X(B, \tilde{B}))$. Thus, the equilibrium allocation, including the entire payment chain network, is a function of $\{B, \tilde{B}\}$.

The debt mapping $\mathcal{B}$ emerges from combining the individual Euler equations of both agents with market clearing conditions. From the creditor's Euler equation, the gross real interest rate can be expressed as a function of B and B' ; namely, $R(B, B') = B'/(B\beta)$. The entrepreneur's Euler equation is more complex due to the discrete nature of payment choices. At any debt level, the entrepreneur faces effective marginal prices that depend on whether they are constrained to use chained purchases. Combining both Euler equations and substituting the equilibrium interest rate yields the fundamental equilibrium condition:

$$\underbrace{\frac{B}{1 - (1 - \beta)B} \cdot Q(B, \tilde{B})}_{\equiv \mathcal{E}(B; \tilde{B})} = \underbrace{\frac{B'}{1 - (1 - \beta)B'} \cdot Q(B', \tilde{B}') \cdot \frac{q^E(B'; B, \tilde{B}, \tilde{B}')}{q^B(B'; \tilde{B}')}}_{\equiv \mathcal{E}'(B'; B, \tilde{B}, \tilde{B}')}$$

where $q^E(B'; B, \tilde{B}, \tilde{B}') = 1 + (q(B, \tilde{B}) - 1)\mathbb{I}_{[B' \geq B^*(\tilde{B})]}$ is the marginal expenditure price and $q^B(B'; \tilde{B}') = 1 + (q(B', \tilde{B}') - 1)\mathbb{I}_{[B' > \tilde{B}]}$ is the marginal borrowing price.

The equilibrium debt mapping is determined by solving $\mathcal{E}(B; \tilde{B}) = \mathcal{E}'(B'; B, \tilde{B}, \tilde{B}')$ for B' . Due to the discontinuous nature of payment choices, this equation may have multiple solutions. However, properties of the individual problems allow us to select the appropriate solution:

$$\mathcal{B}(B, \tilde{B}, \tilde{B}') = \max \left\{ B^*(\tilde{B}), \arg \min_{B'} \{ \mathcal{E}(B; \tilde{B}) = \mathcal{E}'(B'; B, \tilde{B}, \tilde{B}') \} \right\}.$$

This mapping ensures that entrepreneurs never choose debt levels below the threshold $B^*(\tilde{B})$ where they would be unconstrained, and among multiple solutions, they select the one that maintains continuity in their value function. Notice that this recursive representation provides a computational algorithm: first, we solve for the equilibrium

allocation and price functions as a function of $\{B, \tilde{B}\}$, and then, we solve for $\mathcal{B}$.

5.1 Implication I: Transitional Dynamics and Hysteresis

The aggregate Euler condition allows us to characterize the equilibrium dynamics, the possibility of hysteresis, and analyze payments-chain crises in response to shocks.

Steady States. In a steady state with constant exogenous credit limit $\tilde{B}_{ss}$, the endogenous debt level must also be constant: $B_{ss} = B_t$ for all t . From the creditor's solution, we know that a necessary condition for any steady state is that the real interest rate remains constant at $R_{ss} = \beta^{-1}$. A specific debt level constitutes a steady state if, given this interest rate and the resulting equilibrium price q , entrepreneurs optimally choose to maintain that same debt level. Since entrepreneurs face a corner solution at the steady-state rate $R_{ss} = 1/\beta$ —either making all purchases through spot markets or all through chained markets—exactly two types of steady states can emerge.

In an *undisrupted steady state*, debt remains sufficiently low that entrepreneurs exclusively make spot purchases. This yields $q = 1$, immediate trade, and the economy operates at full potential with $\mathcal{Y}_{ss} = 1$. In a *disrupted steady state*, entrepreneurs are forced to rely entirely on chained purchases, resulting in $q > 1$, delayed trade, and output below potential: $\mathcal{Y}_{ss} < 1$.

The inefficiency in the disrupted steady state arises from a debt-overhang equilibrium: high debt levels force entrepreneurs into chained purchases, which raises the effective price q , making the consumption sacrifice required for deleveraging even more costly. This creates a self-reinforcing trap where entrepreneurs find it optimal to remain constrained rather than undertake the costly deleveraging process. Formally, these are given by the following regions:

Corollary 1 (Steady-State Characterization). Fix $\tilde{B}_t = \tilde{B}_{ss} > 0$. Then, for all t :

i The economy is in an *undisrupted steady state* when $B_t \leq B^*(\beta^{-1}, \tilde{B}_{ss})$.

ii The economy is in a *disrupted steady state* when $B_t \geq B^h(\beta^{-1}, \tilde{B}_{ss})$.

The disrupted steady state represents a hysteresis region where excessive debt permanently impairs the economy's payment efficiency. This mechanism provides a theoretical foundation for understanding persistent periods of low productivity observed during extended economic stagnation, such as Japan's lost decades or the prolonged aftermath of the European sovereign debt crisis.

Domain of Attraction toward Efficient Steady States. For debt levels between the two steady-state regions, the economy may experience transitional dynamics toward the undisrupted steady-state region. The domain of attraction toward undisrupted steady states is characterized by the condition that the aggregate Euler equation $\mathcal{E}(B; \tilde{B}) = \mathcal{E}'(B'; B, \tilde{B}, \tilde{B}')$ has a solution where $B' < B$ —that is, entrepreneurs choose to deleverage over time. This domain has an upper bound solving:

$$B^*(\tilde{B}) \equiv \tilde{B} / (S(\tilde{B}, \tilde{B}) + X(\tilde{B}, \tilde{B}) + C(\tilde{B})) \geq \tilde{B},$$

where the inequality follows from total consumption being less than total expenditures when payments are delayed.

Corollary 2 (Convergence to Efficiency). *Let $\tilde{B}_t = \tilde{B}_{ss}$. For any $B_0 < B^*(\tilde{B}_{ss})$:*

1. *If $B_t \in (B^*(\beta^{-1}, \tilde{B}_{ss}), B^*(\tilde{B}_{ss}))$, then $B_{t+1} < B_t$ (deleveraging occurs)*
2. *If $B_t \leq B^*(\beta^{-1}, \tilde{B}_{ss})$, then $B_{t+1} = B_t$ (steady state reached)*
3. *If $\tilde{B}_{ss} > 1$, the economy converges to an undisrupted steady state in finite time*

Thus, a payments-chain crisis is only *temporary* when debt starts in the domain of attraction toward efficient steady states. In this region, entrepreneurs find it optimal to gradually reduce their debt burden despite the short-term consumption sacrifice, because the future benefits of accessing spot markets justify the deleveraging costs.

Figure 9 illustrates these dynamics. Panel (a) plots the functions $\mathcal{E}$ (solid blue) and $\mathcal{E}'$ (dashed gray) against debt levels, holding $\tilde{B}$ fixed. The arrows trace the equilibrium path of debt: when the functions intersect below the 45-degree line, entrepreneurs deleverage ($B' < B$). The convergence region extends from B^* to B^* , while the gray shaded area represents the hysteresis region where no deleveraging occurs. Panel (b) shows how equilibrium interest rates vary across debt levels, remaining at β^{-1} in both steady-state regions but falling below this threshold during transitions.

Finally, we note that the domains of attraction toward efficient steady states and the hysteresis region are disconnected. When debt falls in the intermediate range $\{B^*, B^h\}$, symmetric competitive equilibria do not exist. In this middle region, entrepreneurs individually would want to deleverage, but in general equilibrium, this would require interest rates below β^{-1} , for which no solutions exist to the equilibrium condition. We leave the analysis of asymmetric equilibria for future work.

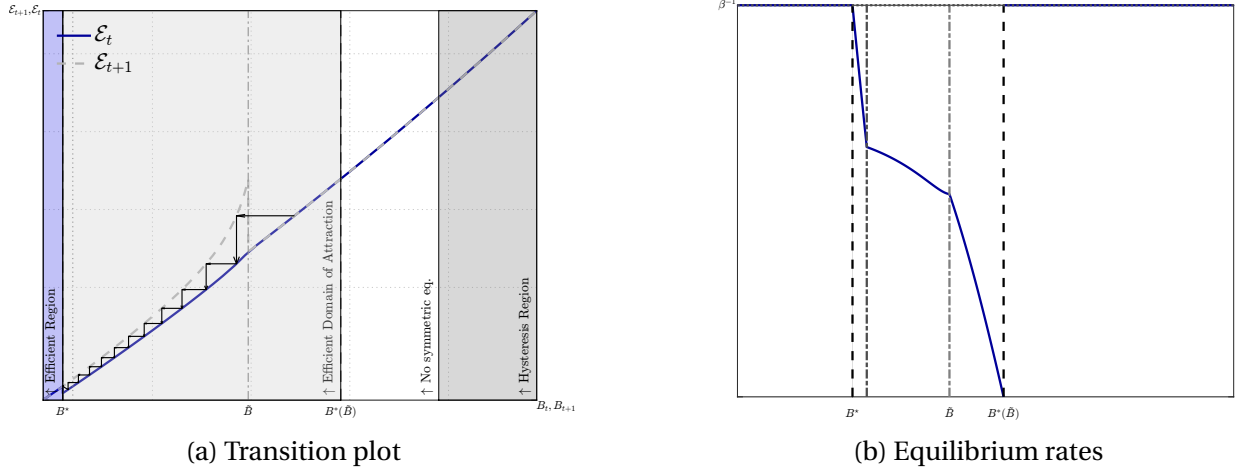


Figure 9: Equilibrium Phase Diagram and Hysteresis Region

Note: Figures are calculated using value function iteration with $\beta = 0.95$, $\delta = 0.9$, and $\tilde{B} = 0.2 \cdot \bar{B}$. Panel (a) shows the phase diagram of B_t constructed using $\mathcal{E}(B'; B)$ and $\mathcal{E}'(B'; B, \tilde{B}, \tilde{B}')$. Panel (b) plots $R(B, \tilde{B}, \tilde{B}')$ over $[B^* - 1, B^* + 2]$. The shaded area corresponds to the hysteresis region.

5.2 Implication II: Credit Crunches and Resilience Paradox

Credit Crunches. We now analyze the economy's response to temporary disruptions in credit availability. Consider a credit crunch modeled as a temporary tightening of borrowing limits $\{\tilde{B}_t\}$ that mean-reverts to its steady-state level. The debt mapping B from the aggregate Euler equation allows us to characterize the full transition path. As long as initial debt satisfies $B_0 \leq B^*(\tilde{B}_{ss})$ and $\tilde{B}_{ss} > 1$, the economy eventually reverts to an undisrupted steady state, but the adjustment process exhibits distinct phases.

Figure 10 illustrates a typical transition following a credit crunch. The solid line shows the competitive equilibrium response, which exhibits four distinct phases: The *extreme phase* occurs when $\tilde{B}_t$ falls to zero, eliminating access to short-term credit. Observed TFP remains depressed at its lowest value as entrepreneurs rely entirely on chained orders. Debt levels remain roughly constant during this phase because entrepreneurs focus on maintaining consumption rather than deleveraging, as the temporary nature of the shock reduces incentives for costly debt reduction. The *smoothing phase* begins once $\tilde{B}_t$ starts recovering toward its steady-state level. Entrepreneurs, anticipating future access to credit that will enable spot expenditures, increase current chained expenditures to smooth consumption across the transition. This forward-looking behavior temporarily raises debt levels despite the long-term goal of deleveraging, reflecting the tension between consumption smoothing and debt management. In the *recovery phase*, the borrowing limit has recovered sufficiently that entrepreneurs

begin trading off consumption smoothing against debt repayment. They gradually deleverage to expand their future credit capacity and regain access to spot markets. This phase sees the most active debt reduction as the benefits of lower debt become tangible. The *convergence phase* sees the economy reach the efficient steady state with permanently lower debt and fully restored TFP. All transactions occur through spot markets, and the final debt level is lower than the pre-shock level, reflecting the entrepreneurs' learned caution about over-leveraging.

The dashed line represents the constrained planner's solution, discussed in the next section, which provides a benchmark for evaluating the efficiency of the market response. This transition pattern holds generically for any credit shock that leaves the economy within the convergence region.

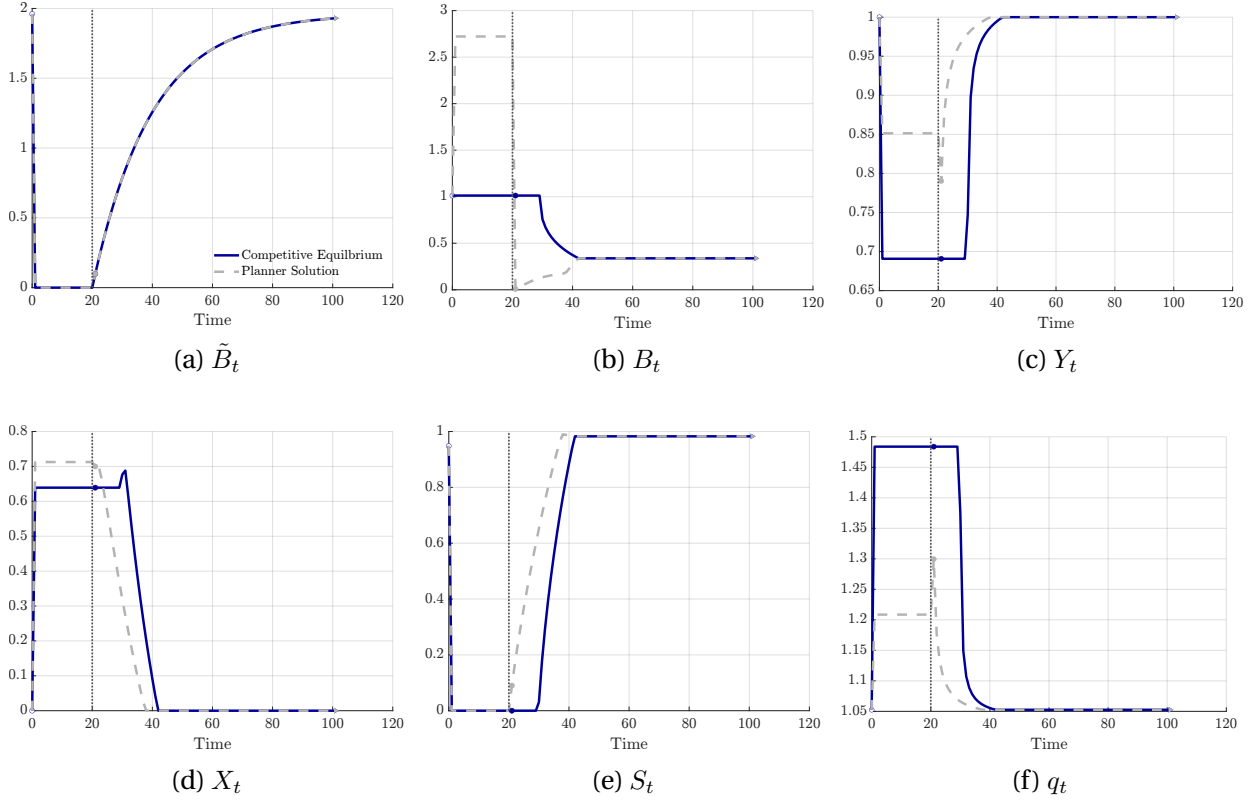


Figure 10: Efficient and Competitive Transition after a Credit Crunch

Note: This figure reports a numerical example of a credit-crunch episode. Figures are calculated using value function iteration with $\beta = 0.95$, $\delta = 0.9$, and $\tilde{B}_{ss} = 0.1 \cdot B^*$. The crunch happens at period $t = 10$. For $t \in \{11, \dots, 29\}$, $\tilde{B}_t = 0$. Then $\tilde{B}_t$ returns to steady state according to an AR(1) process with coefficient 0.9. The transition ends when $\tilde{B}_t = \tilde{B}_{ss}$.

The Resilience Paradox. A natural question arising from the credit crunch analysis is: what determines an economy's resilience to such shocks? We investigate how the

degree of payment delays—captured by parameter δ —affects economic resilience to credit line reductions. This analysis reveals a striking paradox: economies with lower δ (greater payment delays) may exhibit superior resilience to credit shocks, despite experiencing larger losses for any given level of chained purchases.

The mechanism underlying this resilience paradox operates through differential deleveraging incentives. Lower δ means higher payment delays in the payments-chain network, making chained purchases more costly relative to spot purchases. This creates stronger incentives for entrepreneurs to reduce debt levels during normal times to avoid being forced into expensive chained purchases during future shocks. Thus, lower δ induces a fundamental trade-off: greater payment inefficiency in the short run, but reduced vulnerability to credit disruptions through precautionary deleveraging.

Figure 11 demonstrates this paradox through a calibrated comparison. We examine two economies that begin with identical debt levels but differ in their payment efficiency parameter δ . After allowing sufficient time for each economy to reach its preferred debt level, we subject both to an identical credit shock—the reduction in $\tilde{B}_t$ shown in Panel (a).

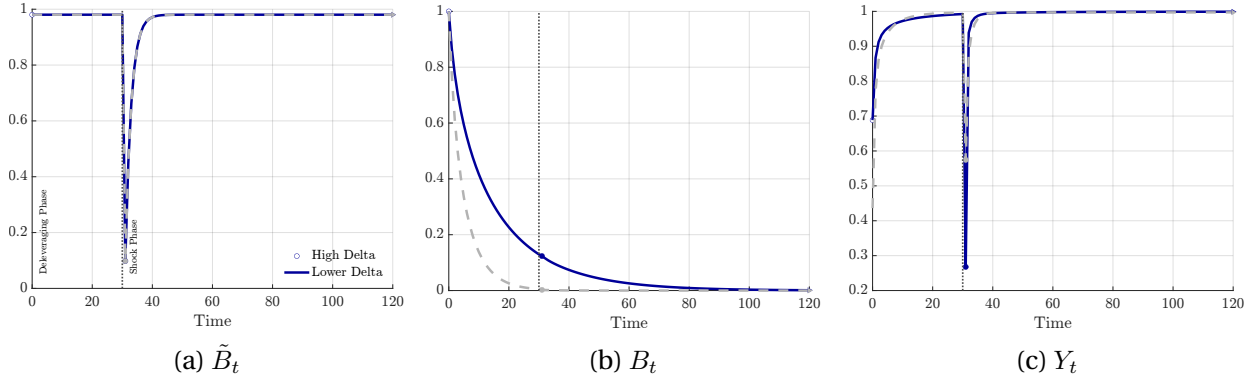


Figure 11: Comparison Resilience: xxx.

Note: This figure reports a numerical example of a credit crunch episode. Figures are calculated using value function iteration: $\beta = xxx$, $\delta = xxx$, and $\tilde{B}_{ss} = xxx \cdot B^*$. The crunch happens at period $t=xxx$. For $t \in \{xxx\}$, $\tilde{B}_t = xxx$. Then $\tilde{B}_t$ returns to steady state according to an AR(1) process with coefficient xxx . The transition ends at $t=xxx$ when $\tilde{B}_t = \tilde{B}_{ss}$.

Panel (b) reveals the key mechanism: when the shock hits, the low- δ economy (solid line) has achieved significantly lower debt levels than the high- δ economy (dashed line) due to stronger deleveraging incentives from costlier chained purchases. Panel (c) shows the paradoxical outcome: despite being less efficient at processing payments, the low- δ economy experiences a smaller output decline during the credit crunch because its lower debt levels reduce reliance on chained purchases when credit tightens.

This resilience advantage comes with important caveats. The paradox only manifests when the low- δ economy has had sufficient time to deleverage following the initial adjustment. For extremely severe shocks that eliminate credit lines entirely ($\tilde{B}_t \rightarrow 0$), the low- δ economy will necessarily perform worse due to its inherent payment inefficiencies. Additionally, both economies must begin in the deleveraging region; if initial debt levels place both in either the hysteresis region (too high) or efficient steady state (too low), no differential deleveraging occurs.

The resilience paradox captures a deeper economic principle: payment system inefficiencies, while costly during normal times, can induce precautionary behavior that builds shock resilience. Economies with more efficient payment systems may become overly comfortable with higher debt levels, leaving them vulnerable when credit conditions deteriorate. This suggests that some degree of "beneficial inefficiency" in payment systems may serve as a natural stabilizer against financial crises.

5.3 Constrained Inefficiency

At this stage, we have identified the sources of inefficiencies in payment-chain economies. In Section 3, we showed that the random nature of the network worsens the delays in the payment chains—a planner would reorganize payments so that each chain is of equal length. While governments cannot directly reorganize payments, they can influence the aggregate spending composition via policy interventions. This section studies such policies, focusing squarely on constrained inefficiencies that arise from expenditure decisions, treating the payment-chain network structure as given. Traffic regulation offers an analogy: governments cannot assign drivers to a lane, but they can tax vehicle purchases to influence traffic patterns. The spirit of our exercise is to clarify the sources and nature of these constrained inefficiencies during credit crunches.

The Ramsey Planner Problem. To study constrained inefficiency, we endow a Ramsey planner with Pareto weights on entrepreneurs, θ , and creditors, $1 - \theta$, and let him to choose sequences of debt taxes $\{\tau_t^k\}$, income taxes $\{\tau_t^\ell\}$, and expenditure taxes $\{\tau_t^c\}$.

Problem 4. (Ramsey Problem): *Given B_0 and $\{\tilde{B}_t\}$:*

$$\max_{\{\tau_{t+1}^k, \tau_t^c, \tau_{t+1}^\ell\}_{t \geq 0}} \sum_{t \geq 0} \beta^t [(1 - \theta) \log(C_t) + \theta \log(S_t + X_t)],$$

subject to the (modified) creditor budget constraint and optimality:

$$(1 + \tau_{t+1}^k) R_{t+1}^{-1} B_{t+1} + (1 + \tau_t^c) C_t = B_t, \forall t \geq 0,$$

the (modified) entrepreneur budget constraint: $B_t + (1 + \tau_t^c) E_t = R_{t+1}^{-1} B_{t+1} + 1 - \tau_t^\ell, \forall t \geq 0$,

the government's budget constraint: $\tau_{t+1}^k R_{t+1}^{-1} B_{t+1} + \tau_t^c (C_t + E_t) + \tau_{t+1}^\ell = 0, \forall t \geq 0$,

and the three conditions of a competitive equilibrium: (i) Optimization, (ii) Asset and Goods market clearing, and (iii) the price consistency.

The Ramsey planner distorts the competitive equilibrium with taxes. This planner cannot distinguish between the spot and chained expenditures, but takes into account the agents' optimal behavior, resource constraint, respects the transactions technology, and satisfies budget balance.

We endow the planner with Pareto weights that guarantee that the planner's allocation reaches the same undisrupted steady-state as does a competitive equilibrium after a credit crunch. Namely, we consider a competitive equilibrium that, as in the previous section, starts from B_0 debt, experiences a credit crunch, and eventually reaches an undisrupted steady-state debt B_{ss} . Since the economy is efficient in an undisrupted steady state, any differences in allocations along the transition path of the credit crunch will reveal the nature of constrained inefficiencies. To guarantee the same terminal condition, the Ramsey planner must assign Pareto weights θ to entrepreneurs:

$$\frac{1 - (1 - \beta) B_{ss}}{(1 - \beta) B_{ss}} = \frac{\theta}{1 - \theta}. \quad (13)$$

The Primal Problem. We solve the Ramsey Problem via an equivalent Primal Problem where the planner directly chooses debt sequences $\{B_t\}_{t \geq 0}$ to maximize:

$$\sum_{t \geq 0} \beta^t \mathcal{P} \left(B_t, \tilde{B}_t \right),$$

$$\text{where } \mathcal{P} \left(B, \tilde{B} \right) \equiv (1 - \theta) \log (C^s(B)) + \theta \log \left(S^w \left(B, \tilde{B} \right) + X^w \left(B, \tilde{B} \right) \right). \quad (14)$$

This equivalence holds as controlling debt pins down all allocations: B_t determines creditor expenditures through log utility, and entrepreneur expenditures follow residually from income minus creditor spending. Implementation requires only capital taxes

to control debt paths and labor taxes for redistribution (proof in Appendix X). As the objective is time-separable, the optimal debt at t depends only on the current SBL $\tilde{B}_t$:

$$B_t = B^p(\tilde{B}_t) = \arg \max_{B \in [0, \tilde{B}]} \mathcal{P}(B, \tilde{B}_t).$$

Planner Solution: Key Results. The planner's solution reveals that during credit crunches, debt may be either inefficiently high or inefficiently low. The planner has two ways to increase productive efficiency: redistribute wealth toward entrepreneurs (reducing debt to free credit lines) or toward creditors (increasing debt to stimulate spot expenditures). This ambivalent nature leads to four qualitatively different regimes depending on the tightness of credit constraints.

When $\tilde{B}$ is sufficiently high, the economy operates efficiently with $\mathcal{A} = 1$ and optimal social insurance. As $\tilde{B}$ falls below the efficiency threshold, the planner faces a trade-off between productive efficiency and social insurance. In moderate constraint regions, the planner sacrifices insurance but maintains efficiency. In severe constraint regions, the planner accepts some productive inefficiency, balancing efficiency gains against insurance costs. Most strikingly, when constraints become extreme, the planner reverses strategy entirely: rather than helping the constrained entrepreneur, the planner redistributes wealth toward creditors. This seemingly draconian policy increases TFP sufficiently to benefit both agents despite appearing regressive. The planner sets debt above steady-state levels because marginal increases in creditor wealth generate more spot expenditures than the efficiency losses from constraining entrepreneurs.

This ambivalent nature distinguishes payment-chain crises from typical models with pecuniary externalities, where social insurance and efficiency typically reinforce each other. Here, the planner may switch between complementary and conflicting policies as constraint severity varies.

Competitive Equilibrium vs. Efficient Transitions: An illustration. Figure 10 illustrates how the planner's solution differs from competitive equilibrium during a payment-chain crisis. Initially, the planner implements regressive policies—taxing constrained entrepreneurs to subsidize creditors—because increased creditor wealth generates spot expenditures that raise TFP. Despite higher taxes on entrepreneurs, their consumption actually increases through lower chained-good prices. As recovery progresses, the planner reverses course, taxing savings to free entrepreneurial credit capacity.

The planner solution also eliminates the hysteresis present in competitive equilibrium, though achieving this may require policies that appear draconian initially.

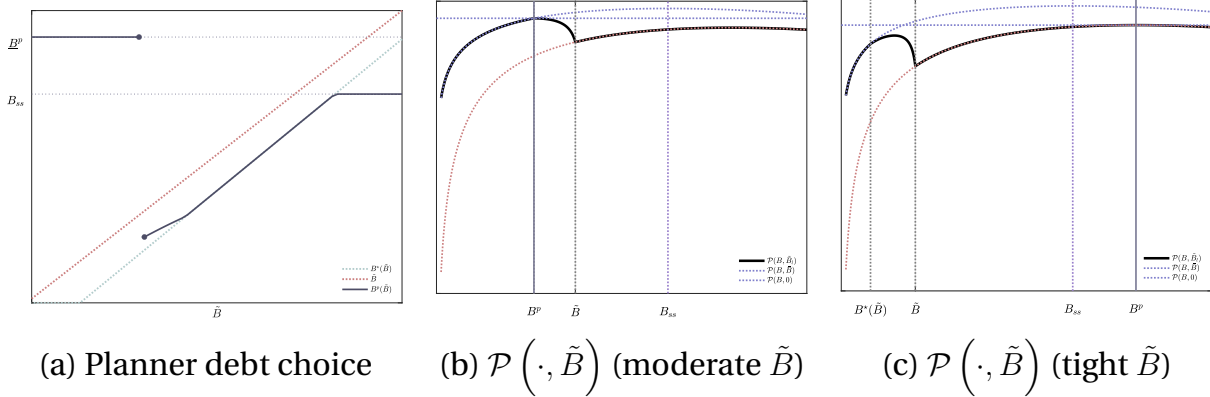


Figure 12: Primal Planner Solution

Note: This figure reports a numerical example of the planner's solution. Figures are calculated using value function iteration: $\beta = 0.95$, $\delta = 0.9$, and $\theta = 0.75$. Panel (a) plots B^p as a function of values of $\tilde{B}$ in the range $[0.1, \theta + 0.1] * B^{star}$. Panel's (b) and (c) plot the objective of the planner for different values of B in the range $[0, 1.8]$. Panel (b) fixes $\tilde{B}$ at 0.15 and Panel (c) at 0.08.

5.4 Fiscal Policy and the Bocola Effect

Government expenditures have strikingly different effects depending on their payment timing. When governments pay for expenditures upfront (spot), they inject immediate liquidity into payment chains. When governments chain expenditures to future tax receipts, they extend payment chains and worsen delays. We call this the Bocola effect.

To formalize this distinction, consider spot government expenditures G^s that require immediate payment, analogous to household spot purchases. The government can also make chained expenditures G^x contingent on receiving tax revenues, treating tax receipts as income flows similar to labor income. For simplicity, assume the government raises labor taxes to finance expenditures under a balanced budget constraint, and that all government spending represents resource waste. The key insight emerges from how these expenditures affect the aggregate fraction of chained transactions:

$$\mu_t = q_t X_t^w + G_t^x \quad \text{and} \quad q_t = \mathcal{A}(\mu_t)^{-1}.$$

Spot government expenditures reduce the chained expenditure ratio by substituting for private chained purchases, while chained government expenditures increase μ_t .

Proposition 5. (Government Spending Multipliers): *Consider infinitesimal government expenditures starting from $(G^x, G^s) = (0, 0)$. Consumption and output multipliers are:*

Efficient Region ($B \leq B^*(\tilde{B})$): Both expenditure types reduce welfare:

$$\frac{\partial C^w}{\partial G^x} = \frac{\partial C^w}{\partial G^s} = -1, \quad \frac{\partial \mathcal{Y}}{\partial G^s} = 0, \quad \frac{\partial \mathcal{Y}}{\partial G^x} = \mathcal{A}(\mu)(1 + \epsilon_\mu^A) - 1 < 0.$$

Crisis Region ($B > B^*(\tilde{B})$): Spot and chained expenditures have opposite effects:

$$\frac{\partial C^w}{\partial G^x} = -\mathcal{A}(\mu), \quad \frac{\partial C^w}{\partial G^s} = -\mathcal{A}(\mu)(1 + \epsilon_\mu^A), \quad \frac{\partial \mathcal{Y}}{\partial G^x} = 0, \quad \frac{\partial \mathcal{Y}}{\partial G^s} = 1 - \mathcal{A}(\mu)(1 + \epsilon_\mu^A) > 0.$$

The economic mechanism is clear. In efficient regions, any government spending wastes resources without improving payment chain efficiency. Spot spending merely transfers efficient expenditures from private to public use, yielding zero output effects. Chained spending worsens efficiency by lengthening payment chains.

During payment-chain crises, the effects reverse dramatically. Chained government expenditures remain harmful, adding to payment delays without productivity benefits. However, spot government expenditures create positive externalities by reducing average chain lengths. The elasticity ϵ_μ^A captures this effect: when sufficiently negative (which occurs in deep crises), spot spending can actually increase entrepreneur consumption despite higher taxes. The mechanism works through reduced prices of chained goods, which more than offset the tax burden.

The output multiplier for spot expenditures is always positive during crises because the government substitutes inefficient private chained expenditures for efficient public spending that shortens payment chains. Welfare increases when this multiplier exceeds unity, which occurs precisely when payment delays are most severe.

Policy Implications and Ricardian Equivalence. This analysis yields a novel perspective on fiscal stimulus. The conventional view that governments can stimulate aggregate demand simply by spending misses a crucial element: the timing of payments matters as much as the amount spent. A government that finances expenditures through future tax receipts—the typical approach—may inadvertently reduce welfare and have no output effects. Effective stimulus requires paying upfront to break payment chains rather than extend them.

The mechanism differs fundamentally from standard aggregate demand arguments. Here, fiscal policy works through a monetarist channel: government expenditures stimulate output by speeding up payments and reducing transaction delays, not by boosting overall spending. Because labor supply is inelastic and creditor expenditures are in-

elastic to capital taxation, the government's problem is equivalent to one involving pure transfers. This reveals that Ricardian equivalence fails in payment-chain economies. Future transfers can have real effects during crises because they influence entrepreneurs' spending composition. If transfers lead entrepreneurs to increase chained expenditures, output may actually decline—the opposite of conventional fiscal intuition.

6. Conclusion

There are many views regarding the nature of financial crises. A classic view is that during financial crises the economy suffers from coordination failures and demand externalities. Yet nothing makes financial crises particularly prone to coordination failures or demand externalities. In fact, the deepest and longest recessions—the Great Depression, Japan's "lost decades," and the Great Recession—were all triggered by disruptions in the financial system.

The contribution of the present paper is to interpret financial crises as arising from delays in payments. We first present empirical evidence on the existence of payment chains and how they can have real economic consequences given a credit crunch. We then develop a theory, where the core economic problem we highlight is that an inefficient distribution of funds causes payment delays that disrupt production. We show that crises of this form feature demand externalities and can lead to coordination failures. Although the policy recommendations resemble those in environments with price rigidity, we underscore the critical importance of accelerating transactions through appropriate transfers of funds.

In building the theory, we make two important simplifications. First, we assume that all transactions are bilateral and of identical size. Second, we assume that non-spot transactions require time to execute. Developing a general version of this model—with a richer sequence of transactions and heterogeneous delays—will be important for empirical applications. Nonetheless, we believe the lessons of this paper are robust to such extensions.

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Appendix (Not intended for publication)

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A. Proofs of Section 3

A.1 Proof of Proposition 2

Part 1. Derivation of TFP. In the body of the paper, we showed that for a chain of three orders, of which two are chained, the vector of production is $\{1, \delta, \delta^2\}$. The following induction argument generalizes: If the k -th node initiates production at time $1 - \delta^{k-1}$, the delay from its inspection is $(1 - \delta) \delta^{k-1}$, which added to previous delays leads to a transfer of funds only by time $1 - \delta^{k-1} + (1 - \delta) \delta^{k-1} = 1 - \delta^k$. This, leaves δ^k time for production to the subsequent unit. Since we computed the delay for $k = 1, 2, \dots$ the productions in a chain of length n are $\{1, \delta, \delta^2, \dots, \delta^n\}$.

It follows that the average output for chained orders, that is excluding the output of the spot order in the chain, in a payments-chain of length n is

$$\bar{y}_n^x = \frac{1}{n} \sum_{m=1}^n \delta^m = \frac{\delta}{n} \left(\frac{1 - \delta^n}{1 - \delta} \right).$$

Proof. Recall that a chained order will necessarily fall in a chain with length $n \geq 1$. Thus, the p.m.f of lengths conditional on this event is

$$G^x(n; \mu) = \frac{(1 - \mu) \mu^n}{\mu}.$$

Next, we turn to our goal of finding the expected output of a chained order:

$$\begin{aligned} \mathbb{E}[\bar{y}^x] &= \sum_{n=1}^{\infty} \bar{y}_n^x G^x(n; \mu), \\ &= \sum_{n=1}^{\infty} \frac{(1 - \mu) \mu^n}{\mu} \cdot \frac{\delta}{n} \left(\frac{1 - \delta^n}{1 - \delta} \right), \\ &= \frac{(1 - \mu)}{\mu} \cdot \frac{\delta}{(1 - \delta)} \cdot \sum_{n=1}^{\infty} \left(\frac{\mu^n}{n} - \frac{(\delta \mu)^n}{n} \right), \\ &= \frac{1 - \mu}{\mu} \cdot \frac{\delta}{1 - \delta} \cdot \ln \left(\frac{1 - \delta \mu}{1 - \mu} \right), \end{aligned}$$

The last equality follows from:

$$\sum_{n=1}^{\infty} a^{n-1} = \frac{1}{1-a} \leftrightarrow \sum_{n=1}^{\infty} \frac{a^n}{n} = \ln \left(\frac{1}{1-a} \right)$$

for $|a| < 1$, which can be shown by simply taking derivatives to both sides. A simple weak law of large numbers yields the desired result. Recall that we can have spot identities $i \in \mathcal{N}^s$ as unique identifiers for payment chains (spot orders for a one-to-one map with chains). Let $\ell(i)$ be the length of the payment chain that starts with spot order i . Given that for each $i \in \mathcal{N}^s$, $\ell(i) \sim i.i.d. G(\mu)$, then

$$\mathcal{A}(\mu; \delta) \equiv \text{plim}_{N \rightarrow \infty} \frac{1}{N^s} \sum_{i \in \mathcal{N}^s} \bar{y}_{\ell(i)}^x = \mathbb{E} [\bar{y}^x].$$

In words, the average output among chained orders converges to $\mathbb{E} [\bar{y}^x]$ as the network gets larger ($N \rightarrow \infty$). $\square$

Next, we derive expected output—the limit as $N \rightarrow \infty$ implicitly. The fraction of spot orders is $(1 - \mu)$. Production is 1 in their case. The fraction of chained orders is μ , and they produce on average $\mathcal{A}(\mu)$. Thus, total output is:

$$\mathcal{Y}(\mu) = (1 - \mu) + \mu \mathcal{A}(\mu).$$

Next, we obtain the derivative and limits of $\mathcal{Y}(\mu)$, $\mathcal{A}(\mu)$.

Part 2. Limits. We first consider the limit as $\mu \rightarrow 0$:

$$\lim_{\mu \rightarrow 0} \mathcal{A}(\mu; \delta) = \frac{\delta}{(1 - \delta)} \lim_{\mu \rightarrow 0} \left(\frac{1}{\mu} - 1 \right) \cdot \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) = \lim_{\mu \rightarrow 0} \frac{\ln \left(\frac{1 - \delta\mu}{1 - \mu} \right)}{\mu}.$$

The last term is the ratio of two variables that converge to zero. Using L'Hospital's rule:

$$\lim_{\mu \rightarrow 0} \frac{\ln \left(\frac{1 - \delta\mu}{1 - \mu} \right)}{\mu} = \frac{\delta}{(1 - \delta)} \frac{\lim_{\mu \rightarrow 0} \left(\frac{1}{1 - \mu} - \frac{\delta}{1 - \delta\mu} \right)}{1} = \delta.$$

where we used

$$\frac{\partial \ln \left(\frac{1-\delta\mu}{1-\mu} \right)}{\partial \mu} = \frac{1-\mu}{1-\delta\mu} \left(\frac{1-\delta\mu}{1-\mu} \right) \left(\frac{1}{1-\mu} - \frac{\delta}{1-\delta\mu} \right) = \left(\frac{1}{1-\mu} - \frac{\delta}{1-\delta\mu} \right).$$

For output, the limit is:

$$\lim_{\mu \rightarrow 0} \mathcal{Y}(\mu; \delta) = \lim_{\mu \rightarrow 0} (1-\mu) \lim_{\mu \rightarrow 0} \left(1 + \frac{\delta}{1-\delta} \ln \left(\frac{1-\delta\mu}{1-\mu} \right) \right) = 1.$$

Next, we consider the limit as $\mu \rightarrow 1$:

$$\lim_{\mu \rightarrow 1} \mathcal{A}(\mu; \delta) = \frac{\delta}{(1-\delta)} \lim_{\mu \rightarrow 1} \left(\frac{1}{\mu} - 1 \right) \lim_{\mu \rightarrow 1} \ln \left(\frac{1-\delta\mu}{1-\mu} \right).$$

This is the product of numbers that go to 0 and infinity. Using L'Hospital's rule:

$$\lim_{\mu \rightarrow 1} \mathcal{A}(\mu; \delta) = \frac{\lim_{\mu \rightarrow 1} \left(-\frac{1}{\mu^2} \right)}{\lim_{\mu \rightarrow 1} \left(\frac{1}{1-\mu} - \frac{\delta}{1-\delta\mu} \right)} = 0.$$

For output, the limit is:

$$\lim_{\mu \rightarrow 1} \mathcal{Y}(\mu) = \lim_{\mu \rightarrow 1} (1-\mu) + \lim_{\mu \rightarrow 1} \mu \lim_{\mu \rightarrow 1} \mathcal{Y}^x(\mu) = 0.$$

Next, we consider the limit as limit as $\delta \rightarrow 0$:

$$\lim_{\delta \rightarrow 0} \mathcal{A}(\mu; \delta) = \left(\frac{1}{\mu} - 1 \right) \lim_{\delta \rightarrow 0} \frac{\delta}{(1-\delta)} \cdot \lim_{\delta \rightarrow 0} \ln \left(\frac{1-\delta\mu}{1-\mu} \right) = 0.$$

For output,

$$\lim_{\delta \rightarrow 0} \mathcal{Y}(\mu) = (1-\mu) + \mu \lim_{\delta \rightarrow 0} \mathcal{Y}^x \mathcal{A}(\mu; \delta) = (1-\mu).$$

Finally, we consider the limit as $\delta \rightarrow 1$:

$$\lim_{\delta \rightarrow 1} \mathcal{A}(\mu; \delta) = \left(\frac{1}{\mu} - 1 \right) \lim_{\delta \rightarrow 1} \delta \cdot \lim_{\delta \rightarrow 1} \frac{1}{(1-\delta)} \cdot \lim_{\delta \rightarrow 1} \ln \left(\frac{1-\delta\mu}{1-\mu} \right).$$

This derivative is of the ratio of two numbers that go to zero. Using L'Hospital's rule:

$$\lim_{\delta \rightarrow 1} \mathcal{A}(\mu; \delta) = \left(\frac{1}{\mu} - 1 \right) \frac{\lim_{\delta \rightarrow 1} \frac{1-\mu}{1-\delta\mu} \left(\frac{-\mu}{1-\mu} \right)}{-1} = \left(\frac{1-\mu}{\mu} \right) \left(\frac{\mu}{1-\mu} \right) = 1.$$

For output,

$$\lim_{\delta \rightarrow 1} \mathcal{Y}(\mu) = (1 - \mu) + \mu \lim_{\delta \rightarrow 1} \mathcal{Y}^x \mathcal{A}(\mu; \delta) = 1.$$

This concludes the derivation of the limits of interest.

Part 3. Monotonicity. Next, we investigate the derivatives of $\mathcal{A}$ and $\mathcal{Y}$. We can write:

$$\mathcal{A}(\mu; \delta) = \left(\frac{1}{\mu} - 1 \right) \cdot \frac{\delta}{(1 - \delta)} \cdot \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right).$$

Thus,

$$\mathcal{A}_\mu = \frac{\delta}{(1 - \delta)} \left(\left(-\frac{1}{\mu^2} \right) \cdot \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) + \left(\frac{1}{\mu} - 1 \right) \left(\frac{-\delta}{1 - \delta\mu} + \frac{1}{1 - \mu} \right) \right).$$

Factoring out $-1/\mu^2$:

$$\begin{aligned} \mathcal{A}_\mu &= -\frac{\delta}{(1 - \delta)} \frac{1}{\mu^2} \left(\ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) - \mu(1 - \mu) \left(\frac{-\delta}{1 - \delta\mu} + \frac{1}{1 - \mu} \right) \right) \\ &= -\frac{\delta}{(1 - \delta)} \frac{1}{\mu^2} \left(\ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) - \mu \left(\frac{1 - \delta}{1 - \delta\mu} \right) \right). \end{aligned}$$

To show that the derivative is negative, we need to show that the term in the parenthesis is positive. Or likewise that

$$\ln(1 - \delta\mu) - \left(\frac{\mu - \delta\mu}{1 - \delta\mu} \right) > \ln(1 - \mu).$$

A concave function $f(x) \equiv \log(1 - x)$ satisfies:

$$f(x) + f'(x) |y - x| > f(y).$$

Let $x = \delta\mu$ and $y = \mu$. Because $\{\delta, \mu\} < 1$,

$$|y - x| = \mu - \delta\mu.$$

Thus, we have:

$$\ln(1 - \delta\mu) + \underbrace{\left(-\frac{1}{1 - \delta\mu}\right)(\mu - \delta\mu)}_{f'(x)|y-x|} > \ln(1 - \mu).$$

which proves the desired inequality. Hence, $\mathcal{A}_\mu < 0$ for any $\mu > 0$. At $\mu = 0$, the derivative is zero.

We also obtain that:

$$\mathcal{Y}_\mu = -1 + \mu\mathcal{A}_\mu + \mathcal{A} = -\underbrace{(1 - \mathcal{A})}_{>0} - \frac{\delta}{(1 - \delta)} \frac{1}{\mu} \left(\underbrace{\left(\ln\left(\frac{1 - \delta\mu}{1 - \mu}\right) - \mu \left(\frac{1 - \delta}{1 - \delta\mu}\right) \right)}_{>0} \right) \text{ for } \mu > 0.$$

The derivative is also zero at $\mu = 0$.

Part 4. Concavity. Next we perform the convexity analysis. $\mathcal{A}_{\mu\mu}$ is

$$\frac{\delta}{(1 - \delta)} \left[2 \frac{1}{\mu^3} \left(\ln\left(\frac{1 - \delta\mu}{1 - \mu}\right) - \mu \left(\frac{1 - \delta}{1 - \delta\mu}\right) \right) - \frac{1}{\mu^2} \left(\frac{1}{1 - \mu} - \frac{\delta}{1 - \delta\mu} - \frac{1 - \delta}{1 - \delta\mu} - \mu\delta \frac{1 - \delta}{(1 - \delta\mu)^2} \right) \right]$$

After some algebraic manipulations, the second term in parenthesis simplifies to:

$$\frac{1}{1 - \mu} - \frac{\delta}{1 - \delta\mu} - \frac{1 - \delta}{1 - \delta\mu} - \mu\delta \frac{1 - \delta}{(1 - \delta\mu)^2} = \mu \frac{(\delta - 1)^2}{(1 - \mu)(1 - \delta\mu)^2}.$$

Thus:

$$\mathcal{A}_{\mu\mu} = \frac{\delta}{(1 - \delta)} \frac{1}{\mu^3} \left[2 \ln\left(\frac{1 - \delta\mu}{1 - \mu}\right) - 2\mu \left(\frac{1 - \delta}{1 - \delta\mu}\right) - \frac{\mu^2 (1 - \delta)^2}{(1 - \mu)(1 - \delta\mu)^2} \right].$$

We can add the second and third terms to obtain:

$$\mathcal{A}_{\mu\mu} = \frac{\delta}{(1 - \delta)} \frac{1}{\mu^3} \left[\ln\left(\frac{1 - \delta\mu}{1 - \mu}\right)^2 - \mu \left(\frac{1 - \delta}{1 - \delta\mu}\right) \left(\frac{2 - \mu - 3\delta\mu + 2\delta\mu^2}{(1 - \mu)(1 - \delta\mu)} \right) \right].$$

This function is strictly concave if:

$$\ln(1 - \delta\mu)^2 < \ln(1 - \mu)^2 + \frac{1}{(1 - \mu)^2} \left(\frac{\mu(1 - \mu)(1 - \delta)}{(1 - \delta\mu)^2} (2 - \mu + 2\delta\mu^2 - 3\delta\mu) \right) \quad (15)$$

Let $F(x) = \ln(x)$. Set $x_0 = (1 - \mu)^2$ and $x_1 = (1 - \delta\mu)^2$ so that

$$x_0 - x_1 = -\mu(1 - \delta)(2 - \mu(1 + \delta)).$$

By strict concavity of $\ln(x)$ we have

$$\ln(1 - \delta\mu)^2 < \ln(1 - \mu)^2 - \frac{1}{(1 - \mu)^2} \mu(1 - \delta)(2 - \mu(1 + \delta))$$

so to prove that $\mathcal{A}_{\mu\mu}$ is strictly negative, we need to prove that the right hand side of the expression above is smaller than the condition needed for concavity, condition (15),

$$-\frac{1}{(1 - \mu)^2} \mu(1 - \delta)(2 - \mu(1 + \delta)) \leq \frac{1}{(1 - \mu)^2} \left(\frac{\mu(1 - \mu)(1 - \delta)}{(1 - \delta\mu)^2} (2 - \mu + 2\delta\mu^2 - 3\delta\mu) \right).$$

Cancelling common terms and rearranging, this condition is equivalent to:

$$-(1 - \delta\mu)^2(2 - \mu - \mu\delta) \leq (1 - \mu)(2 - \mu + 2\delta\mu^2 - 3\delta\mu). \quad (16)$$

The term on the left is negative—and strictly negative for $\delta, \mu < 1$.²⁹ Hence, the inequality above is verified as long as:

$$2 \geq \alpha(\mu, \delta) \equiv \mu - 2\delta\mu^2 + 3\delta\mu.$$

Hence, as long as

$$2 \geq \alpha^* = \max_{\{\mu, \delta\} \in [0, 1]^2} \alpha(\mu, \delta)$$

the condition for concavity holds for all $\{\mu, \delta\} \in [0, 1]^2$. We study this max function. Fix any μ . Since

$$\delta(3\mu - 2\mu^2) \geq 0,$$

the maximal value of α is achieved when $\delta = 1$. Hence, the objective is

$$\alpha^* = \max_{\{\mu, \delta\} \in [0, 1]^2} \alpha(\mu, \delta) = \max_{\{\mu\} \in [0, 1]} \alpha(\mu, 1) = \max_{\{\mu\} \in [0, 1]} 4\mu - 2\mu^2.$$

Maximizing the last expression over μ yields $\mu = 1$ as a solution and $\alpha^* = 2$ as the value. Hence, the inequality holds and guarantees (16). This suffices to prove concavity.

²⁹This follows immediately because μ and δ are fractions.

Next, we verify the concavity of total output. We have that $\mathcal{Y}_{\mu\mu}$ is

$$\begin{aligned}
&= \mathcal{A}_\mu + \mu \mathcal{A}_{\mu\mu} \\
&= -\frac{\delta}{(1-\delta)} \frac{1}{\mu^2} \left(\ln \left(\frac{1-\delta\mu}{1-\mu} \right) - \mu \left(\frac{1-\delta}{1-\delta\mu} \right) \right) + \frac{\delta}{(1-\delta)} \frac{2}{\mu^2} \mu \left(\ln \left(\frac{1-\delta\mu}{1-\mu} \right) - \mu \left(\frac{1-\delta}{1-\delta\mu} \right) \right) \\
&\quad - \frac{\mu}{\mu^2} \left(\frac{1}{1-\mu} - \frac{\delta}{1-\delta\mu} - \frac{1-\delta}{1-\delta\mu} - \mu\delta \frac{1-\delta}{(1-\delta\mu)^2} \right) \\
&= \mathcal{A}_{\mu\mu} - \frac{\delta}{(1-\delta)} \frac{\mu}{\mu^3} \left(\ln \left(\frac{1-\delta\mu}{1-\mu} \right) - \mu \left(\frac{1-\delta}{1-\delta\mu} \right) \right) \\
&< 0.
\end{aligned}$$

A.2 Related Results Used Elsewhere

In this section, we derived properties that are used later in the text.

Part 5. Inverse productivity. Now, we study the inverse of productivity. Let

$$q(\mu; \delta) = \mathcal{A}^{-1}(\mu; \delta).$$

Clearly, the function has the limits:

$$\lim_{\mu \rightarrow 0} q(\mu; \delta) = \delta^{-1} \text{ and } \lim_{\mu \rightarrow 1} q(\mu; \delta) = \infty \text{ and } \lim_{\delta \rightarrow 0} q(\mu; \delta) = \infty \text{ and } \lim_{\delta \rightarrow 1} q(\mu; \delta) = 1.$$

We also have that:

$$q_\mu = -\frac{\mathcal{A}_\mu}{\mathcal{A}^2} > 0.$$

We use the limit of the derivative of this function:

$$q_\mu(\mu) = \frac{\frac{\delta}{(1-\delta)} \frac{1}{\mu^2} \left(\ln \left(\frac{1-\delta\mu}{1-\mu} \right) - \mu \left(\frac{1-\delta}{1-\delta\mu} \right) \right)}{\left(\frac{(1-\mu)}{\mu} \cdot \frac{\delta}{(1-\delta)} \cdot \ln \left(\frac{1-\delta\mu}{1-\mu} \right) \right)^2}$$

Next, we check the convexity of function:

$$q_{\mu\mu} = -\frac{\mathcal{A}_{\mu\mu}}{\mathcal{A}^2} + \frac{\mathcal{A}_\mu}{\mathcal{A}^3} > 0.$$

Hence, q is convex in μ .

Part 6. Elasticity of $\mathcal{A}$. A useful object in later derivations is the elasticity of $\mathcal{A}$. Consider the derivative of

$$\mathcal{A}(\mu) \mu.$$

We have:

$$\mathcal{A}'(\mu) \mu + \mathcal{A}(\mu) = \mathcal{A}(\mu) [1 + \epsilon_\mu^{\mathcal{A}}].$$

Recall that,

$$\mathcal{A}(\mu) \mu = (1 - \mu) \left(\frac{\delta}{1 - \delta} \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) \right).$$

Hence,

$$\begin{aligned} \mathcal{A}'(\mu) \mu + \mathcal{A}(\mu) &= \frac{\delta}{1 - \delta} \left(-\ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) + (1 - \mu) \left(\frac{-\delta}{1 - \delta\mu} + \frac{1}{1 - \mu} \right) \right) \\ &= \frac{\delta}{1 - \delta} \left(-\ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) + \left(\frac{1 - \delta}{1 - \delta\mu} \right) \right). \end{aligned}$$

Hence, we obtain that:

$$\begin{aligned} [1 + \epsilon_\mu^{\mathcal{A}}] &= \frac{\frac{\delta}{1 - \delta} \left(-\ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) + \left(\frac{1 - \delta}{1 - \delta\mu} \right) \right)}{\mathcal{A}(\mu)} \\ &= \frac{\mu}{1 - \mu} \left(\frac{\frac{1 - \delta}{1 - \delta\mu}}{\ln \left(\frac{1 - \delta\mu}{1 - \mu} \right)} - 1 \right). \end{aligned}$$

We are interested in the sign of $1 + \epsilon_\mu^{\mathcal{A}}$ and its limits. We know $\epsilon_\mu^{\mathcal{A}} < 0$. Thus, the sign of $1 + \epsilon_\mu^{\mathcal{A}}$ is the sign of:

$$\left(\frac{1 - \delta}{1 - \delta\mu} \right) - \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right).$$

The limits of the function that governs the sign are:

$$\lim_{\mu \rightarrow 0} \left(\frac{1 - \delta}{1 - \delta\mu} \right) - \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) = (1 - \delta) > 0,$$

and

$$\lim_{\mu \rightarrow 1} \left(\frac{1 - \delta}{1 - \delta\mu} \right) - \ln \left(\frac{1 - \delta\mu}{1 - \mu} \right) = -\infty.$$

Since the function is continuous in μ , the sign is ambiguous.

Next, to establish monotonicity, notice that

$$\begin{aligned} \delta \frac{1-\delta}{(1-\delta\mu)^2} - \frac{1-\delta}{(1-\delta\mu)(1-\mu)} &= (1-\delta) \left(\frac{-\delta-1+\delta(1-\delta\mu)}{(1-\delta\mu)^2(1-\mu)} \right) \\ &= -(1-\delta) \left(\frac{1+\delta^2\mu}{(1-\delta\mu)^2(1-\mu)} \right) \\ &< 0. \end{aligned}$$

Hence, there's a unique crossing point where the function $1 + \epsilon_\mu^A$ is negative.

Finally, we compute relevant limits. First, we compute:

$$\lim_{\mu \rightarrow 0} 1 + \epsilon_\mu^A = (1-\delta) \lim_{\mu \rightarrow 0} \frac{\mu}{\ln\left(\frac{1-\delta\mu}{1-\mu}\right)} = \frac{1-\delta}{\lim_{\mu \rightarrow 0} \frac{1-\mu}{1-\delta\mu} \frac{1-\delta\mu}{1-\mu} \left(\frac{-\delta}{1-\delta\mu} - \frac{-1}{1-\mu} \right)} = 1.$$

where the first equality are the surviving terms after taking limits, the second equality follows from L'Hospital's rule. For the limit $\mu \rightarrow 1$, by L'Hospital's rule:

$$\lim_{\mu \rightarrow 1} \frac{1}{1-\mu} \left(\frac{1}{\ln\left(\frac{1-\delta\mu}{1-\mu}\right)} \right) = \lim_{\mu \rightarrow 1} \frac{-\frac{1}{1-\mu^2}}{\frac{1-\delta}{(1-\delta\mu)(1-\mu)}} = \lim_{\mu \rightarrow 1} -\frac{(1-\delta\mu)}{(1-\mu)(1-\delta)}.$$

Adding terms the missing term:

$$\begin{aligned} \lim_{\mu \rightarrow 1} [1 + \epsilon_\mu^A] &= \lim_{\mu \rightarrow 1} -\frac{(1-\delta\mu)}{(1-\mu)(1-\delta)} - \frac{\mu}{1-\mu} \\ &= \lim_{\mu \rightarrow 1} \frac{-1+2\delta\mu-\mu}{(1-\mu)(1-\delta)} \\ &= -2 \lim_{\mu \rightarrow 1} \frac{1}{(1-\mu)} \\ &= -\infty. \end{aligned}$$

We finally compute the elasticity. From

$$\frac{\mathcal{A}'(\mu)}{\mathcal{A}(\mu)} \mu < 0,$$

we know that the elasticity ϵ_μ^A is decreasing. Hence, $1 + \epsilon_\mu^A$ starts at zero and falls continuously until diverging at $\mu \rightarrow 1$.

B. Model Extensions

B.1 Markov Chain Representation

- μ_x fraction of chained
- $\mu_s = 1 - \mu_x$ fraction of spot
- the column vector is μ with the first entry μ

Payments chain have a Markov chain representation:

- It behaves as a history of states
- $S^t = \{s, x, s, x, s\}$

Thus, must be possible to write it as a Markov Chain with transition matrix

$$Q = \begin{bmatrix} p_{ss} & 1 - p_{xx} \\ 1 - p_{ss} & p_{xx} \end{bmatrix}.$$

Since Q is a probability matrix, all the information is contained in one row. Then, we must have that populations are maintained:

$$\mu = Q\mu.$$

We have special restrictions.

- iid case, considered in current version:

$$p_{ss} = \mu_s, p_{xx} = \mu_x.$$

Thus:

$$Q^{iid} = \begin{bmatrix} \mu_s & \mu_s \\ \mu_x & \mu_x \end{bmatrix}$$

The, clearly:

$$\mu = Q^{iid}\mu$$

since:

$$\begin{bmatrix} \mu_s \\ \mu_x \end{bmatrix} = \begin{bmatrix} \mu_s & \mu_s \\ \mu_x & \mu_x \end{bmatrix} \begin{bmatrix} \mu_s \\ \mu_x \end{bmatrix} = \begin{bmatrix} \mu_s (\mu_s + \mu_x) \\ \mu_x (\mu_s + \mu_x) \end{bmatrix}.$$

- Deviations from the iid case:

$$p_{ss} = \mu_s \kappa_s, p_{xx} = \mu_x \kappa_x.$$

Thus:

$$Q = \begin{bmatrix} \mu_s \kappa_s & 1 - \mu_x \kappa_x \\ 1 - \mu_s \kappa_s & \mu_x \kappa_x \end{bmatrix}$$

Which we conveniently write:

$$Q = \begin{bmatrix} \mu_s + \mu_s (\kappa_s - 1) & 1 - \mu_x - \mu_x (\kappa_x - 1) \\ 1 - \mu_s - \mu_s (\kappa_s - 1) & \mu_s + \mu_x (\kappa_x - 1) \end{bmatrix} = (Q^{iid} + \tilde{Q})$$

where

$$\tilde{Q} = \begin{bmatrix} \mu_s (\kappa_s - 1) & -\mu_x (\kappa_x - 1) \\ -\mu_s (\kappa_s - 1) & \mu_x (\kappa_x - 1) \end{bmatrix}$$

- Next, we have that:

$$\mu = Q\mu = (Q^{iid} + \tilde{Q})\mu$$

since

$$\mu = Q^{iid}\mu$$

then:

$$0 = \tilde{Q}\mu$$

$$\mu_s^2 (\kappa_s - 1) = \mu_x^2 (\kappa_x - 1)$$

Hence:

$$\mu_s^2 (\kappa_s - 1) = \mu_x^2 (\kappa_x - 1).$$

Thus, if

$$\mu_s (\kappa_s - 1) = \frac{\mu_x}{\mu_s} \mu_x (\kappa_x - 1).$$

Define:

$$q = \mu_s (\kappa_s - 1).$$

Then:

$$Q = \begin{bmatrix} \mu_s + q & 1 - \mu_x - \frac{\mu_s}{\mu_x} q \\ 1 - \mu_s - q & \mu_x + \frac{\mu_s}{\mu_x} q \end{bmatrix},$$

or more convenient to do change of variables:

$$\frac{\mu_s}{\mu_x} q = \rho.$$

- Bounded probabilities:

- if $\mu_s > 1/2 \rightarrow q \leq 1 - \mu_s, q \geq -\mu_s$
- if $\mu_x > 1/2 \rightarrow q \leq 1 - \mu_x, q \geq -\mu_x$
- coupling: $q \leq 1 - \max \{ \mu_x, \mu_s \}, q \geq -\max \{ \mu_x, \mu_s \}.$

- This allows for a correlated structure.

- Moments:

- fraction of spots in a chain:

$$1 - \mu_s - q$$

- fraction of spots alone:

$$\mu_s + q$$

- distribution of chain length, conditional on chain of size at least one:

$$F(n) = \left(1 - \mu_x - \frac{\mu_x}{\mu_s} q \right) \left(\mu_x + \frac{\mu_x}{\mu_s} q \right)^{n-1}$$

- Then:

$$F(n) = (1 - (\mu_x + \rho)) (\mu_x + \rho)^{n-1}.$$

That, is, chain distribution is Geometric with probability $p = \mu_x + \rho$.

- Output of chained orders in chain of length n

$$y^x(n) = \delta (1 + \delta + \delta^2 \dots \delta^{n-1}) = \delta \left(\frac{1}{1-\delta} - \frac{\delta^n}{1-\delta} \right),$$

for $n \geq 1$. Clearly,

$$y^x(1) = \delta$$

- Average output in chain of length n is:

$$\bar{y}^x(n) = \frac{y(n)}{n}.$$

- Integrating the first term with respect to the Geometric distribution:

$$\mathbb{E}[\bar{y}^x(n) | n \geq 1] = \frac{1}{p} \frac{\delta}{1-\delta} \sum_{n=1}^{\infty} \frac{1}{n} (1-\delta^n) F(n).$$

- Then, average output is given by:

$$\mathbb{E}[\bar{y}^x(n) | n \geq 1] = \frac{\delta}{p} \frac{1-p}{1-\delta} \sum_{n=1}^{\infty} \frac{1}{n} (p^n - (\delta p)^n).$$

We use that:

$$\vartheta(p) = \sum_{n=1}^{\infty} \frac{p^n}{n} = -\log(1-p),$$

which is shown by taking derivatives on both sides:

$$\vartheta'(p) = \sum_{n=1}^{\infty} p^{n-1} = \frac{1}{1-p},$$

and integrating back evaluating at $p = 0$ to obtain the constant to be equal to zero.

Then, we have that:

$$\begin{aligned} \mathbb{E}[\bar{y}^x(n) | n \geq 1] &= -\delta \frac{1-p}{1-\delta} \log(1-p) + \delta \frac{1-p}{1-\delta} \log(1-\delta p) \\ &= \delta \frac{1-p}{1-\delta} \log\left(\frac{1-\delta p}{1-p}\right). \end{aligned}$$

- This is the average output of chained orders conditional on falling in a chain.

- A very similar formula to the one employed in the paper, except that we have only consider production
- Overall production among spot is:

$$\mathbb{E} [y^s(n) | n \geq 0] = 1$$

Thus, overall production is:

$$\begin{aligned} & \mathbb{E} [y^s(n) | n \geq 0] (1 - \mu^x) + \mu^x \mathbb{E} [\bar{y}^x(n) | n \geq 1] \\ &= (1 - \mu^x) + \mu^x \delta \frac{1-p}{1-\delta} \log \left(\frac{1-\delta p}{1-p} \right) \\ &= (1 - \mu^x) + \mu^x \frac{1 - \mu_x + \rho}{\mu_x + \rho} \frac{\delta}{1-\delta} \log \left(\frac{1 - \delta (\mu_x + \rho)}{1 - \mu_x + \rho} \right). \end{aligned}$$

- Using the notation in the paper:

$$(1 - \mu) + \mu \frac{1 - \mu + \rho}{\mu + \rho} \frac{\delta}{1-\delta} \log \left(\frac{1 - \delta (\mu + \rho)}{1 - \mu + \rho} \right).$$

B.2 Truncation

- We verified that the formula generalized to the case with correlation across chained orders and spot orders.
- Simplified notation:

$$\mu_x = \mu$$

- Now, we investigate the role of truncation.
- The chain length is as before, distributed

$$F(n) = \frac{(1-p)}{p} (p)^n,$$

which integrates to 1.

- Truncation states that if n is above a threshold value η , the chained order is dropped.

Thus, production is given by:

$$y^x(n) = \delta(1 + \delta + \delta^2 \dots \delta^{n-1}) = \delta \left(\frac{1}{1-\delta} - \frac{\delta^n}{1-\delta} \right), \text{ if } n \leq \eta,$$

and otherwise:

$$y^x(n) = \delta(1 + \delta + \delta^2 \dots \delta^{n-1}) = \delta \left(\frac{1}{1-\delta} - \frac{\delta^\eta}{1-\delta} \right) = \delta \left(\frac{1}{1-\delta} - \frac{\delta^n}{1-\delta} - \frac{\delta^\eta}{1-\delta} (1 - \delta^{n-\eta}) \right).$$

Thus, with truncation,

$$\bar{y}^x(n) = \frac{y(n)}{n} = \frac{\delta}{n} \left(\frac{1}{1-\delta} - \frac{\delta^n}{1-\delta} \right) - \bar{\gamma}(n)$$

where

$$\bar{\gamma}(n) = \begin{cases} 0 & \text{if } n \leq \eta \\ \frac{1}{n} \frac{\delta^\eta}{1-\delta} (1 - \delta^{n-\eta}) & \text{if } n > \eta. \end{cases}$$

Thus, from our earlier derivation:

$$\mathbb{E}[\bar{y}^x(n) | n \geq 1] = \frac{1 - \mu_x + \rho}{\mu_x + \rho} \frac{\delta}{1-\delta} \log \left(\frac{1 - \delta(\mu_x + \rho)}{1 - \mu_x + \rho} \right) - \mathbb{E}[\bar{\gamma}(n) | n \geq 1].$$

We compute:

$$\mathbb{E}[\bar{\gamma}(n) | n \geq 1] = \mathbb{E} \left[\frac{1}{n} \frac{\delta^\eta}{1-\delta} (1 - \delta^{n-\eta}) | n \geq \eta \right] \Pr(n \geq \eta).$$

- Working by pieces:

$$\begin{aligned}
\Pr(n \geq \eta) &= \sum_{n=\eta}^{\infty} F(n) \\
&= \sum_{n=\eta}^{\infty} \frac{(1-p)}{p} (p)^n \\
&= \frac{(1-p)}{p} p^{\eta} \left(\sum_{n=\eta}^{\infty} (p)^{n-\eta} \right) \\
&= \frac{(1-p)}{p} p^{\eta} \left(\sum_{j=0}^{\infty} (p)^j \right) \\
&= \frac{(1-p)}{p} \frac{p^{\eta}}{(1-p)} = p^{\eta-1}.
\end{aligned}$$

- Next, we compute:

$$\begin{aligned}
\mathbb{E} \left[\frac{1}{n} \frac{\delta^{\eta}}{1-\delta} (1 - \delta^{n-\eta}) \mid n \geq \eta \right] &= \sum_{n=\eta}^{\infty} \frac{1}{n} \frac{\delta^{\eta}}{1-\delta} (1 - \delta^{n-\eta}) \frac{F(n)}{\Pr(n \geq \eta)} \\
&= \frac{(1-p)}{p} \frac{\delta^{\eta}}{1-\delta} \sum_{n=\eta}^{\infty} \frac{1}{n} (1 - \delta^{n-\eta}) (p)^{n-\eta} \\
&= \frac{(1-p)}{p} \frac{\delta^{\eta}}{1-\delta} \sum_{n=\eta+1}^{\infty} \frac{1}{n} (1 - \delta^{n-\eta}) (p)^{n-\eta} \\
&= \frac{(1-p)}{p} \frac{\delta^{\eta}}{1-\delta} \sum_{j=1}^{\infty} \frac{1}{\eta+j} (1 - \delta^j) (p)^j \\
&= \frac{(1-p)}{p} \frac{\delta^{\eta}}{1-\delta} \sum_{j=1}^{\infty} \frac{1}{\eta+j} (p^j - (p\delta)^j) \\
&= \frac{(1-p)}{p} \frac{\delta^{\eta}}{1-\delta} \left(\frac{1}{p^{\eta}} \sum_{j=1}^{\infty} \frac{1}{\eta+j} p^{j+\eta} - \frac{1}{(p\delta)^{\eta}} \sum_{j=1}^{\infty} \frac{1}{\eta+j} (p\delta)^{j+\eta} \right) \\
&= \frac{(1-p)}{p} \frac{\delta^{\eta}}{1-\delta} \left(\frac{1}{p^{\eta}} \sum_{j=1}^{\infty} \frac{1}{\eta+j} p^{j+\eta} - \frac{1}{(p\delta)^{\eta}} \sum_{j=1}^{\infty} \frac{1}{\eta+j} (p\delta)^{j+\eta} \right) \\
&= \frac{(1-p)}{p} \frac{\delta}{1-\delta} \left(\frac{\delta^{\eta}}{\delta p^{\eta}} \sum_{j=0}^{\infty} \frac{1}{\eta+j} p^{j+\eta} - \frac{1}{\delta (p)^{\eta}} \sum_{j=0}^{\infty} \frac{1}{\eta+j} (p\delta)^{j+\eta} \right) \\
&= \frac{(1-p)}{p} \frac{\delta^{\eta}}{1-\delta} (\Phi(p, 1, \eta) - \Phi(p\delta, 1, \eta))
\end{aligned}$$

The term:

$$\Phi(z, 1, \eta) = \sum_{j=0}^{\infty} \frac{z^j}{\eta + j} = \int_0^{\infty} \frac{\exp(-\eta t)}{1 - z \exp(-t)} dt,$$

is the Lerch transcendent. Notice that for $\eta = 1$:

$$\begin{aligned} \int_0^{\infty} \frac{\exp(-t)}{1 - z \exp(-t)} dt &= \int_0^{\infty} \frac{1}{\exp(t) - z} dt = \frac{1}{z} \log(1 - z \exp(-t)) \Big|_{t=0}^{t=\infty} \\ &= -\frac{1}{z} \log(1 - p) = \frac{1}{z} \sum_{k=1}^{\infty} \frac{z^k}{k} = \frac{1}{z} \sum_{j=0}^{\infty} \frac{z^{j+1}}{1+j} = \frac{1}{z} \sum_{j=1}^{\infty} \frac{z^j}{1+j}. \end{aligned}$$

- and thus:

$$= \frac{(1-p)}{p} \frac{\delta^\eta}{1-\delta} \left(\frac{\delta^\eta}{\delta p^\eta} \Phi(p, 1, \eta) - \frac{1}{\delta p^\eta} \Phi(p\delta, 1, \eta) \right)$$

- This formula allows us to generalize to the case with truncation:

$$\mathbb{E} \left[\frac{1}{n} \frac{\delta^\eta}{1-\delta} (1 - \delta^{n-\eta}) \mid n \geq \eta \right] p^{\eta-1} = \frac{(1-p)}{p} \frac{\delta}{1-\delta} \left(\frac{\delta^\eta}{\delta p} \Phi(p, 1, \eta) - \frac{1}{\delta p} \Phi(p\delta, 1, \eta) \right)$$

Thus, we obtain:

$$(1 - \mu^x) + \mu^x \frac{1 - \mu_x + \rho}{\mu_x + \rho} \frac{\delta}{1-\delta} \left(\log \left(\frac{1 - \delta(\mu_x + \rho)}{1 - \mu_x + \rho} \right) - \left(\frac{\delta^\eta}{\delta p} \Phi(p, 1, \eta) - \frac{1}{\delta p} \Phi(p\delta, 1, \eta) \right) \right).$$

In the special case of $\eta \rightarrow \infty, \rho = 0$ we have the original formula.

B.3 Multinomial Case

Now consider the situation where instead of experiencing delays, a particular transaction is executed but the receiver fails to receive the funds with probability $1 - q$ without a delay. Otherwise, it faces the same delay structure as above. This probability is completely random.

Given a chain of length n , the chances that there are k delays is:

$$M(k, n) = \binom{n}{k} q^k (1 - q)^{n-k}.$$

While we can compute average number of delays, they have to be ordered: meaning, delays occur starting from k-th unit. Delays have to be sequential.

Production is:

$$y(k, n) = n - k + \delta \frac{1 - \delta^k}{1 - \delta}.$$

Then, we integrate across all k :

$$\sum_{k=0}^n y(k, n) \times M(k, n) = (n - k) \sum_{k=0}^n M(k, n) + \frac{\delta}{1 - \delta} \sum_{k=0}^n \frac{1}{1 - \delta} M(k, n) - \frac{\delta}{1 - \delta} \sum_{k=0}^n \delta^k \binom{n}{k} q^k (1 - q)^{n-k}$$

But we know that the multinomial integrates to 1:

$$\sum_{k=0}^n \binom{n}{k} q^k (1 - q)^{n-k} = 1.$$

We also know that its expectation is:

$$\sum_{k=0}^n k \binom{n}{k} q^k (1 - q)^{n-k} = \sum_{k=0}^n k \frac{n!}{(n-k)!k!} q^k (1 - q)^{n-k}$$

but by property of factorials:

$$n - k! = n - k \times n - k - 1 \times n - k - 2 \dots$$

$$n - (k - 1)! = n + 1 - k \times n - k - 1 \times n - k - 2 \dots$$

$$\binom{n}{k} = \frac{n}{k} \times \binom{n-1}{k-1}$$

$$\begin{aligned}
\sum_{k=1}^n k \binom{n}{k} q^k (1-q)^{n-k} &= n \sum_{k=1}^n \binom{n-1}{k-1} q^k (1-q)^{n-k} \\
&= nq \left(\sum_{k=1}^{n-1} \binom{n-1}{k-1} q^{k-1} (1-q)^{n-k} \right) \\
&= nq \left(\sum_{k=1}^{n-1} \binom{n-1}{k-1} q^{k-1} (1-q)^{n-1-(k-1)} \right)
\end{aligned}$$

change of variables guarantees:

$$nq.$$

Hence:

$$\sum_{k=0}^n k M(k, n) = qn.$$

Thus:

$$(n-k) \sum_{k=0}^n M(k, n) = (1-q)n.$$

And, moreover:

$$\frac{1}{1-\delta} \sum_{k=0}^n M(k, n) = \frac{1}{1-\delta}.$$

Finally, we have that:

$$\sum_{k=0}^n \delta^k \binom{n}{k} q^k (1-q)^{n-k} = \sum_{k=0}^n \binom{n}{k} (\delta q)^k (1-q)^{n-k}$$

which by the binomial theorem equals:

$$(1-q+\delta q)^n$$

Hence, all in all, production is:

$$\mathbb{E}[y(k, n) | n] = (1-q)n + \delta \frac{1 - (1-q+\delta q)^n}{1-\delta}.$$

Thus, for $q = 0$ we have:

$$\mathbb{E} [y (k, n) | n] = n$$

and when $q = 1$

$$\mathbb{E} [y (k, n) | n] = \delta \frac{1 - \delta^n}{1 - \delta}.$$

corresponding to the formula above for chained production. Then, average production is:

$$\mathbb{E} [\bar{y} (k, n) | n] = (1 - q) + \frac{1}{n} \frac{\delta - \delta (1 - q + \delta q)^n}{1 - \delta}.$$

Thus:

$$\begin{aligned} \mathbb{E} [\mathbb{E} [\bar{y} (k, n) | n]] &= (1 - q) + \sum_{n=1}^{\infty} \frac{1}{n} \frac{\delta - \delta (1 - q + \delta q)^n}{1 - \delta} F(n, p) \\ \sum_{n=1}^{\infty} \frac{1}{n} \frac{\delta - \delta (1 - q + \delta q)^n}{1 - \delta} F(n, p) &= \frac{1 - p}{p} \frac{\delta}{1 - \delta} (\log (1 - p (1 - q + \delta q)) - \log (1 - p)). \end{aligned}$$

Thus, we obtain a generalized formula:

$$\mathbb{E} [\bar{y}^x] = (1 - q) + \frac{1 - p}{p} \frac{\delta}{1 - \delta} (\log (1 - p (1 - q + \delta q)) - \log (1 - p)).$$

Which again, collapses to the general formula.

B.4 Resampling

- Suppose that after observing their own draw, of place in a Chain, chained types can try to resample.
- Potentially there is a cost, but for now assume there is not, and only assume that types above $n > \eta$, instead of dropping out, resample.
- If they resample, they contact another chain at random and they are placed somewhere in the chain, their own chain shrinks to size η .
- They can again match with a chain of size η , which would force another resampling.
- The process repeats itself indefinitely until there are no further resampling.
- The question is, to what distribution does the process converge too?

- We've already seen that the distribution of chain sizes will be Geometric:
 - $F(n, p)$ with $(1 - p) p^n$ be the distribution of chain size with $n \geq 0$
- Let $z \in \{0, 1, \dots, \infty\}$ be the number of rounds.
 - it is better to write this as a continuous-time process so we avoid problems with integers
 - matches occur at rate intensity λ
- Let $G_0(n) = F(n, p)$ be the initial distribution
- After the initial distribution, the fraction

$$\mu_0 = \sum_{n > \eta} G_0(n)$$

will seek to resample. They randomly match with some other agent. A match is successful if it falls within a chain of length less than η , because otherwise we ran into the same problem again. The available chains are:

$$A_0 = \sum_{n=0}^{\eta-1} G_0(n)$$

out of the pool:

$$1 - \mu_0 = \sum_{n=0}^{\eta} G_0(n).$$

Thus, a match is useful with probability:

$$p_0 = \frac{\alpha_0}{1 - \mu_0} \rightarrow p(z) = \frac{\sum_{n=0}^{\eta-1} G_0(n)}{\sum_{n=0}^{\eta} G_0(n)}$$

and

$$p(n, z) = \frac{G_z(n)}{\sum_{j=0}^{\eta} G_z(j)}.$$

- Thus, we can think of the mass of unsuccessful matches as shrinking at rate:

$$\dot{\mu}(z) = -\lambda p(z) (1 - \mu(z)) \mu(z).$$

If matches occur, chains add one more type.

- Thus, they satisfy the law of motion:

- inflow probabilities are:
- for $n = 0$

$$\dot{G}(0, z) = -\lambda p(0, z) \mu(z) (1 - \mu(z)) = -\lambda G(0, z) \mu(z).$$

and for $n \in \{1, \eta - 1\}$

$$\dot{G}(n, z) = -\lambda (G(n, z) \mu(z) - 2G(n - 1, z) \mu(z))$$

and for $n = \eta$

$$\dot{G}(\eta, z) = \lambda 2G(\eta - 1, z) \mu(z).$$

- Now, we verify things by adding masses:

$$\sum_{n=0}^{\eta} \dot{G}(n, z) = \lambda G(0, z) \mu(z) + \lambda G(1, z) \mu(z) + G(1, z) \mu(z) \dots + G(\eta - 1, z) \mu(z)$$

which accounts for the preservation of mass probability mass since:

$$\lambda G(0, z) \mu(z) + \lambda G(1, z) \mu(z) + G(1, z) \mu(z) \dots + G(\eta - 1, z) \mu(z) = \lambda p(z) (1 - \mu(z)) \mu(z)$$

Hence:

$$\sum_{n=0}^{\eta} \dot{G}(n, z) + \dot{\mu} = 0.$$

- Next, we try to find a way to characterize the evolution of the system. The key challenge is the presence of η . In matrix form we can write:

$$\mathbf{G}(z) = \begin{bmatrix} G(1, z) \\ G(2, z) \\ \vdots \\ G(\eta, z) \end{bmatrix}$$

$$\dot{\mathbf{G}} = -\lambda \mathbf{A} \mathbf{G}(z) \mu(z)$$

where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -2 & 1 & 0 & & \\ 0 & -2 & 1 & & \\ & & & \ddots & \\ \vdots & & & -2 & 1 & \vdots \\ 0 & & & & -2 & 0 \end{bmatrix}.$$

- We write the system as in non-linear form as:

$$\dot{\mathbf{G}} = -\lambda \mathbf{A}(z) \mathbf{G}(z)$$

$$\mathbf{A}(z) = \mathbf{A} \mu(z)$$

which in Matrix form solves:

$$\mathbf{G}(z) = \exp \left(-\lambda \int_0^z \mathbf{A}(\tau) d\tau \right) \mathbf{G}(0),$$

where

$$\mu(z) = -\lambda \left(1 - \frac{G(\eta, z)}{\sum_{n=0}^{\infty} G(n, z)} \right) \mu(z) (1 - \mu(z)).$$

- The solution to $\mu(z)$ can also be written as a time-varying problem:

$$\dot{\mu}(z) = -\lambda(z) \mu(z) (1 - \mu(z))$$

where

$$\lambda(z) = \lambda \left(1 - \frac{G(\eta, z)}{\sum_{n=0}^{\infty} G(n, z)} \right)$$

$$\dot{\mu}(z) = -\lambda(z) \mu(z) (1 - \mu(z)).$$

Guess and verify that:

$$\mu(z) = \frac{1}{1 + \exp \left(\int_0^z \lambda(\tau) d\tau \right)}.$$

Derivative:

$$\dot{\mu}(z) = -\lambda(\tau) \frac{\exp\left(\int_0^z \lambda(\tau) d\tau\right)}{\left(1 + \exp\left(\int_0^z \lambda(\tau) d\tau\right)\right)^2}$$

$$-\lambda(\tau) \mu(z) (1 - \mu(z)) = -\lambda(\tau) \frac{1}{1 + \exp\left(\int_0^z \lambda(\tau) d\tau\right)} \left(1 - \frac{1}{1 + \exp\left(\int_0^z \lambda(\tau) d\tau\right)}\right),$$

proving the solution.

- Thus, we need to solve the system:

$$\mathbf{G}(z) = \mathbf{G}(0) \exp\left(-\mathbf{A} \int_0^z \lambda \mu(z) d\tau\right)$$

where k is some constant.

$$\mu(z) = \frac{1}{1 + \exp\left(\int_0^z \lambda(\tau) d\tau\right)}.$$

- Thus, suppose that the process does indeed converge in finite time.

$$\mathbf{G}(\infty) = \exp(-k\mathbf{A}) \mathbf{G}(0)$$

- Then:

$$\mathbf{e} \mathbf{G}(\infty) = 1 = \mathbf{e} \exp(-k\mathbf{A}) \mathbf{G}(0).$$

Thus,

$$\exp\left(-\lambda \int_0^z \mathbf{A}(\tau) d\tau\right).$$

Hence, we look for k such that

$$1 = \mathbf{e} \exp(-k\mathbf{A}) \mathbf{G}(0).$$

If we find such a k we have the solution.

C. Proofs of Section 5

D. Business Cycle Model: Details

The following lemma identifies two threshold points that characterize the worker's expenditure mix in terms of their debt.

Lemma 2. (Expenditure Threshold Points): *Define the **efficiency threshold**, $B_{t+1}^* \equiv R_{t+1} (\tilde{B}_t - 1)$. Then, $S_{t+1}^w = 0$ if and only if $B_{t+1} > \tilde{B}_{t+1}$. In addition, $X_t^w > 0$ if and only if $B_{t+1} > B_{t+1}^*$.*

Proof. The proof is immediate. If $B_{t+1} > \tilde{B}_{t+1}$, the worker cannot spend spot. In turn, $B_{t+1} > R_{t+1} (\tilde{B}_t - 1)$ happens if and only if $E_t^w = B_{t+1}/R_{t+1} - B_t + 1 > \tilde{B}_t - B_t$. In that case, the worker's chosen expenditures are sufficiently high that he makes chained orders. □

It is convenient to define some relevant objects derived from these threshold points.

Definition 4. (Marginal Prices and Marginal inflation):

I. The **average price** per unit of worker expenditure is $Q_t \equiv E_t^w / C_t^w$.

II. Given B' , the price of the good at t bought with a marginal increase in B' is:

$$\tilde{q}_t^E(B') \equiv 1 + (q_t - 1) \mathbb{I}_{[B' \geq B_{t+1}^*]},$$

the price of the good purchased at $t + 1$ after a marginal decrease in B' is

$$\tilde{q}_{t+1}^B(B') \equiv 1 + (q_{t+1} - 1) \mathbb{I}_{[B' > \tilde{B}_{t+1}]}$$

These terms define **marginal prices**.

III. **Marginal inflation** is $\Pi_{t+1}(B') \equiv \tilde{q}_{t+1}^B(B') / \tilde{q}_t^E(B')$, a continuous function of B' except at discontinuity points $\{B_{t+1}^*, \tilde{B}_{t+1}\}$.

The average goods price is the ratio of expenditures to the quantity of goods the worker buys. Marginal prices have the following interpretation: If at t the worker spends on chained goods, the reduction in her debt is financed with a reduction in chained expenditures. Otherwise, if she does not spend on chained goods, a reduction in her debt is financed with a reduction in spot goods. Since the price of chained goods is q_t and

the price of spot goods is 1, the worker sacrifices $1/\tilde{q}_t^E$ units of consumption per unit of debt reduction. Likewise, if she spends spot at $t + 1$, any past savings can translate into spot expenditure at $t + 1$. Otherwise, if she only spends in chained goods at $t + 1$, any past savings translate into chained expenditures. Hence, the worker can buy $1/\tilde{q}_{t+1}^B$ additional goods by reducing her debt at t on the margin. Marginal inflation is the ratio of marginal prices, a definition that enters the following generalized Euler equation.

Proposition 6. (Workers's First-Order Condition): *Fix a sequence $\{\tilde{B}_t, R_{t+1}, q_t\}_{t \geq 0}$ such that $\tilde{B}_t$ is an increasing and $\beta R_{t+1} \leq 1$. Then, any solution $\{B_{t+1}\}_{t \geq 0}$ to the worker's problem satisfies the following generalized Euler equation:*

$$\frac{E_{t+1}^w}{E_t^w} \frac{Q_t}{Q_{t+1}} = \beta \frac{R_{t+1}}{\Pi_{t+1}(B_{t+1})} \quad \text{if } B_{t+1} \neq B_{t+1}^* \quad (17)$$

and

$$\beta q_t R_{t+1} \geq \frac{E_{t+1}^w}{E_t^w} \geq \beta R_{t+1} \quad \text{if } B_{t+1} = B_{t+1}^*.$$

This Euler equation is unconventional, but the interpretation is standard. The left-hand side is the ratio of marginal rates of substitution between t and $t+1$ consumption—expressed in expenditures over average prices.³⁰ The right-hand side captures the relation between discounting and rate of return, βR_{t+1} ; the novelty is that it is deflated by marginal inflation. Marginal inflation enters the expression because this is the relevant ratio of prices that deliver marginal utilities at t and $t + 1$, respectively $\tilde{q}_t^E$ and $\tilde{q}_{t+1}^B$. This generalized Euler equation holds exactly except at the discontinuity point B_{t+1}^* . The inequalities that are satisfied when $B_{t+1} = B_{t+1}^*$ correspond to a sub-differential optimality condition: the condition says that increasing debt is optimal to the left of B_{t+1}^* , but not to its right.³¹

Equilibrium Dynamics. We now analyze the equilibrium dynamics of B_t . The equilibrium is recursive with state variable $B_t \times \tilde{B}_t \times \tilde{B}_{t+1} \in [0, \bar{B}]^3$. A recursive representation requires that we have a relationship between a sequence and a recursive formulation: a variable m_t , can be expressed as an equilibrium function, $m : [0, \bar{B}]^3 \rightarrow \mathbb{R}_+$ such that $m_t = m(B_t, \tilde{B}_t, \tilde{B}_{t+1})$. Hence, from now we use m to represent the function that

³⁰The ratio of expenditures to average prices, is the ratio of marginal utilities under log preferences.

³¹There is also a discontinuity in the Euler equation at $B_{t+1} = \bar{B}_{t+1}$, but the corresponding sub-differential does not yield an optimality condition.

maps the state into its equilibrium value m_t . We also adopt the convention of using m' to refer to m_{t+1} .

Using the recursive formulation, from (??), (??), and (??) we have:

$$C^s(B) = (1 - \beta) B, \quad E^w(B) = 1 - (1 - \beta) B, \quad (18)$$

and

$$S^w(B, \tilde{B}) = \min \left\{ \bar{S}(B, \tilde{B}), E^w(B) \right\}.$$

Several other equilibrium functions are deduced from these functions directly: $\mu = E^w - S^w$, $q \equiv \mathcal{A}^{-1}(\mu)$, $X^w \equiv \mu/q$, $Q \equiv E^w/C^w$, and, finally, $C^w \equiv S^w + X^w$.

The only endogenous argument of the state is B_t . Thus, we need a map $\mathcal{B}$ from the current state to its future value, $B' = \mathcal{B}(B, \tilde{B}, \tilde{B}')$. If we obtain that map, the equilibrium rate will satisfy $R_{t+1} = \mathcal{R}(B_t, \tilde{B}_t, \tilde{B}_{t+1})$ where $\mathcal{R}(B, \tilde{B}, \tilde{B}') \equiv \beta^{-1} \mathcal{B}(B, \tilde{B}, \tilde{B}')/B$, following the saver's optimal rule. With that equilibrium function, we can define the threshold function $B^*(B, \tilde{B}, \tilde{B}') \equiv \mathcal{R}(B, \tilde{B}, \tilde{B}') \cdot (\tilde{B} - 1)$. We solve for $\mathcal{B}$ below, after defining a functional representation for the equilibrium prices:

$$\Pi(B'; B, \tilde{B}, \tilde{B}') \equiv \tilde{q}^B(B'; \tilde{B}') / \tilde{q}^E(B'; B, \tilde{B}, \tilde{B}'),$$

where

$$\tilde{q}^E(B'; B, \tilde{B}, \tilde{B}') \equiv 1 + \left(q(B, \tilde{B}) - 1 \right) \cdot \mathbb{I}_{[B' \geq B^*(B, \tilde{B}, \tilde{B}')]}$$

and

$$\tilde{q}^B(B'; \tilde{B}') \equiv 1 + \left(q(B', \tilde{B}') - 1 \right) \cdot \mathbb{I}_{[B' > \tilde{B}']}.$$

To find $\mathcal{B}$, we combine the worker and saver Euler equations, (17) and (??). We then substitute out R_{t+1} and use the equilibrium worker expenditures (18), to obtain an equilibrium condition B' :

$$\underbrace{\frac{B}{1 - (1 - \beta) B} \cdot Q(B, \tilde{B})}_{\equiv \mathcal{E}(B; \tilde{B})} = \underbrace{\frac{B'}{1 - (1 - \beta) B'} \frac{Q(B', \tilde{B}')}{\Pi(B'; B, \tilde{B}, \tilde{B}')}}_{\equiv \mathcal{E}'(B'; B, \tilde{B}, \tilde{B}')} \quad (19)$$

The left and right hand sides define functions $\mathcal{E}(B'; B)$ and $\mathcal{E}'(B'; B, \tilde{B}, \tilde{B}')$. At equal-

ity, these functions define the law of motion of debt. Because Q is discontinuous, there may be multiple solutions B' . The following result tells us that in a transition, the lowest value is valid.

Proposition 7. (Equilibrium Rates and Expenditures): *Consider a weakly monotone increasing sequence of spot borrowing lines $\tilde{B}_t \rightarrow \tilde{B}_{ss}$. For any $B_0 < B^h(\tilde{B}_{ss})$, if an equilibrium exist, then $B_{t+1} = \mathcal{B}(B_t, \tilde{B}_t, \tilde{B}_{t+1})$ where*

$$\mathcal{B}(B, \tilde{B}, \tilde{B}') = \max \left\{ B^*(\tilde{B}), \arg \min_{B'} \left\{ \mathcal{E}(B'; B, \tilde{B}, \tilde{B}') = \mathcal{E}'(B'; B, \tilde{B}, \tilde{B}') \right\} \right\}.$$

Proposition 7 is key to describing the dynamics.³²

D.1 Analysis: Preliminary Observations

We begin with a set of identities that are convenient to prove the main results.

Average Price: Definitions and Identities. Recall that the average price of a worker as it's expenditures relative to total consumption: $Q \equiv E^w / C^w$. The saver's expenditure is shown to be $E^s = (1 - \beta) B$. By the income expenditure identity:

$$E^w = 1 - (1 - \beta) B. \quad (20)$$

Total consumption by the worker, given its optimal expenditures, is given by:

$$C^w = \frac{E^w - S^w}{q} + S^w = \frac{E^w}{q} - \left(\frac{1}{q} - 1 \right) S^w.$$

where

$$S^w = \min \left\{ \max \left\{ 0, \tilde{B} - B \right\}, 1 - (1 - \beta) B \right\},$$

$$X^w = (E^w - S^w) / q.$$

³²Implicitly, the proposition yields an algorithm to compute equilibria. Starting from, B_0 , the sequence of debt generated in equilibrium is given by $\mathcal{B}(B, \tilde{B}, \tilde{B}')$, which is the smallest solution B' to the equation $\mathcal{E}(B'; B, \tilde{B}, \tilde{B}') = \mathcal{E}'(B'; B, \tilde{B}, \tilde{B}')$. For each B , we obtain B' and update the state accordingly.

Thus, we have that:

$$Q = \left(\frac{1}{q} - \left(\frac{1}{q} - 1 \right) \frac{S^w}{E^w} \right)^{-1} = \left(\frac{1}{q} \left(1 - \frac{S^w}{E^w} \right) + \frac{S^w}{E^w} \right)^{-1}.$$

Also, we have that:

$$Q = \frac{E^w - S^w}{C^w} + \frac{S}{C^w} = q \frac{X^w}{C^w} + \frac{S}{C^w}.$$

The last two proof the following result.

Lemma 3. *Q is average of the goods prices weighted by the worker's consumption shares and Q is the harmonic mean of the goods prices weighted by the expenditure share.*

Marginal Expenditure and Borrowing Prices. For convenience, we define again the two prices that enter in marginal decisions. First, we define the *marginal expenditure price*:

$$\tilde{q}_t^e \equiv q_t \tilde{\approx}_{[B_t \geq B^*(\tilde{B}_t)]} + \left(1 - \tilde{\approx}_{[B_t \geq B^*(\tilde{B}_t)]} \right).$$

Next, we define the *marginal borrowing price*:

$$\tilde{q}_{t+1}^b \equiv q_{t+1} \tilde{\approx}_{[\tilde{B}_{t+1} \leq B_{t+1}]} + \left(1 - \tilde{\approx}_{[\tilde{B}_{t+1} \leq B_{t+1}]} \right).$$

Analysis of the Marginal Expenditure Price. Next, we describe the behavior of $\tilde{q}_t^e$ in equilibrium. We have that $\tilde{q}_t^e = \tilde{q}^e(B_t, \tilde{B}_t)$ where:

$$\tilde{q}^e(B, \tilde{B}) \equiv q \left(\mu(B, \tilde{B}) \right) \tilde{\approx}_{[B \geq B^*(\tilde{B})]} + \left(1 - \tilde{\approx}_{[B \geq B^*(\tilde{B})]} \right).$$

In this expression, we are using:

$$\mu(B, \tilde{B}) = 1 - (1 - \beta) B - \min \left\{ \max \left\{ 0, \tilde{B} - B \right\}, 1 - (1 - \beta) B \right\}.$$

Observe that

$$\mu(B, \tilde{B}) = \begin{cases} 0 & B < B^*(\tilde{B}), \\ 1 + \beta B - \tilde{B} & B \in [B^*(\tilde{B}), \tilde{B}], \\ 1 - (1 - \beta) B & B > \tilde{B}. \end{cases}$$

$\mu(B, \tilde{B})$ is continuous, starts at zero and increases up to $B = \tilde{B}$, starting from that point, the function is decreasing. Since q_t is monotone in μ_t , the function $\tilde{q}^e(B, \tilde{B})$ must follow the same pattern in the interior of the middle segment of the function and in the last segment. Also notice that when $B < B^*(\tilde{B})$, $q^e = 1$. Next, we have that since μ is continuous, μ goes to 0 from above as $B \downarrow B^*(\tilde{B})$ so we can write,

$$\lim_{B \downarrow B^*(\tilde{B})} \tilde{q}^e(B, \tilde{B}) = \lim_{\mu \downarrow 0} q(\mu) \cdot 1 = \delta^{-1}$$

where we used the fact that $\lim_{\mu \downarrow 0} \mathcal{A}(\mu) = \delta$. Hence, the function $\tilde{q}^e$ is discontinuous at $B = B^*$ because $\lim_{B \uparrow B^*(\tilde{B})} \tilde{q}^e = 1 \neq \delta^{-1}$, the function is also not monotonic. Then³³

$$\tilde{q}^e(B, \tilde{B}) = \begin{cases} 1 & B < B^*(\tilde{B}) \\ q & B \in [B^*(\tilde{B}), \tilde{B}] \\ q & B > \tilde{B}. \end{cases}$$

Next, we are interested in the behavior of $\frac{Q}{q^e}$, for reasons that become clear in the main text. We have that for $B < B^*(\tilde{B})$, since $q = 1$, it must be that $\frac{Q}{q^e} = 1$. For $B \geq \tilde{B}$ also $q^e = Q = q$. Therefore, $\frac{Q}{q^e} = 1$. In the middle range of values, we have that:

$$\frac{Q}{q^e} = \frac{1}{q} \cdot \frac{q}{\left(1 - \frac{S^w}{E^w}\right) + q \frac{S^w}{E^w}} = \frac{1}{\left(1 - \frac{S^w}{E^w}\right) + q \frac{S^w}{E^w}}.$$

Thus, we have the following formula:

$$\frac{Q}{q^e} = \begin{cases} 1 & B < B^*(\tilde{B}) \\ \frac{1}{1 - \frac{\min\{\max\{0, \tilde{B} - B\}, 1 - (1 - \beta)B\}}{1 - (1 - \beta)B} + q(\mu(B, \tilde{B})) \frac{\min\{\max\{0, \tilde{B} - B\}, 1 - (1 - \beta)B\}}{1 - (1 - \beta)B}} & B \in [B^*(\tilde{B}), \tilde{B}] \\ 1 & B > \tilde{B}. \end{cases}$$

³³In fact, the value of q^e at $B = B^*$ will not be used since the function is discontinuous at that point and optimality conditions will be characterized via the right and left limit. However, we choose to define it according to the intuition presented lines above.

Since at $B = \tilde{B}$ we have $\frac{S^w}{E^w} = 0$, the function is continuous at that point. However,

$$B \downarrow B^* \left(\tilde{B} \right) \implies \frac{S^w}{E^w} \downarrow 1, \text{ and } q \downarrow \delta^{-1} > 1,$$

at that point. Thus, $\frac{Q}{q^e} = \delta$ at $B = B^* \left(\tilde{B} \right)$ (and at its right limit) and $Q/q^e = 1$ at the left limit of this point. Namely, the function Q/q^e is discontinuous at B^* .

Analysis of the Marginal Borrowing Price. Next, we investigate the behavior of the marginal borrowing price. Recall that this is the price of consumption at which the worker trades off future consumption when he borrows marginally. Let B_{t+1} be the debt level the worker chooses today for next period and $\tilde{B}_{t+1}$ the next period's SBL. We have that $\tilde{q}_{t+1}^b = \tilde{q}^b \left(B_{t+1}, \tilde{B}_{t+1} \right)$ where:

$$\tilde{q}^b \left(B, \tilde{B} \right) \equiv q \left(\mu \left(B, \tilde{B} \right) \right) \lesssim_{[B \geq \tilde{B}]} + \left(1 - \lesssim_{[B \geq \tilde{B}]} \right).$$

Then,

$$\tilde{q}^b \left(B, \tilde{B} \right) = \begin{cases} 1 & B < B^* \left(\tilde{B} \right), \\ 1 & B \in \left[B^* \left(\tilde{B} \right), \tilde{B} \right], \\ q & B > \tilde{B}. \end{cases}$$

We have observed that for $B = B^* \left(\tilde{B} \right)$, all consumption is spot and thus $q = 1$. However, at $B = \tilde{B}$, the function features a discontinuity since:

$$\lim_{B \downarrow \tilde{B}} \tilde{q}^b \left(B, \tilde{B} \right) = q \left(\mu \left(\tilde{B}, \tilde{B} \right) \right) = q \left(1 - (1 - \beta) \tilde{B} \right) > 1.$$

Next we investigate the behavior of $\frac{Q}{q^b}$.

For any $B < B^* \left(\tilde{B} \right)$, both q^b and Q must equal 1, thus, $\frac{Q}{q^b} = 1$. Then, we have that for $B \in \left[B^* \left(\tilde{B} \right), \tilde{B} \right]$, it must be the case that $\frac{Q}{q^b} = Q$ because $q^b = 1$. Finally, when $B > \tilde{B}$,

$\tilde{q}^b(B, \tilde{B}) = Q = q$. Thus, we have:

$$\frac{Q}{\tilde{q}^b} = \begin{cases} 1 & B < B^*(\tilde{B}), \\ Q & B \in [B^*(\tilde{B}), \tilde{B}], \\ 1 & B > \tilde{B}, \end{cases}$$

where

$$Q = \left(\frac{1}{q} \left(1 - \frac{S^w}{E^w} \right) + \frac{S^w}{E^w} \right)^{-1}$$

and $\lim_{B \downarrow B^*(\tilde{B})} Q = 1$.

Average Price Elasticity. Next, we derive the elasticity of the average price with respect to total debt, in equilibrium—i.e., after replacing $E^w = 1 - (1 - \beta)B$. This elasticity is critical for the Ramsey policy analysis.

We have that the average price is the harmonic mean:

$$Q = \frac{1}{\frac{1}{q} \left(1 - \frac{S^w}{E^w} \right) + \frac{S^w}{E^w}}.$$

where both the price q , E^w , and S^w are functions of B .

Thus, we have that:

$$\frac{\partial Q}{\partial B} = -Q \cdot \frac{\left(1 - \frac{S^w}{E^w} \right) \frac{\partial}{\partial B} \left[\frac{1}{q} \right] - \left(\frac{1}{q} - 1 \right) \frac{\partial}{\partial B} \left[\frac{S^w}{E^w} \right]}{\left(\frac{1}{q} \left(1 - \frac{S^w}{E^w} \right) + \frac{S^w}{E^w} \right)}.$$

The numerator has two additional derivatives. The first one is:

$$\frac{\partial}{\partial B} \left[\frac{S^w}{E^w} \right] = \begin{cases} 0 & B < B^*(\tilde{B}) \\ \left[\frac{\tilde{B} - B}{1 - (1 - \beta)B} \right] \left(\frac{(1 - \beta)}{1 - (1 - \beta)B} - \frac{1}{\tilde{B} - B} \right) & B \in [B^*(\tilde{B}), \tilde{B}) \\ 0 & B \geq \tilde{B} \end{cases}$$

where the term in the intermediate region follows from:

$$\frac{\partial}{\partial B} \left[\frac{S^w}{E^w} \right] = \frac{S^w}{E^w} \left(\frac{1}{S^w} \frac{\partial}{\partial B} [S^w] - \frac{1}{E^w} \frac{\partial}{\partial B} [E^w] \right),$$

but then for $B \in [B^*(\tilde{B}), \tilde{B})$

$$\frac{\partial}{\partial B} [S^w] = -1,$$

and

$$\frac{\partial}{\partial B} [E^w] = -(1 - \beta).$$

Hence:

$$\frac{\partial}{\partial B} \left[\frac{S^w}{E^w} \right] = \left[\frac{\tilde{B} - B}{1 - (1 - \beta) B} \right] \left(\frac{(1 - \beta)}{1 - (1 - \beta) B} - \frac{1}{\tilde{B} - B} \right) < 0.$$

We evaluate the limits of this function. Clearly, at

$$\frac{\partial}{\partial B} \left[\frac{S^w}{E^w} \right] \Big|_{B \uparrow \tilde{B}} = -\frac{1}{1 - (1 - \beta) \tilde{B}} < -1,$$

and at

$$\begin{aligned} \frac{\partial}{\partial B} \left[\frac{S^w}{E^w} \right] \Big|_{B \downarrow B^*} &= \frac{(1 - \beta) (\tilde{B} - B^*(\tilde{B}))}{\left(1 - (1 - \beta) B^*(\tilde{B}) \right)^2} - \frac{1}{\left(1 - (1 - \beta) B^*(\tilde{B}) \right)} \\ &= \frac{(1 - \beta) (\tilde{B} - B^*(\tilde{B})) - \left(1 - (1 - \beta) B^*(\tilde{B}) \right)}{\left(1 - (1 - \beta) B^*(\tilde{B}) \right)^2} \\ &= \frac{(1 - \beta) \tilde{B} - 1}{\left(1 - (1 - \beta) B^*(\tilde{B}) \right)^2} \\ &= -1. \end{aligned}$$

The last steps follows from the definition of B^* . Hence, this derivative is discontinuous.

For the second derivative of interest, recall that:

$$\frac{\partial}{\partial B} [q] = \frac{\partial}{\partial \mu} [q] \cdot \frac{\partial \mu}{\partial B},$$

and

$$\frac{\partial \mu}{\partial B} = \frac{\partial}{\partial B} [E^w] - \frac{\partial}{\partial B} [S^w]$$

since $\mu = qX = E^w - S^w$. Then, we have that:

$$\frac{\partial}{\partial B} [q] = q_\mu \mu \frac{\frac{\partial}{\partial B} [E^w] - \frac{\partial}{\partial B} [S^w]}{E^w}.$$

We can express this as:

$$\frac{\partial}{\partial B} [q^{-1}] = \frac{1}{q} \epsilon_\mu^q \frac{\frac{\partial}{\partial B} [E^w] - \frac{\partial}{\partial B} [S^w]}{E^w - S^w}.$$

Hence, we obtain:

$$\frac{\partial Q}{\partial B} \frac{1}{Q} = -Q \cdot \left\{ \left(1 - \frac{S^w}{E^w} \right) \frac{1}{q} \epsilon_\mu^q \frac{\frac{\partial}{\partial B} [E^w] - \frac{\partial}{\partial B} [S^w]}{E^w - S^w} - \left(\frac{1}{q} - 1 \right) \frac{S^w}{E^w} \left[\frac{\frac{\partial}{\partial B} [S^w]}{S^w} - \frac{\frac{\partial}{\partial B} [E^w]}{E^w} \right] \right\}.$$

E. Empirical Appendix

E.1 Additional Results

Table 3: Heterogeneous Effect by Size of Amounts Sent/Received

Dependent variable: Prob. firm j makes a payment at time t

	(1)
$received_{j,t-1} \times smaller\ sent_{jt}$	0.526 (0.002)***
$received_{j,t-1}$	-0.109 (0.001)***
Adjusted R ²	0.535
Observations	10,001,248
clusters	222,306
$t, j \times d$ FE	Yes

Notes: The independent variable $received_{j,t}$ is an indicator equal to one if firm j received a transfer from another firm within the last 24 hours. This variable is interacted with $smaller\ sent_{jt}$, which equals one if the amount received by firm j at time $t - 1$ is less than the amount sent by firm j at time t , and zero otherwise. Robust standard errors, clustered by firm, are in parentheses. We include time and firm $\times$ weekday fixed-effects. Data is daily and spans 2024.

E.2 Intensive Margin Regressions

We also study the intensive margin relationship between payments made and received. We follow equation (1), but now considering the (inverse hyperbolic sine of the) amounts sent and received by firm j . Results are presented in Table 4 consistent with those of the extensive margins; firms become more likely to make payments once they receive money transfers. The elasticity across all firms in column (1) implies that a 1% increase in the amounts received by a firm on a given day increases the amounts sent the next day by approximately 0.04%.

Table 4: Triggered Payments Across Firms—Intensive Margin*Dependent variable: Payments made by firm j at time t (asinh)*

	(1)	(2)	(3)	(4)	(5)	(6)
$\ln \text{amount received}_{j,t-1}$	0.036	0.017	0.045	0.044		0.036
(asinh)	(0.0005)***	(0.001)***	(0.001)***	(0.001)***		(0.0005)***
$\text{Small}_j \times \ln \text{amount received}_{j,t-1}$		0.016				
		(0.001)***				
$\frac{\text{Assets}_j}{\text{Sales}_j} \times \ln \text{amount received}_{j,t-1}$			-0.019			
			(0.001)***			
$\frac{\text{Liabilities}_j}{\text{Sales}_j} \times \ln \text{amount received}_{j,t-1}$				-0.018		
				(0.001)***		
$\ln \text{amount received}_{j,t-5}$					0.012	0.012
(asinh)					(0.0004)***	(0.0004)***
Adjusted R ²	0.496	0.464	0.497	0.497	0.495	0.496
Observations	10,001,248	5,290,408	8,399,599	8,399,599	10,001,248	10,001,248
Clusters	222,306	74,566	174,896	174,896	222,306	222,306
$t, j \times \text{d FE}$	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The independent variable in column (1) is the total amount (in colones) received by firm j from another firm within the last 24 hours, transformed using the inverse hyperbolic sine function. This variable is interacted with firm j 's size, assets-to-sales and liabilities-to-sales in columns (2), (3), and (4) respectively. Columns (5) and (6) add an independent variable equal to the total amount (in colones) received by firm j from another firm 5 days before time t , transformed using the inverse hyperbolic sine function. Robust standard errors, clustered by firm, are in parentheses. We include time and firm $\times$ weekday fixed-effects. Data is daily and spans 2024.

E.3 Placebo Exercises

Table 5: Placebos—Receiving Payments Right After Making a Transfer

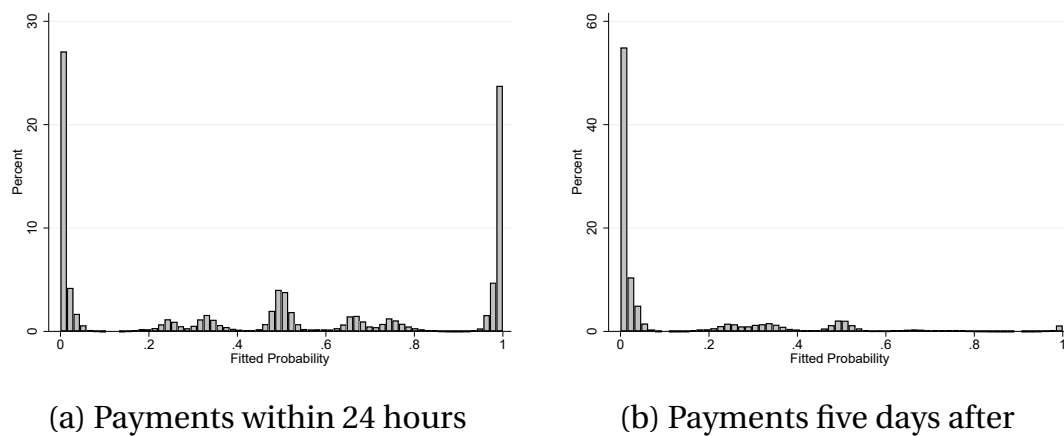
Dependent variable: Payment made by firm j at time t

	A. Extensive margin				B. Intensive margin (<i>asinh</i>)		
	$h = 1$	$h = 2$	$h = 3$		$h = 1$	$h = 2$	$h = 3$
	(1)	(2)	(3)		(4)	(5)	(6)
$received_{j,t+h}$	-0.001 (0.0004)***	-0.003 (0.0004)***	-0.001 (0.0004)	$\ln amount$	-0.001 (0.0004)	-0.002 (0.0004)***	0.000 (0.0004)
Adjusted R ²	0.483	0.483	0.483		0.495	0.495	0.495
Observations	10M	10M	10M		10M	10M	10M
Clusters	222,306	222,306	222,306		222,306	222,306	222,306
$t, j \times d$ FE	Yes	Yes	Yes		Yes	Yes	Yes

Notes: In panel A, the independent variables are indicators equal to one if firm j received a payment in $t + h$. In panel B, the independent variables are the total amounts (in colones) received by firm j from another firm in $t + h$, transformed using the inverse hyperbolic sine function. Robust standard errors, clustered by firm, are in parentheses. We include time and firm $\times$ weekday fixed-effects. Data is daily and spans 2024.

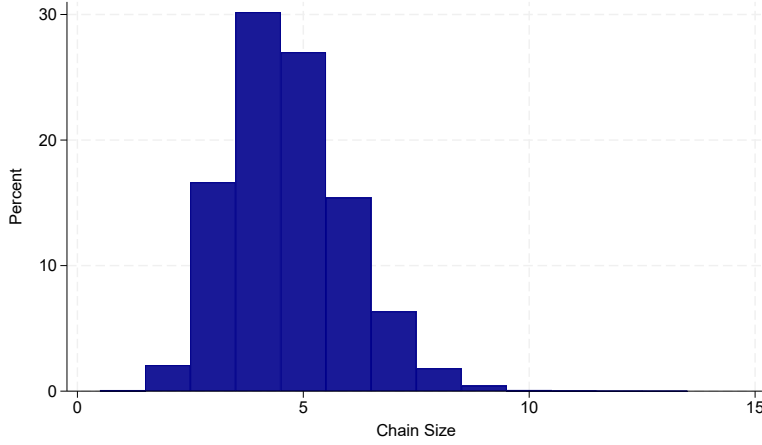
E.4 Results on Firm Classification and Chains

Figure 13: Histogram of Fitted Probabilities of Triggered Payments by Firm



Notes: The figures show the distribution of monthly average triggered payments by firm based on an extensive margin estimation. Firms with a value of zero have a payment behavior that is orthogonal to receiving money transfers, while firms with a value of one have a perfect correlation between receiving a transfer and making a payment either the day after (panel a) or five days later (panel b). Data is monthly and spans 2024.

Figure 14: Distribution of Chain Sizes



Notes: The figure shows the distribution of chain sizes based on a 5% random sample of spot firms in the economy used as initial nodes. Data is collapsed for December 2024 to construct the chains.

E.5 Chained Payments and Prices

We consider the following regression to explore the role of deferred payments on prices:

$$\ln Price_{vpjt} = \delta + \gamma Deferred_{vpjt} + \lambda_p + \lambda_j + \lambda_t + \varepsilon_{vpjt}$$

for transaction v , where $Deferred_{vpjt} = 1$ if sale v has a deferred payment and zero otherwise, and we include product, firm, and time fixed effects. We can also interact $Deferred_{vpjt}$ with an indicator, $Chained_{kt} = 1$, which depends on whether firm k —which made payment v to firm j —was classified as chained.

Column (1) of Table 6 shows that prices for a given product are 21% higher in transactions with a deferred payment. Column (2) shows this effect is larger for firms classified as chained. Column (3) further breaks down transactions depending on the type of both the receiver (j) and sender (k) of the payment. Reassuringly, a higher price is charged only if the payment is made by a chained firm, regardless of the receiver's type.

E.6 Payments and Timing of Delivery

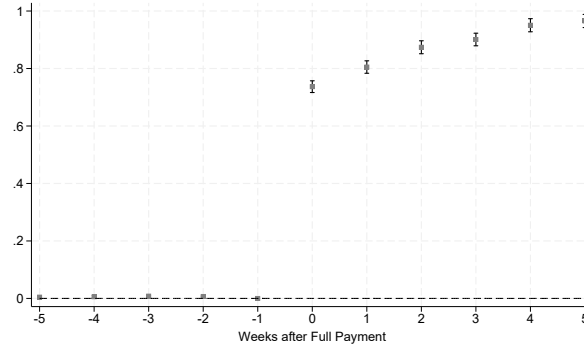


Figure 15: Timing of Product Delivery

Notes: The figure shows an event study for orders with deferred payment in which product sales become positive after the *full* payment is due. Time zero corresponds with the date in which the full payment is due and the coefficient represents the sales of the product at the store.

Table 6: Chained Payments and Prices

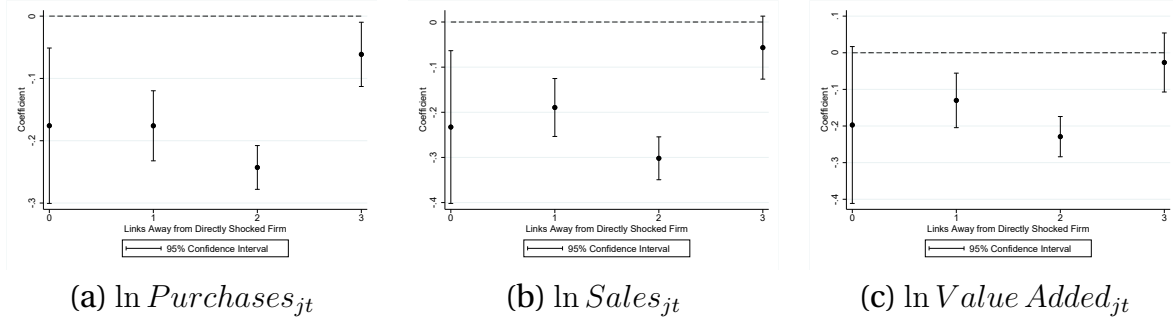
Dependent variable: Price of transaction v for good p sold by firm j to firm k at time t (log)

	(1)	(2)	(3)
$Deferred_{vpjt}$	0.189 (0.006)***	0.111 (0.006)***	0.181 (0.006)***
$Chained_{kt} \times Deferred_{vpjt}$		0.091 (0.006)***	
$Chained_{jt} \& Chained_{kt} \times Deferred_{vpjt}$			0.065 (0.002)***
$Spot_{jt} \& Chained_{kt} \times Deferred_{vpjt}$			0.058 (0.003)***
$Spot_{jt} \& Spot_{kt} \times Deferred_{vpjt}$			0.006 (0.004)
Adjusted R ²	0.682	0.682	0.682
Observations	45,801,998	45,801,998	45,801,998
Clusters	175,829	175,829	175,829
p, j, t FE	Yes	Yes	Yes

Notes: The independent variable in column (1) is an indicator equal to 1 if transaction v 's payment was deferred, as opposed to having an immediate payment. This variable is interacted with a dummy that depends on firm k 's classification as chained or spot in column (2). Column (3) further decomposes transactions depending on whether firm j and k were chained or spot at t ; the omitted category has firm j as chained and firm k as spot. Robust standard errors, clustered by firm, are in parentheses. We include product, firm, and time fixed-effects. Data is by sale and spans 2024.

E.7 Persistence of Desyfin Effect Along the Chain

Table 7: Desyfin Effect Along the Chain: Firms Upstream From Directly Hit Firm



Notes: The figure shows how the impact of being upstream from a Desyfin firm evolves as increase the number of links of separation along the chain. The dependent variables in panels (a), (b), and (c) are the (log) of purchases, sales, and value added, respectively, of firm j in month t . The plotted coefficients correspond with the triple interaction of $Links(n)_{jt} \times Switch2chained_{it} \times Desyfin_i \times Shock_t$, and captures the impact of being n links away from a Desyfin firm i ($Links(n)_{jt}$) and firm i switching to being classified as chained *after* the freeze. Robust standard errors are clustered by firm. We include province $\times$ industry (ISIC-4) and time fixed-effects. Data is monthly and spans 2024.